

An application of Newton–Puisseux charts to the Jacobian problem

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ABSTRACT

We study 2-dimensional Jacobian maps using so-called Newton–Puisseux charts. These are multi-valued coordinates near divisors of resolutions of indeterminacies at infinity of the Jacobian map in the source space as well as in the target space. The map expressed in these charts takes a very simple form, which allows us to detect a series of new analytical and topological properties. We prove that the Jacobian Conjecture holds true for maps (f, g) whose topological degree is ≤ 5 , for maps with $\gcd(\deg f, \deg g) \leq 16$ and for maps with $\gcd(\deg f, \deg g)$ equal to 2 times a prime.

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1. Introduction

A polynomial map $\mathbb{C}^2 \rightarrow \mathbb{C}^2$

$$(x, y) \rightarrow (f(x, y), g(x, y))$$

which satisfies the Jacobian equation

$$\text{Jac}(f, g) = \begin{vmatrix} \partial f / \partial x & \partial f / \partial y \\ \partial g / \partial x & \partial g / \partial y \end{vmatrix} = 1$$

is called the Jacobian map. The Jacobian Conjecture states that any Jacobian map is invertible.

There are two major approaches to the 2-dimensional Jacobian Conjecture: algebraic and topological. In fact, these approaches are not that different; they represent different sides of a wider common approach.

The algebraic approach was developed mainly by Abhyankar [1]. It relies on expansions of the component polynomials f, g and of the Jacobian $\text{Jac}(f, g)$ with respect to quasi-homogeneous gradations defined by means of edges of the Newton diagrams of the polynomials f, g . One obtains similarity of the Newton diagrams, reduction of their sizes and obstructions to the highest quasi-homogeneous parts of f and g . Further development in this direction was continued in the works of Moh [19], Appelgate and Onishi [2], Heitmann [14], Oka [25], Nagata [20,21], Nowicki and Nakai [24] and others. In particular, the Jacobian Conjecture was proved for maps with $\deg P \leq 101$, with $\gcd(\deg f, \deg g)$ being a prime number and with $\gcd(\deg f, \deg g) < 16$.

The topological approach was initiated by Vitushkin [31,32]. It essentially relies on resolution of indeterminacies of f and g at infinity. One obtains a holomorphic map between compact algebraic surfaces, which should be ramified (for a counterexample to the Jacobian Conjecture). In [31] an example of such a map is constructed in the topological category; it is 3-fold and is ramified along some divisor at ‘infinity’. The paper [31] is very technical, but later Orevkov [29] explained Vitushkin’s construction in terms of knots and fundamental groups (see also [34]). Recently, Egorov [10] has repeated Vitushkin’s arguments in constructing a 5-fold map with two divisors of ramification.

Orevkov continued the study of the topology of Jacobian maps. In [28,30] he constructed examples of holomorphic maps from an open surface to $\mathbb{C}^2$ which are non-ramified outside a divisor with self-intersection $+1$ (like the line at infinity).

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Notations

f, g	polynomials
P	Jacobian map
$\hat{f}, \hat{g}, \hat{P}$	corresponding rational maps
$\mathcal{X}^0 = \mathbb{C}^2$, $\mathcal{X} = \mathbb{CP}^2$, $\mathcal{X}_\infty$	the source space, its closure and the line at infinity
$\mathcal{Y}^0 = \mathbb{C}^2$, $\mathcal{Y} = \mathbb{CP}^2$, $\mathcal{Y}_\infty$	the target space, its closure and the line at infinity
$\pi : \mathcal{Z} \rightarrow \mathcal{X}$	resolution of indeterminacies
$\theta, \tilde{\theta}$	N–P chart and N–P alteration chart
u, v	variables in $\theta, \tilde{\theta}$
γ_j, γ	exponents in θ
l_j, l, k	exponents in $\tilde{\theta}$
(r_j, k_j)	characteristic pairs for θ
$k^{(j)} = k_1 \dots k_j$	
a_j	coefficients in θ
$\tilde{f} = f \circ \theta$, $\tilde{g} = g \circ \theta$	
p, q	leading exponents on $\tilde{f}, \tilde{g}$
m, n	relatively prime and proportional to $\deg f, \deg g$
φ, ψ	leading coefficients in $\tilde{f}, \tilde{g}$
$\chi = \varphi^{1/m}$, $p_0 = p/m$	
$\Delta(f)$, $\Delta(f_d)$, $\Gamma(f)$, $\Gamma(f_d)$	Newton polygons and Newton diagrams
Δ_0, Γ_0	web Newton polygon and web Newton diagram
(α, β)	root vertex of Γ_0
l, l', j	edges in Newton diagrams
f_l	polynomial associated with l
$\mathfrak{A}, \mathfrak{S}, \mathfrak{N}$	intersection graph, splice diagram and Newton–Puiseux graph
$\Theta, \tilde{\Theta}$	N–P charts in the target space
U, V	variables in $\Theta, \tilde{\Theta}$
δ_j, δ	exponents in Θ and in quasi-Puiseux expansion
(ξ_j, η_j)	characteristic pairs for Θ
$\eta^{(j)} = \eta_0 \dots \eta_j$	
$b_j, c(u), \tilde{c}(z)$	coefficients in Θ and in quasi-Puiseux expansion
$\tilde{g}_N$	non-constant part of quasi-Puiseux expansion
$q_N, \tilde{\psi}_N$	leading exponent and leading coefficient in $\tilde{g}_N$
ϵ_i	exponents in φ
s	number of ‘essential’ factors in φ
$\nu = [(1 - \gamma) \deg \varphi / p - 1] / k_d$	quantity in Proposition 4.2
$S(P), S_j$	the non-properness set and its components
$\mu_z(\cdot)$	multiplicity of a map at point z
μ_D	typical ramification index along divisor D
$\deg_{\text{top}} D$	multiplicity of divisor D

He and A. Domrına proved that the Jacobian Conjecture holds true for maps of geometrical degree ≤ 4 [27,8,7]. Orevkov’s approach uses so-called splice diagrams (introduced by Eisenbud and Neumann [11] in knot theory).

There was an attempt to prove the Jacobian Conjecture by showing topological equisingularity (or C^0 -sufficiency) of the family of curves $\{f = \lambda\}$, see the works of Lê Dũng Tráng, Michel and Weber [16–18]. They also used diagrams to encode the resolution of singularities of the foliation $\{f = \lambda\}$.

It seems that the fundamental problem with the Jacobian Conjecture is the lack of a simple and effective formalism for the investigation of polynomial maps near infinity.

The aim of the present paper is to introduce new tools which unify the algebraic and topological approaches. We define the so-called Newton–Puiseux charts. In the source space the Newton–Puiseux charts are charts near some special divisors obtained after resolution of indeterminacies of P at infinity. On the other hand, they generalize the notion of weighted gradation associated with edges of the Newton diagrams of f and g . Their advantage relies on the simplicity of the Jacobian equation. Moreover, they are not overloaded with abstract formalism, they are only multi-valued changes of coordinates.

We use the Newton–Puiseux charts also in the target space. Then the representation of the Jacobian map leads to the so-called quasi-Puiseux expansion, which is an expansion of g in powers of f with many constant coefficients. The first non-constant coefficient is expressed via some Schwarz–Christoffel integral. Asymptotic analysis of these integrals yields certain

restrictions on the polynomials f and g ; (in the algebraic approach these restrictions are obtained in a series of technical lemmas).

Using the Newton–Puiseux expansions (in the source space) and the quasi-Puiseux expansions (in the target space) we can easily calculate the ramification indices of dicritical divisors of resolution. Moreover, it is possible to obtain a subtler characterization of the divisors of resolution; we shall divide them into four types: Abhyankar, non-properness, transitory and infinite.

The effectiveness of our method is demonstrated in a direct proof of the Orevkov–Domrina theorem and its generalization to 5-sheeted maps. As another application we present an elementary proof of the Jacobian Conjecture in the cases $\gcd(\deg f, \deg g)$ equals 2 times a prime (improvement of the Appelgate–Onishi theorem) and $\gcd(\deg f, \deg g) \leq 16$ (improvement of Heitmann’s result).

We hope that the present paper will form a solid basis in the search for a counter-example to this conjecture. (After failing to get a contradiction in the analysis of Jacobian maps, the author began to be more and more convinced about the existence of such a counter-example).

Here we do not present the history and main results of the Jacobian problem. The interested reader is referred to the survey articles by Bass, Connell and Wright [4], by Drużkowski [9], by van den Essen [12], by the author [34] and to the recent book of van den Essen [13].

The plan of the paper is the following. In Section 2 we define Newton–Puiseux charts and use them in the resolution of indeterminacies at infinity of one polynomial. In Section 3 we resolve the indeterminacies of a non-invertible Jacobian map; there we introduce the quasi-Puiseux expansion and give a classification of the Newton–Puiseux charts (Theorem 3.14). Sections 2 and 3 form a basis of our study of the Jacobian maps. So, at the end of Section 3 we outline the results proved in the further sections. Here we only give localization of the main theorems: the generalizations of the Appelgate–Onishi and Heitmann theorems (Theorems 4.7, 4.12 and 4.16) are given in Section 4 and in Section 6 we generalize the Domrina–Orevkov’s result (Theorem 6.12). In the Appendix we prove some lemmas about Schwarz–Christoffel integrals and we provide a list of the main notations.

2. Resolution of indeterminacies of a polynomial via Newton–Puiseux charts

Let $f = f(x, y) : \mathbb{C}^2 \rightarrow \mathbb{C}$ be a polynomial. In applications we shall consider the case when

$$f = x^a y^b + \text{lower order terms.} \quad (2.1)$$

We fix this system of coordinates; (in fact, the changes $(x, y) \rightarrow (x - x_0, y - y_0)$ are admissible, but later we shall remove this non-uniqueness). We assume also that

$$f(0, 0) \neq 0. \quad (2.2)$$

We shall denote $\mathcal{X}^0 = \mathbb{C}^2$, $\mathcal{X} = \mathbb{CP}^2$ (the closure of $\mathcal{X}^0$) and $\mathcal{X}_\infty = \mathcal{X} \setminus \mathcal{X}^0$ (the line at infinity).

The polynomial f defines a rational map $\hat{f} : \mathcal{X} \rightarrow \mathbb{CP}^1$. However $\hat{f}$ is not a map in the usual set-theoretic sense. It is defined by means of its graph $G(\hat{f}) \subset \mathcal{X} \times \mathbb{CP}^1$, which is the closure of the graph $G(f) \subset \mathcal{X}^0 \times \mathbb{C}^1$. It is possible that the image of a point is the whole of $\mathbb{CP}^1$.

In case (2.1) there are two candidates for the indeterminacy points: $A_{\text{left}} = (0 : 1 : 0)$, $A_{\text{right}} = (1 : 0 : 0)$. For example, near the point A_{left} we can introduce the projective coordinates $z = \frac{1}{y}$, $u = \frac{x}{y}$. Then we get

$$\hat{f}(u, z) = f(u/z, 1/z) = \frac{f_1(u, z)}{z^{a+b}} = \frac{u^a + O(z)}{z^{a+b}},$$

i.e. a ratio of polynomials which vanish simultaneously at the point $A_{\text{left}} : u = z = 0$. The indeterminacy of $\hat{f}$ means that, when we approach A_{left} along different curves, then f tends to different values.

Definition 2.1. The **resolution of indeterminacy** of $\hat{f}$ consists of a 2-dimensional complex manifold $\mathcal{Z}$ and a modification map

$$\pi : \mathcal{Z} \rightarrow \mathcal{X} \quad (2.3)$$

(which is a composition of elementary blow-ups of points) such that the composition

$$\hat{f} \circ \pi : \mathcal{Z} \rightarrow \mathbb{CP}^1$$

is a holomorphic definite map. (Sometimes this resolution is called the *compactification* of f .)

The space $\mathcal{Z}$ is obtained from $\mathcal{X}^0$ by adding divisors $D_i \simeq \mathbb{CP}^1$, components of the algebraic curve $\pi^{-1}(\mathcal{X}_\infty)$. One component is the strict transform of the line at infinity $\mathcal{X}_\infty = \mathcal{X} \setminus \mathbb{C}^2$ (i.e. the closure of $\pi^{-1}(\mathcal{X}_\infty \setminus \{A_{\text{left}}, A_{\text{right}}\})$); we denote it by D_0 . Other components are divisors in $\pi^{-1}(A_{\text{left}})$ (the **left divisors**) and in $\pi^{-1}(A_{\text{right}})$ (the **right divisors**).

One defines the **intersection graph** $\mathfrak{A}(f)$ of the resolution $(\mathcal{Z}, \pi)$ as follows (see [27]). Its vertices correspond to the divisors D_i and two vertices are connected by an edge if the corresponding divisors have a point of intersection. Of course, $\mathfrak{A}(f)$ is a tree graph. Below we slightly modify the intersection graph by adding some auxiliary external vertices.

A divisor $D \in \mathfrak{A}(f)$ is called **dicritical for f** if $\hat{f} \circ \pi|_D \neq \text{const}$. This means that the foliation defined by the curves $C_\lambda := \{\hat{f} \circ \pi = \lambda\}_{\lambda \in \mathbb{C}}$ is generically transversal to D .

The following result is well known.

Theorem 2.2 ([27, 17]). For any polynomial f there is a resolution $(\mathbb{Z}, \pi)$ of its indeterminacies which has the following property.

The subspace $(\hat{f} \circ \pi)^{-1}(\infty)$ is a connected union of some divisors D_j , defining a subtree $\mathfrak{A}_\infty(f)$ of $\mathfrak{A}(f)$, and each connected component $\mathfrak{B}$ of $\mathfrak{A}(f) \setminus \mathfrak{A}_\infty(f)$ is a bamboo (a tree without ramifications) containing a unique vertex-divisor D , which is dicritical for f and intersects $(\hat{f} \circ \pi)^{-1}(\infty)$ at one point (which we shall denote simply by ∞).

By the bamboo we mean a linear chain $-\circ-\circ-\cdots-\circ-\circ-$.

We complete Definition 2.1 and Theorem 2.2 with the following

Proposition 2.3. With any divisor D in $\mathfrak{A}(f)$ we can associate in a canonical way two rational functions $v_D(x, y)$ and $w_D(x, y)$ such that:

- $\mathbb{C}(x, y) = \mathbb{C}(v_D, w_D)$,
- $w_D \circ \pi|_D : D \rightarrow \mathbb{CP}^1$ is one-to-one,
- $D \subset \{v_D \circ \pi = 0\}$ and $v_D \circ \pi$ is typically regular on D .

Proof. Consider the resolution of A_{left} (the A_{right} case is analogous). Assume that the divisors $D = D_i \in \pi^{-1}(A_{\text{left}})$ of the resolution are ordered in such a way that: if $D_i \in \mathfrak{A}_\infty(f)$ and D_j belongs to a subtree of $\mathfrak{A}(f)$ with root at D_i (i.e. in the resolution of a point in D_i), then $D_i < D_j$; (it means that the order of the pole of $f \circ \pi$ at D_i is greater than the order of this pole at D_j). Moreover, along any bamboo $\mathfrak{B}$ from Theorem 2.2 the order is linear with the dicritical divisor D at the first place.

For D_0 (the strict transform of $\mathcal{X}_\infty$) we put $w_{D_0} = \frac{x}{y}$, $v_{D_0} = \frac{1}{y}$. When we blow-up a point $v_{D_1} = 0$, $w_{D_1} = a_1$, which is not a corner point, and we obtain a divisor D_j , then we put $w_{D_j} = (w_{D_1} - a_1)/v_{D_1}$, $v_{D_j} = v_{D_1}$. When we blow-up a corner point $D_i \cap D_j$, $D_i < D_j$, giving a divisor D_m , then we put $w_{D_m} = w_{D_j} w_{D_i}$, $v_{D_m} = w_{D_i}$. $\square$

Example 2.4. Consider the function

$$f = xy^2 - y + 1.$$

The corresponding functions v_D, w_D are given below:

D_j	$w_j = w_{D_j}$	$v_j = v_{D_j}$	f
D_0	xy^{-1}	y^{-1}	$(w_0 - v_0^2 + v_0^3)v_0^{-3}$
D_1	$w_0 v_0^{-1} = x$	$v_0 = y^{-1}$	$(w_1 - v_1 + v_1^2)v_1^{-2}$
D_2	$w_1 v_1^{-1} = xy$	$v_1 = y^{-1}$	$(w_2 - 1 + v_2)v_2^{-1}$
D_3	$(w_2 - 1)v_2^{-1} = y(xy - 1)$	$v_2 = y^{-1}$	$w_3 + 1$
D_4	$v_0^{-1} = y$	$v_0 w_0^{-1} = x^{-1}$	$(w_4^2 - w_4 v_4 + v_4)v_4^{-1}$
D_5	$w_4 v_4^{-1} = xy$	$v_4 = x^{-1}$	$1 + (w_5^2 - w_5)v_5$
D_6	$w_4 w_5 = xy^2$	$w_5 = xy$	$(w_6 + 1) - v_6$.

Generally, one has $f \circ \pi = (v_D \circ \pi)^p \Phi(w_D \circ \pi) + \cdots$ near D where Φ is a polynomial and p is an integer (equal to 0 for dicritical D).

By Proposition 2.3 each divisor $D \in \mathfrak{A}(f)$ has two distinguished points $0 = 0_D = D \cap \{w_D = 0\}$ and $\infty = \infty_D = D \cap \{w_D = \infty\}$. Any curve germ $w_D \circ \pi = \infty$ near D coincides with the germ of another divisor intersecting D at ∞ .

A curve germ $w_D \circ \pi = 0$ near D can be also a germ of another divisor, but it can lie outside the exceptional set as well.

This can be encoded in the graph $\mathfrak{A}(f)$; in the second case we attach the *boundary vertex* $-\bullet$ to the vertex D in $\mathfrak{A}(f)$.

If D is dicritical for f and λ is a typical value for f , then we attach to D as many *arrowhead vertices* $\rightarrow$ as exist local components near D of the curve $f \circ \pi = \lambda$.

Namely, such a modified graph will be treated as the **intersection graph**. In this way we obtain a diagram which we denote by $\mathfrak{A}(f)$. The vertices of $\mathfrak{A}(f)$ incident to at least three edges are called **nodes**.

The intersection graph $\mathfrak{A}(f)$ for the function $f = xy^2 - y + 1$ from Example 2.4 is presented at Fig. 1(a).

Now, following Orevkov [30, 8] we replace the latter diagram by the so-called **splice diagram** $\mathfrak{S}(f)$ defined via the rule: any maximal chain $-\circ-\cdots-\circ-$ in $\mathfrak{A}(f)$, consisting of edges and non-node vertices $\neq D_0$, is replaced by one edge.

$\mathfrak{S}(f)$ for $f = xy^2 - y + 1$ is presented at Fig. 1(b).

The splice diagram defined above would be the same as the splice diagram defined by Neumann [22] if we had added weights to some ends of edges.

In [11] and in [22] the splice diagrams were introduced to encode ‘links at infinity’ defined via the intersection of a large sphere $S_R \subset \mathbb{C}^2$ (of radius R) with an algebraic curve. In our situation this curve is $f = \lambda$ (for typical λ). The link $L = S_R \cap \{f = \lambda\}$ consists of knots L_j , which are iterated torus knots.

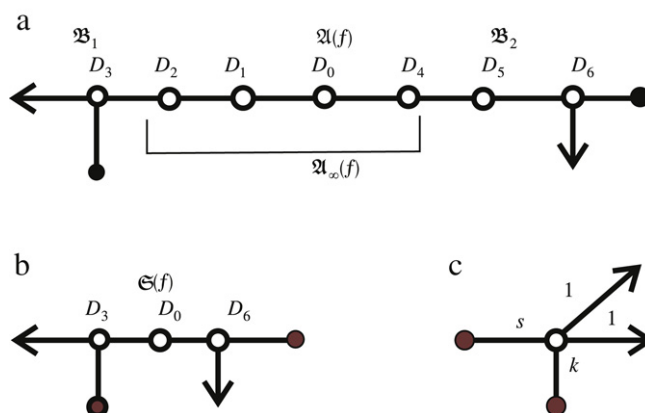


Fig. 1.

Recall that a knot K' is (s, k) -cable of a knot K if K' is a knot in the boundary $\partial N(K)$ of a tubular neighborhood $N(K)$ of K , such that K' is homologically equivalent to kK , $k > 0$, in $N(K)$ and has the linking number with K equal to $l(K', K) = s$, $s \in \mathbb{Z}$, $\gcd(s, k) = 1$. A knot K is an *iterated torus knot* if there is a series $K_0, \dots, K_n = K$ of knots such that K_0 is trivial and K_{j+1} is (s_{j+1}, r_{j+1}) -cable of K_j .

If a knot K is the trace in S_R of a local component C of the curve $f = \lambda$ near infinity, then the numbers s_j, k_j are calculated from the Puiseux expansion of the curve C . We can assume that $S_R = \{\max(|x|, |y|) = R\}$ and $K \subset \{|y| = R, |x| \ll |y|\}$. We have

$$x = x(y) = y^{r_1/k_1} \left[d_1 + \dots + y^{-r_2/k_1 k_2} \left[d_2 + \dots + y^{-r_n/k_1 \dots k_n} [d_n + \dots] \right] \right], \quad (2.4)$$

where $\gcd(r_j, k_j) = 1$, $r_2, r_3, \dots > 0$, $d_j \neq 0$ and the dots in the i -th summand contain only powers of $y^{-1/k_1 \dots k_i}$. Then we have

$$s_1 = r_1, \quad s_{j+1} = -r_j + k_j k_{j+1} s_j \quad (2.5)$$

(see [11]).

The integers k_j, s_j appear as weights of ends of edges at the nodes of the splice diagram. The whole splice diagram is obtained from elementary splice diagrams as in Fig. 1(c). The latter are Seifert links and the whole link L is the result of ‘splicing’ such elementary links. The details are in [11].

It is worthwhile noting that the integers k_j, s_j can be computed from the intersection matrix $(D_i \cdot D_j)_{i,j}$ of the exceptional divisors. Suitable formulas are given in [11] and in [30].

We shall avoid using splice diagrams with their rather complicated system of weights. Instead, we shall deal with initial parts of the Puiseux expansions associated with local (near infinity) components of the curve $f = \lambda$. These finite Puiseux series lead naturally to the Newton–Puiseux charts, which were used already by Newton [23] and which we define below.

Definition 2.5. The **left Newton–Puiseux chart** (or left N–P chart, or N–P chart) is a change of the form

$$\theta : (u, y) \rightarrow (x, y) = (a_1 y^{-\gamma_1} + a_2 y^{-\gamma_2} + \dots + a_{d-1} y^{-\gamma_{d-1}} + u y^{-\gamma}, y), \quad (2.6)$$

where $y \in (\widehat{\mathbb{C}}, \infty)$ (i.e. $y \rightarrow \infty$) and $u \in \mathbb{C}$. The **right Newton–Puiseux chart** is a change of the form

$$\theta : (x, u) \rightarrow (x, y) = (x, a_1 x^{-\gamma_1} + \dots + a_{d-1} x^{-\gamma_{d-1}} + u x^{-\gamma}), \quad (2.7)$$

where $x \rightarrow \infty$ and $u \in \mathbb{C}$.

In (2.6) and (2.7) a_i are constants and the rational exponents satisfy $-1 \leq \gamma_1 < \gamma_2 < \dots < \gamma_{d-1} < \gamma$. The natural number $d =: d(\theta)$ is called the **depth** of the chart θ , $\gamma =: \gamma(\theta)$ is called the **slope** of the left chart θ (respectively the **inverse slope** of the right chart θ).

Sometimes, in order to avoid multivaluedness of the Newton–Puiseux charts, we shall use the so-called Newton–Puiseux alteration charts. As a **left Newton–Puiseux alteration chart** (or left N–P alteration chart), associated with the multi-valued map θ defined in (2.6), we take the meromorphic map

$$\tilde{\theta} : (u, v) \rightarrow (x, y) = (a_1 v^{l_1} + a_2 v^{l_2} + \dots + a_{d-1} v^{l_{d-1}} + u v^l, v^{-k}), \quad (2.8)$$

where k is the smallest common denominator of the ratios $\gamma_1 = \frac{l_1}{k}, \dots, \gamma_{d-1} = \frac{l_{d-1}}{k}, \gamma = \frac{l}{k}$ and $\gcd(l_1, \dots, l_{d-1}, l, k) = 1$. Sometimes we shall denote $k =: k(\theta)$ and $l =: l(\theta)$. Analogously one defines **right Newton–Puiseux alteration charts**.

If θ is a Newton–Puiseux chart (left or right), then we can represent the quantities $\Delta\gamma_1 = \gamma_1$, $\Delta\gamma_2 = \gamma_2 - \gamma_1$, $\Delta\gamma_3 = \gamma_3 - \gamma_2, \dots$ as follows

$$\Delta\gamma_i = \frac{r_i}{k_1 \dots k_i}, \quad \gcd(r_j, k_j) = 1. \quad (2.9)$$

The pairs $(r_1, k_1), (r_2, k_2), \dots, (r_d, k_d)$ are called the **characteristic pairs** (or Newton–Puiseux pairs, see [11]) for θ . Therefore we have

$$\begin{aligned} k &= k_1 \dots k_d, \\ l_i &= r_1 k_2 \dots k_d + \dots + r_i k_{i+1} \dots k_d. \end{aligned} \quad (2.10)$$

Let us present some basic properties of the N–P charts.

Lemma 2.6. Any Newton–Puiseux alteration chart is a multivalued holomorphic map $\tilde{\theta} : \mathbb{C} \times (\mathbb{C}, 0) \rightarrow \mathbb{CP}^2$ of topological degree $k = k_1 \dots k_d$. Its multiplicities $\mu_{(u,v)} \tilde{\theta}$ (see Definition 6.3 in Section 6) of $\tilde{\theta}$ at points (u, v) are the following:

$$\begin{aligned} \mu_{(u,v)} \tilde{\theta} &= 1 \quad \text{if } v \neq 0, \\ \mu_{(u,0)} \tilde{\theta} &= k_1 \dots k_{d-1} \quad \text{if } u \neq 0, \\ \mu_{(0,0)} \tilde{\theta} &= k. \end{aligned}$$

In particular, $\tilde{\theta}$ restricted to $v = 0$ is k_d to 1.

Moreover, the restrictions $\tilde{\theta}|_{u=u_0} : v \rightarrow (x(u_0, v), y(u_0, v))$, $u_0 \neq 0$, define primitive parametrizations of the corresponding curves in the image, i.e. $\tilde{\theta}|_{u=u_0}$ is an embedding near $v = 0$.

Proof. The first statement is clear, there are k preimages of a typical point (x, y) (see (2.8)).

To compute other multiplicities we fix (u_0, v_0) and solve the equation $\tilde{\theta}(u, v) = \tilde{\theta}(u_0, v_0)$. We get $v = \zeta_j v_0$, $\zeta_j = e^{2\pi i j/k}$, $j = 0, \dots, k-1$, and

$$a_1(1 - \zeta_j^{l_1})v_0^{l_0} + \dots + a_{d-1}(1 - \zeta_j^{l_{d-1}})v_0^{l_{d-1}} + (u_0 - u\zeta_j^{l_d})v_0^{l_d} = 0. \quad (2.11)$$

If $v_0 \neq 0$, then we get k distinct solutions.

In the limit $v_0 \rightarrow 0$ we should have $1 - \zeta_j^{l_1} = \dots = 1 - \zeta_j^{l_{d-1}} = 0$. Since $\gcd(l_1, \dots, l_{d-1}) = k_d$, (see (2.10)), it should be $j = j_0 k_1 \dots k_{d-1}$, $j_0 = 0, \dots, k_{d-1}$ and we get k_d values for u (when $u_0 \neq 0$).

In the proof of the second statement of the lemma we use the same Eq. (2.11) with $u = u_0$. We get the conditions $1 - \zeta_j^{l_1} = \dots = 1 - \zeta_j^{l_d} = 0$ which imply $\zeta_j = 1$. $\square$

Remark 2.7. N–P alteration charts are examples of maps called alterations (see [26]). They differ from the modification maps (used in Hironaka’s resolution theorem), which are typically one-to-one. Using alterations one can get a very simple resolution of singularities of a subvariety.

We shall use the N–P charts to resolve indeterminacies at infinity of polynomials (and of Jacobian maps).

Definition 2.8. Let $f(x, y)$ be a polynomial satisfying (2.1) and (2.2) and let $\lambda \in \mathbb{C}$ be a typical value for f , i.e. the curve $f = \lambda$ is homeomorphic to the curves $f = \mu$ for μ close to λ . We regard λ as varying.

Let $C_\lambda^{(r)}$, $r = 1, \dots, R$, be the components of the curve $f = \lambda$ near infinity. Each $C_\lambda^{(r)}$ has Puiseux expansion of the form

$$x = a_1^{(r)} y^{-\gamma_1^{(r)}} + a_2^{(r)} y^{-\gamma_2^{(r)}} + \dots, \quad y \rightarrow \infty$$

or of the form

$$y = a_1^{(r)} x^{-\gamma_1^{(r)}} + a_2^{(r)} x^{-\gamma_2^{(r)}} + \dots, \quad x \rightarrow \infty.$$

Here some coefficients $a_1^{(r)}, \dots, a_{d_r-1}^{(r)}$ do not depend on λ ; let $a_{d_r}^{(r)}(\lambda)$ be the first which essentially depends on λ .

We take the following collection of N–P charts:

$$\begin{aligned} \theta_0 : x &= uy \\ \theta_1^{(r)} : x &= uy^{-\gamma_1^{(r)}}, \quad \dots, \quad \theta_{d_r}^{(r)} : x = a_1^{(r)} y^{-\gamma_1^{(r)}} + \dots + uy^{-\gamma_{d_r}^{(r)}} \end{aligned} \quad (2.12)$$

(and corresponding right N–P charts) for $r = 1, \dots, R$. We call the above system of charts the **Newton–Puiseux resolution of indeterminacies of f** .

There is a relation between the standard resolution of definiteness of f (see Definition 2.1) and the above resolution. However, this relation is not direct.

With any exceptional divisor $D \in \mathfrak{A}(f)$ we can associate a N–P chart θ_D as follows. By Proposition 2.3 we have the rational function $w_D(x, y)$ such that $w_D \circ \pi$ is non-constant on D . Then θ_D is defined as the corresponding finite Puiseux series of the Puiseux expansion of the curves $w_D = \mu$.

But not all N–P charts arising in this way belong to the set defining the Newton–Puiseux resolution. Certainly, the nodes in $\mathfrak{A}(f)$ have their analogues in the list (2.12). But the divisors from the bamboos in $\mathfrak{A}(f)$ do not have relatives in (2.12). The divisors which are present in the splice diagram $\mathfrak{S}(f)$ have their charts in (2.12).

Nevertheless, with any N–P chart from (2.12) we can associate a divisor $D_\theta \in \mathfrak{A}(f)$; if necessary, we enlarge $\mathfrak{A}(f)$ by some additional blow-ups.

Using the above relation $\theta \leftrightarrow D$ we can give an interpretation of the variables (u, y) in θ (respectively, (u, x) , (u, v)) as ‘coordinates’ near the affine part $D_\theta \setminus \infty$ of the divisor D_θ . Thus $u \circ \pi : (D_\theta \setminus \infty) \times (\mathbb{C}, 0) \rightarrow \mathbb{C}$, $v \circ \pi : (D_\theta \setminus \infty) \times (\mathbb{C}, 0) \rightarrow \mathbb{C}$ define a multi-valued chart near $D_\theta \setminus \infty$.

Moreover, by Lemma 2.6, the natural parameter on $D_\theta \setminus \infty$ is u^{k_d} , not just u .

Note also that if $\theta_D : x = a_1 y^{-\gamma_1} + a_2 y^{-\gamma_2} + \dots + a_{d-1} y^{-\gamma_{d-1}} + u y^{-\gamma}$ is associated with D , then the charts $x = a'_1 y^{-\gamma_1} + a'_2 y^{-\gamma_2} + \dots + a'_{d-1} y^{-\gamma_{d-1}} + u y^{-\gamma}$ such that $(a'_1)^{k_1} = a_1^{k_1}, \dots, (a'_{d-1})^{k_{d-1}} = a_{d-1}^{k_{d-1}}$ are also associated with D . We fix one of them, denoted θ_D .

It is easy to find the N–P charts which resolve indeterminacies of f using the Newton diagram of f .

Definition 2.9. Let $f = \sum_{(i,j) \in \text{Supp}(f)} f_{ij} x^i y^j$ be the Taylor expansion of the polynomial f . Here $f_{ij} \neq 0$ and the set $\text{Supp}(f) \subset \mathbb{Z}^2 \subset \mathbb{R}^2$ is called the **Newton support** of the polynomial f ; (by the assumption (2.2) $(0, 0) \in \text{Supp}(f)$). The convex hull (in $\mathbb{R}^2$) of the support $\text{Supp}(f)$ is called the **Newton polygon** of f ; we denote it by $\Delta(f)$. The boundary of the Newton polygon forms the **Newton diagram** of f , denoted by $\Gamma(f)$.

For any edge $I \subset \Gamma(f)$ we define the polynomial

$$f_I = \sum_{(i,j) \in I} f_{ij} x^i y^j,$$

i.e. the part of the expansion of f associated with I .

Assuming the form (2.1) of f , we call the vertex $(a, b) \in \Gamma(f)$ the **root** of $\Gamma(f)$. The edges I of $\Gamma(f)$ which lie on the left of the root and are not vertical are called **left edges**. Analogously the non-horizontal edges on the right of the root are called the **right edges**.

Algorithm 2.10 (Of the Newton–Puiseux Resolution of Indeterminacy of a Polynomial). Firstly we resolve the indeterminacy at A_{left} of the rational function f using the left Newton–Puiseux charts (2.6). This resolution is based on the Newton–Puiseux expansions of local irreducible components $C_\lambda^{(r)}$ of the typical curve $f = \lambda$ near the accumulation point A_{left} .

Step 0. We begin with the polynomial $f_1 := f = \sum f_{1;ij} x^i y^j$, $f_{1;ij} := f_{ij}$, $x_1 := x$ and its Newton diagram $\Gamma(f_1) = \Gamma(f)$.

Step 1. Choose a left edge $I_1 \subset \Gamma(f_1)$ with slope γ_1 . One has $(f_1)_{I_1} = \sum_{(i,j) \in I_1} f_{1;ij} x^i y^j = \varphi_{\theta_1}(u_1) y^{p(\theta_1)}$, where $u_1 = x_1 y^{\gamma_1}$ is given by the left N–P chart $\theta_1 : (u_1, y) \rightarrow (x, y)$, and $p(\theta_1) \geq 0$.

If $p(\theta_1) = 0$, then the resolution of a suitable component $C_\lambda^{(r)}$ is finished; in fact, here several components corresponding to roots $a_i(\lambda)$ of the equation $\varphi_{\theta_1}(u_1) = \lambda$ are resolved altogether. Then one takes another local component, i.e. one takes another edge of $\Gamma(f_1)$.

Assume that $p(\theta_1) > 0$. Choose $a_1 \neq 0$, a zero of the polynomial $\varphi_{\theta_1}(u_1)$; (the choice of the zero $a_1 = 0$, if it is a zero, would correspond to another choice of the edge I_1). Introduce a new variable x_2 by the formula $u_1 = a_1 + x_2$, i.e. $x_1 = (a_1 + x_2) y^{-\gamma_1}$, and define the Puiseux polynomial $f_2(x_2, y) := f_1((a_1 + x_2) y^{-\gamma_1}, y) = \sum f_{2;ij} x_2^i y^j$. We have the Newton diagram $\Gamma(f_2)$ which is contained in the half-plane $\{(i, j) : j \leq p(\theta_1)\}$. The horizontal line $\{j = p(\theta_1)\}$ contains the edge $I'_1 \subset \Gamma(f_2)$ associated with the edge $I_1 \subset \Gamma(f_1)$; I'_1 is obtained from I_1 by means of an affine transformation and (eventually) cutting or stretching.

After Step 1 one passes to Step 2.

Step m , $m > 1$. Assume that we have: a Puiseux polynomial $f_m = \sum f_{m;ij} x_m^i y^j$ its Newton diagram $\Gamma(f_m)$ with a distinguished horizontal edge $I'_{m-1} \subset \{j = p(\theta_{m-1})\}$, $p(\theta_{m-1}) > 0$ and separated from the vertical axis.

Choose an edge $I_m \subset \Gamma(f_m)$ on the left of I'_{m-1} (there are such edges) with (relative) slope $\Delta\gamma_m$, $0 < \Delta\gamma_m < \infty$, and put $\gamma_m = \gamma_{m-1} + \Delta\gamma_m$. We get $(f_m)_{I_m} = y^{p(\theta_m)} \varphi_{\theta_m}(u_m)$, where $p(\theta_m) < p(\theta_{m-1})$ and $u_m = x_m y^{\Delta\gamma_m}$. The variables (u_m, y) define the N–P chart θ_m of depth m . We can assume also that $p(\theta_m) \geq 0$; (this holds because $f_{m;00} = f_{00} = f(0, 0) \neq 0$ by assumption).

If $p(\theta_m) > 0$, then choose a_m , a nonzero root of the equation $\varphi_{\theta_m}(u_m) = 0$. We make the change $x_m = (a_m + x_{m+1}) y^{-\Delta\gamma_m}$. We get a Puiseux polynomial $f_{m+1}(x_{m+1}, y) = f_m(x_m, y)$ with its Newton diagram $\Gamma(f_{m+1})$ and a horizontal edge $I'_m \subset \Gamma(f_{m+1})$ corresponding to I_m . Next, one goes to Step $m + 1$.

If $p(\theta_m) = 0$, then the resolution of the component $C_\lambda^{(r)}$ (for typical λ 's) is finished, and one can return to Step 1 for resolution of another branch of $f = \lambda$.

The algorithm ends up after resolving all the components $C_\lambda^{(r)}$.

Analogously one defines the Newton–Puiseux resolution of indeterminacy of f at the point A_{right} .

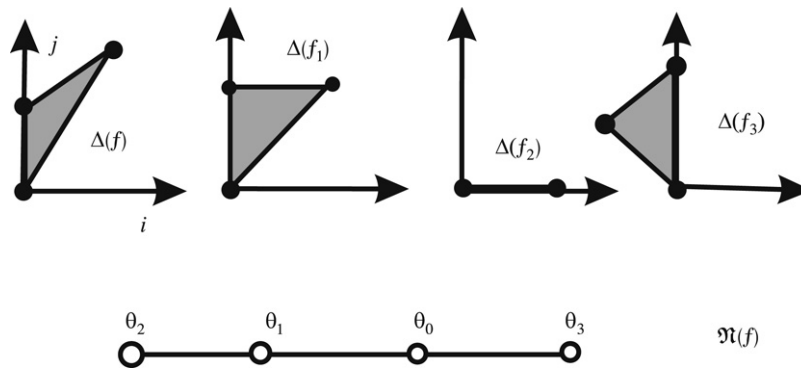


Fig. 2.

Using this algorithm we can encode the Newton–Puiseux resolution of indeterminacies of f by means of the following **Newton–Puiseux graph** $\mathfrak{N}(f)$:

- The vertices of $\mathfrak{N}(f)$ correspond to the N–P charts; θ_0 is called the **root**.
- Two vertices θ, θ' are joined by an edge in $\mathfrak{N}(f)$ if the corresponding edges l, l' in a Newton diagram $\Gamma(f_m)$ are adjacent; if l lies on the left of l' , then θ lies on the left of θ' . The root θ_0 is joined with the N–P charts corresponding to the two edges in $\Gamma(f)$ incident to the root $(m\alpha, m\beta)$.

It is easy to see that $\mathfrak{N}(f)$ is a tree graph with end-vertices corresponding to the dicritical divisors in $\mathfrak{N}(f)$.

For example, in the case $f = y(xy - 1) + 1$, we have two left Newton–Puiseux charts $\theta_1 : x = u_1 y^{-1}$ and $\theta_2 : x = y^{-1} + u_2 y^{-2}$ with $f \circ \theta_2 = u_2 + 1$ and one right Newton–Puiseux chart $\theta_3 : y = u x^{-1/2}$ with $f \circ \theta_3 = u^2 + 1 + O(x^{-1/2})$. Fig. 2 shows the corresponding Newton diagrams and the Newton–Puiseux graph. We have $\theta_1 = \theta_{D_2}$, $\theta_2 = \theta_{D_3}$ and $\theta_3 = \theta_{D_6}$ in terms of Fig. 1.

Observe that we can interpret the depth $d = d(\theta)$ as a measure of how deeply we enter into the interior of the Newton polygon $\Delta(f)$.

Lemma 2.11. Algorithm 2.10 provides simultaneous Puiseux expansions for the components $C_\lambda^{(r)}$ for typical λ 's. Any such local component and its expansion is determined by the system $[(l_1, a_1), \dots, (l_d, a_d(\lambda))]$ of edges and zeros of the polynomials $\varphi_{\theta_1}, \dots, \varphi_{\theta_d}$.

Proof. In the case $p(\theta_d) = 0$ one has $f \circ \theta_d = \varphi_{\theta_d}(u_d) + o(1)$ as $y \rightarrow \infty$ for a polynomial φ_{θ_d} . If λ is not a critical value of the polynomial φ_{θ_d} , then the Implicit Function Theorem gives existence of a solution $u_d = u_d(y; \lambda)$ to the equation $f_d(u_d y^{-\Delta y_d}, y) = 0$ in the form of a Puiseux expansion in y , $u_d = a_d(\lambda) + \sum c_{k_j}(\lambda) y^{k_j}$, with coefficients $c_{k_j}(\lambda)$ depending on λ (continuously on the set of non-critical values of φ_{θ_d}). Since x is expressed by means of u_d and powers of y , we get a function $x = x(y; \lambda)$ in the form of a Puiseux expansion. This is the Puiseux expansion of the component $C_\lambda^{(r)}$, locally simultaneous for all family. $\square$

Lemma 2.12. The variable v of the corresponding (final) Newton–Puiseux alteration charts $\tilde{\theta}_d$ defines the primitive parametrization of the local curves $C_\lambda^{(r)}$ (for typical λ 's).

Proof. The curves $C_\lambda^{(r)}$, λ – typical, behave in a similar way to the curves $u = \text{const}$. Their preimages are also transversal to the dicritical divisor D , defined as $v = 0$. Since v is a primitive parameter on $u = \text{const}$ (see Lemma 2.6), it is also primitive on $C_\lambda^{(r)}$. $\square$

Remark 2.13. In a traditional (algebraic) approach to the resolution of indeterminacies of a polynomial at infinity people use a series of modifications, similar to the changes $(v_D, w_D) \rightarrow (x, y)$ such that $\mathbb{C}(x, y) = \mathbb{C}(v_D, u_D)$ from Proposition 2.3; compare [14] for example. Theoretically it is possible to interpret our changes θ in such terms. Unfortunately, the formulas are so complicated that there is no real possibility to provide them here. Example 2.4 demonstrates this complexity in a simplest nontrivial case.

3. Newton–Puiseux charts in the Jacobian problem

Definition 3.1. A polynomial map $P : \mathbb{C}^2 \rightarrow \mathbb{C}^2$,

$$(x, y) \rightarrow (X, Y) = (f(x, y), g(x, y))$$

which satisfies the **Jacobian equation** $\text{Jac } P(x, y) = \text{Jac } (f, g) = \det P' = 1$ is called the **Jacobian map** and the pair (f, g) the **Jacobian pair**. The function g is called also the **Jacobian mate** for f .

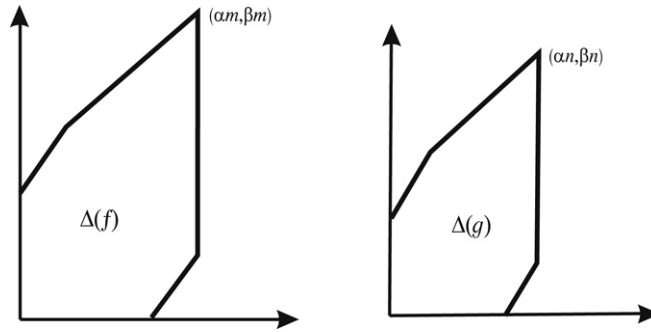


Fig. 3.

Beginning from this section we assume that $P = (f, g)$ is a Jacobian map. Recall that Jacobian Conjecture states that any such map is invertible.

We assume that the map P is not linear,

$$\deg f + \deg g > 2$$

and that

$$f(0, 0)g(0, 0) \neq 0.$$

We recall some results belonging mainly to Abhyankar [1] (see also [34]). (The property (3.4) is achieved after a suitable shift $(x, y) \rightarrow (x - x_0, y - y_0)$.)

Theorem 3.2. (a) The Newton polygons $\Delta(f)$ and $\Delta(g)$ are similar, $\Delta(g) = \delta_0 \cdot \Delta(f)$, $\delta_0 = \frac{\deg g}{\deg f}$.

(b) These polygons intersect the coordinate axes along edges of positive length.

(c) For any edge $I \subset \Gamma(f)$ not contained in the coordinate axes and a similar edge $J = \delta_0 \cdot I \subset \Gamma(g)$ one has $g_J \equiv cf_I^{\delta_0}$, where the constant c does not depend on the edge (it can be normalized to $c = 1$).

(d) In search of a counter-example it is enough to consider Jacobian maps of the form

$$f = (x^\alpha y^\beta)^m + \dots, \quad g = (x^\alpha y^\beta)^n + \dots \quad (3.1)$$

(see Fig. 3), where

$$m > n, \quad m/n \notin \mathbb{Z}, \quad \gcd(m, n) = 1 \quad (3.2)$$

and

$$0 < \alpha < \beta. \quad (3.3)$$

We can also assume that

$$f_{0, \beta m} = f_{\alpha m, 0} = 0. \quad (3.4)$$

Denote $\mathcal{Y}^0 := \mathbb{C}^2$ (the target space) and $\mathcal{Y} = \mathbb{C}\mathbb{P}^2$, $\mathcal{Y}_\infty = \mathcal{Y} \setminus \mathcal{Y}^0$. The polynomial map $P : \mathcal{X}^0 \rightarrow \mathcal{Y}^0$ defines a rational map $\hat{P} : \mathcal{X} \rightarrow \mathcal{Y}$. It is defined by means of its graph $G(\hat{P}) \subset \mathcal{X} \times \mathcal{Y}$, which is the closure of the graph $G(P) \subset \mathbb{C}^2 \times \mathbb{C}^2$. It is possible that the image of a point is a 1-dimensional curve. The indefiniteness of $\hat{P}$ means: either the indeterminacy of $\hat{f}$ or of $\hat{g}$ (with finite values), or the indeterminacy of $\hat{g}/\hat{f}$ when $f, g \rightarrow \infty$, but g/f becomes finite and not constant. The cases when $\hat{g}$ becomes finite and $\hat{f}$ infinite (i.e. definite), or conversely, do not correspond to indeterminacy of $\hat{P}$.

Definition 3.3. The **resolution of indeterminacy** of $\hat{P}$ consists of a 2-dimensional complex manifold $\mathcal{Z}$ and a modification map $\pi : \mathcal{Z} \rightarrow \mathcal{X}$ (which is a composition of elementary blowing-ups of points) such that the compositions $\hat{f} \circ \pi, \hat{g} \circ \pi : \mathcal{Z} \rightarrow \mathbb{C}\mathbb{P}^1$ and $\hat{P} \circ \pi : \mathcal{Z} \rightarrow \mathcal{Y}$ are holomorphic definite maps.

From Theorem 2.2 it follows that such a resolution exists; (sometimes it is called the *compactification* of P). One defines the **intersection graph** $\mathfrak{A}(P)$ of the resolution $(\mathcal{Z}, \pi)$ analogously as in Definition 2.1.

As in the previous section one can realize the resolution of indeterminacies of P by means of Newton–Puiseux charts. The **Newton–Puiseux resolution of the rational map $\hat{P}$** is defined by means of the union of N–P charts obtained in the Newton–Puiseux resolutions of indeterminacies of the functions $\hat{f}, \hat{g}$ and $\hat{f}/\hat{g}$ (where we have ∞/∞).

One defines the Newton–Puiseux graph $\mathfrak{A}(P)$ analogously as in Algorithm 2.10.

Recall that N–P charts take the form $\theta : (u, y) \rightarrow (x, y) = (a_1 y^{-\gamma_1} + \dots + a_{d-1} y^{-\gamma_{d-1}} + u y^{-\gamma}, y)$ (left, of depth $d =: d(\theta)$ and with slope $\gamma =: \gamma(\theta)$) or $\theta : (x, u) \rightarrow (x, y) = (x, a_1 x^{-\gamma_1} + \dots + a_{d-1} x^{-\gamma_{d-1}} + u x^{-\gamma})$ (right, of depth d and with the inverse slope γ).

The Newton–Puiseux charts of depth $d = 1$ are related to the left and right edges of the Newton diagram $\Gamma(f)$ (see Definition 2.9 and Algorithm 2.10). If $I \subset \Gamma(f)$ is a left edge of slope γ , then we have the left N–P chart $\theta = \theta(I) : (u, y) \rightarrow (uy^{-\gamma}, y)$, and if I is a right edge with slope $1/\gamma$, then we have the right N–P chart $\theta = \theta(I) : (x, u) \rightarrow (x, ux^{-\gamma})$.

Lemma 3.4. (a) If θ is a left N–P chart, then we have the expansions

$$\begin{aligned}\tilde{f} &:= f \circ \theta = \varphi(u)y^p + \varphi_1 y^{p_1} + \dots \\ \tilde{g} &:= g \circ \theta = \psi(u)y^q + \psi_1 y^{q_1} + \dots \\ (\text{Jac } P) \circ \theta &= y^{p+q+\gamma-1} \begin{vmatrix} \varphi' & p\varphi \\ \psi' & q\psi \end{vmatrix} + \dots\end{aligned}\quad (3.5)$$

and if θ is a right N–P chart, then

$$\begin{aligned}\tilde{f} &:= f \circ \theta = \varphi(u)x^p + \dots \\ \tilde{g} &:= g \circ \theta = \psi(u)x^q + \dots \\ (\text{Jac } P) \circ \theta &= -x^{p+q+\gamma-1} \begin{vmatrix} \varphi' & p\varphi \\ \psi' & q\psi \end{vmatrix} + \dots;\end{aligned}\quad (3.6)$$

here $\varphi, \varphi_1, \varphi_2, \dots$ and $\psi, \psi_1, \dots$ are polynomials and the leading rational exponents $p =: p(\theta), q =: q(\theta) \geq 0$.

(b) We have always $p + q + \gamma - 1 \geq 0$.

(c) If $p + q + \gamma - 1 > 0$, then $\psi \equiv \text{const} \cdot \varphi^{q/p}$;

(d) if $p = q = 0$, then $\gamma > 1$.

Proof. The part (a) of the lemma is a calculation of the leading term in the Jacobian.

Since, by the assumption, $\text{Jac } P \equiv 1$ we find that $(\text{Jac } P) \circ \theta$ cannot tend to 0 as $y \rightarrow \infty$ (respectively as $x \rightarrow \infty$). This implies the inequality in (b).

In the situation when the exponent $p + q + \gamma - 1 > 0$ the coefficient in the leading term of $(\text{Jac } P) \circ \theta$ must vanish. This leads to the differential equation $\psi' = \frac{q}{p} \frac{\varphi'}{\varphi} \psi$ with a general solution like in (c).

Finally, for $p = q = 0$ the determinant $\begin{vmatrix} \varphi' & p\varphi \\ \psi' & q\psi \end{vmatrix}$ vanishes and hence the exponent $p + q + \gamma - 1$ should be positive. $\square$

Remark 3.5. (a) In the case of charts of depth $d = 1$ associated with edges $I \subset \Gamma(f)$ the exponents in (3.5) and (3.6) have the following geometrical interpretation. Let I be a left edge; then p (respectively q) is the j coordinate of the intersection of the straight line through I (respectively through $J = \delta_0 \cdot I \subset \Gamma(g)$, $\delta_0 = \frac{q}{p} = \frac{\deg g}{\deg f}$) with the j axis. If I is a right edge, then we take the intersection with the i axis.

In the case of depth $d > 1$ the power p has an analogous interpretation with respect to the diagram $\Gamma(f_d)$ (see Algorithm 2.10).

(b) In Orevkov's approach [30], via splice diagrams, an analogue of the power $p + q + \gamma - 1$ in the above expressions for the Jacobians takes a very complicated form. He expresses the divisor of zeros and poles of the pull-back form $\pi^*(dx \wedge dy)$ in terms of the divisors of resolution of indeterminacies of P . The numbers $k \cdot (p + q + \gamma - 1)$ (where $k = k_1 \dots k_d$ from Definition 2.5) are just the orders of poles of $\pi^*(dx \wedge dy)$ at the divisors D_θ corresponding to the charts θ .

The N–P charts can be defined also in the target space $\mathcal{Y}$, i.e. in some cone-like domains with vertex at infinity. Usually such charts are associated with concrete N–P charts in the source space. The expression of the Jacobian map in such charts (in $\mathcal{X}$ and in $\mathcal{Y}$) leads to special expansions of $\tilde{g}$ in rational powers of $\tilde{f}$.

To see this, consider the case when $p + q + \gamma - 1 > 0$ for a left chart. We try to eliminate the variable y from the first two equations in (3.5). Since, by Lemma 3.4(b), $\psi \equiv b_0 \cdot \varphi^{\delta_0}$, $\delta_0 = \frac{q}{p}$, we get $\tilde{g} = b_0 \tilde{f}^{\delta_0} + \tilde{g}_1$ where $\tilde{g}_1(u, y)$ is of lower order as $y \rightarrow \infty$. Here $\tilde{f}^{\delta_0}$ as well as $\tilde{g}_1$ must be treated as a formal power series with rational coefficients, i.e.

$$\begin{aligned}\tilde{f}^{\delta_0} &= (\varphi(u)y^p)^{q/p} \left[1 + \frac{\varphi_1}{\varphi} y^{p_1-p} + \dots \right]^{q/p} \\ &= \frac{1}{b_0} \psi(u)y^q \left[1 + \frac{q}{p} \frac{\varphi_1}{\varphi} y^{p_1-p} + \dots \right].\end{aligned}$$

We have $\tilde{g}_1 = \tilde{f}_1(u)y^{q_1} + \dots$. If $p + q_1 + \gamma - 1 > 0$, then again we get that $\tilde{f}_1(u) \equiv b_1 \tilde{f}^{\delta_1}(u)$ and $\tilde{g}_1 = b_1 \tilde{f}^{\delta_1} + \tilde{g}_2$. We again consider the pair $(\tilde{f}, \tilde{g}_2)$, and so on.

We continue this process until we arrive to a pair

$$(\tilde{f}, \tilde{g}_N) = (\varphi y^p + \dots, \psi_N y^{q_N} + \dots), \quad (3.7)$$

where $N =: N(\theta)$ depends on the chart and

$$q_N := 1 - \gamma - p. \quad (3.8)$$

We underline the following important

Proposition 3.6 (Rationality Condition). *The function $\tilde{\psi}_N(u)$ is rational.*

From the above we obtain the following expansion

$$\tilde{g} = b_0 \tilde{f}^{\delta_0} + b_1 \tilde{f}^{\delta_1} + \cdots + b_{N-1} \tilde{f}^{\delta_{N-1}} + c(u) \tilde{f}^{\delta} + \cdots, \quad (3.9)$$

where $\delta_0 > \delta_1 > \cdots > \delta_{N-1} > \delta$ are rational exponents, $b_0, b_1, \dots, b_{N-1}$ are constant coefficients and

$$\delta = \frac{1-\gamma}{p} - 1 \quad (3.10)$$

depends on the chart θ , $\delta =: \delta(\theta)$.

Definition 3.7. The expansion (3.9) is called the **quasi-Puiseux expansion associated with the N–P chart θ** (shortly, quasi-Puiseux expansion). The part $b_0 \tilde{f}^{\delta_0} + b_1 \tilde{f}^{\delta_1} + \cdots + b_{N-1} \tilde{f}^{\delta_{N-1}}$ is called the **constant part** (of the quasi-Puiseux expansion (3.9)) and the term $c(u) \tilde{f}^{\delta}$ is called the **first non-constant term** (of this quasi-Puiseux expansion). (Heitmann [14] calls the constant part an approximation to g).

Remark 3.8. The expansion (3.9) means rewriting the map $P \circ \theta$ in the following Newton–Puiseux chart in the space $\mathcal{Y}$:

$$\Theta : (X, U) \rightarrow (X, Y) = (X, b_0 X^{\delta_0} + \cdots + b_{N-1} X^{\delta_{N-1}} + UX^{\delta}). \quad (3.11)$$

We have

$$X = \varphi(u) y^{p\theta} + \cdots, U = c(u) + \cdots. \quad (3.12)$$

The expansion (3.9) is valid where $\tilde{f} \rightarrow \infty$, i.e. in domains of the form $\{(u, y) : |\varphi(u)| > \varepsilon_1, |y| > 1/\varepsilon_2\}$ for $\varepsilon_1 > 0$ and small ε_2 . The coefficients b_i and the branches of the roots $\tilde{f}^{1/p}$ do not depend on (y, u) and are defined in a unique way.

Another treatment of the formula (3.9) is the following. We take germs of curves $\Gamma : (\mathbb{C}, 0) \rightarrow (\mathcal{X}, \mathcal{X}_{\infty})$ such that $\Gamma(0) \in \mathcal{X}_{\infty} \setminus \{\varphi = 0\}$; (here we can assume that Γ lies in the domain $\{|\varphi| > \varepsilon_1, |y| > 1/\varepsilon_2\}$). The expansion (3.9) is the Puiseux expansion along the curve $P(\Gamma)$. We treat $Y|_{P(\Gamma)}$ as a multivalued (algebraic) function of X : $Y = Y(X)$. The formula (3.9) says that the above Puiseux expansion is locally uniform with respect to the curve Γ . The coefficients $b_1, \dots, b_{N-1}$ and the roots do not change when we slightly vary Γ .

Lemma 3.9. *The first non-constant coefficient $c_{\theta}(u) := c(u) = \tilde{\psi}_N(u)/\varphi^{\delta}(u)$ is given by the following formula*

$$c(u) = \pm \frac{1}{p} \int^u \varphi(v)^{-\delta-1} dv = \pm \frac{1}{p} \int^u \varphi(v)^{(\gamma-1)/p} dv, \quad (3.13)$$

+ in the case of right N–P chart and – in the case of left N–P chart.

Moreover, the condition that the function $\tilde{\psi}_N$ is rational (Proposition 3.6) leads to the condition that the Schwarz–Christoffel integral in (3.13) is a Darboux function (see Appendix).

Proof. Consider the left case. The Jacobian condition $\text{Jac } P = 1$ takes the form of the following (linear and non-homogeneous) differential equation

$$(\tilde{\psi}_N)' = \delta \frac{\varphi'}{\varphi} \tilde{\psi}_N - \frac{1}{p\varphi}. \quad (3.14)$$

The general solution is given by the formula

$$\tilde{\psi}_N(u) = -\frac{1}{p} \varphi^{\delta}(u) \int^u \varphi(v)^{-\delta-1} dv.$$

We see that $c(u)$ is indeed not constant.

The second statement follows from the definition of a Darboux function (a product of powers of linear factors) given in the Appendix. $\square$

Remark 3.10. The problem of determining the rational function $\tilde{\psi}_N$ is equivalent to finding a quasi-homogeneous rational mate $\tilde{\psi}_N(xy^{\gamma})y^{q_N}$ to the quasi-homogeneous polynomial $\varphi(xy^{\gamma})y^p$. In Appendix (Section 7) we present necessary conditions that the right-hand side of (3.13) defines a Darboux function. Using some of them, item (d) of Theorem 3.2 is proved.

Moreover, an accurate use of the Rationality Condition allows us to give elementary proofs of many important results; we discuss this in the next section.

Note also that the constant of integration in (3.13) is not fixed. It will be determined in further sections.

Other investigators (see [1,2,13,25]) also encountered the problem of determining the function $\tilde{\psi}_N$. They use algebraic tools. The notions of the quasi-Puiseux expansion and Schwarz–Christoffel integral seem to be very natural here.

To proceed further with the analysis of the Newton–Puiseux resolution of Jacobian maps we need some generalization of the Abhyankar theorem. In the following we shall assume the situation as in the next theorem (whose proof is postponed to the next section).

Theorem 3.11 (Heitmann [14]). The polygon $\Delta(f)$ is contained in the quadrangle with vertices: $(0, 0)$, $(\alpha m, 0)$, $(\alpha m, \beta m)$, $(0, (\beta - \alpha)m)$ (see Theorem 3.2 and Fig. 3).

Corollary 3.12. We have

$$1 \leq \gamma_1 < \dots < \gamma_{d-1} < \gamma$$

for any left N–P chart (associated with resolution of indeterminacies of P) and

$$0 \leq \gamma_1 < \dots < \gamma$$

for any right N–P chart.

We shall finish this section by the presentation of a certain classification of divisors of the resolution of indeterminacies of P . This classification seems to be very important in the Jacobian problem, e.g. in constructing counter-examples.

Definition 3.13. A divisor $D \subset \pi^{-1}(\{A_{\text{left}}, A_{\text{right}}\})$ (and the associated chart $\theta = \theta_D$) is said to be of **Abhyankar type** if

$$p, q > 0, \quad p + q + \gamma - 1 > 0.$$

The divisor D (and the chart θ) is of **non-properness type** if

$$p = q = 0.$$

The divisor (and the chart) is of **transitory type** if

$$p + q > 0, \quad p + q + \gamma - 1 = 0, \quad \frac{p}{q} = \frac{\deg f}{\deg g}.$$

The divisor (and the chart) is of **infinite type** if

$$p, q \geq 0, \quad p + q > 0, \quad p + q + \gamma - 1 = 0, \quad \frac{p}{q} \neq \frac{\deg f}{\deg g}.$$

Theorem 3.14. (a) The Newton–Puiseux resolutions of indeterminacies of $\hat{f}$ and $\hat{g}$ at A_{left} are the same and define the Newton–Puiseux resolution of indeterminacy of $\hat{P}$ at A_{left} . The corresponding divisors are either of Abhyankar type, or are of non-properness type (both types are realized).

This implies that $\hat{P}(A_{\text{left}}) = \{(1 : 0 : 0)\} \cup \bigcup_{k=1}^{M_1} S_k$, where $S_k = \hat{P} \circ \pi(D_{ik} \setminus \infty) \subset \mathcal{Y}^0(D_{ik} \text{ of non-properness type})$ are the so-called left components of the non-properness set (see Definition 6.1).

(b) In the Newton–Puiseux resolution of indeterminacy of $\hat{P}$ at the point A_{right} all four types of N–P charts (and divisors) may occur: of Abhyankar type (they do occur), of non-properness type (they may be absent), of transitory type (they do occur) and of infinite type (they do occur). Among the divisors of infinite type there exists a divisor sent to the line at infinity $\mathcal{Y}_\infty$ by means of $\hat{P} \circ \pi$.

This implies that $\hat{P}(A_{\text{right}}) = \mathcal{Y}_\infty \cup \bigcup_{k=M_1+1}^M S_k$, where S_k are the right components of the non-properness set.

Proof. (a) Recall that we have $\gamma \geq 1$ for any left chart (see Corollary 3.12). Therefore, whenever we have $p \geq 0$ and $q \geq 0$ but $p + q > 0$, then $p + q + \gamma - 1 > 0$ and hence $\psi = \text{const} \cdot \varphi^{q/p}$ (Lemma 3.4(c)). Therefore this point is proved in the same way as Theorem 3.2 (where only charts of Abhyankar type are met), until one encounters a chart θ of depth $d > 1$ such that $p = 0$, i.e. dicritical for $\hat{f}$. Such a θ must be associated with a left-most left edge I of $\Gamma(f_d)$ (see Algorithm 2.10). By similarity of the remaining left parts of $\Gamma(f_d)$ and $\Gamma(g_d)$ (proved in the previous steps of the algorithm), one gets that $J = \frac{\deg g}{\deg f} I$ is the left-most edge of $\Gamma(g_d)$. This gives $q = 0$. We note also the asymptotic behavior $\psi(u) \sim \varphi(u)^{\deg g / \deg f} \rightarrow \infty$ as $u \rightarrow \infty$. This chart must be of non-properness type.

Note that in all the cases when $p, q > 0$ we have $(g/f) \circ \theta \sim O(y^{q-p}) \rightarrow 0$ as $y \rightarrow \infty$, because we have assumed $\deg g < \deg f$ (see (3.2)). This means that the corresponding non-dicritical divisor is mapped to the point $(1 : 0 : 0) \in \mathcal{Y}_\infty$.

(b) The difference with the resolution of the point A_{left} relies on the fact that it is possible (and it does occur) that γ becomes < 1 for a right N–P chart. So, in the process of resolution we must control carefully the exponents $p + q + \gamma - 1$ appearing in the expansion of $(\text{Jac } P) \circ \theta$.

Let us consider a series of charts $\theta_1, \theta_2, \dots, \theta_d$ defined by systems $[I_1], [(I_1, a_1), I_2], [(I_1, a_1), (I_2, a_2), I_3], \dots$ leading to separation of local components of the curves $\hat{f} = \lambda$ and/or $\hat{g} = \mu$, i.e. $p(\theta_d) = 0$ and/or $q(\theta_d) = 0$. Of course, θ_1 is of Abhyankar type, here $p(\theta_1) + q(\theta_1) + \gamma(\theta_1) - 1 > 0$.

It is possible that for all the charts $\theta_1, \dots, \theta_{d-1}$ we have $p(\theta_j) + q(\theta_j) + \gamma(\theta_j) - 1 > 0$ and $\gamma(\theta_d) > 1$. Then $\theta_1, \dots, \theta_{d-1}$ are of Abhyankar type and θ_d is of non-properness type.

But it may occur that some $p(\theta_j) + q(\theta_j) + \gamma(\theta_j) - 1 = 0$. Let θ_j be the first such chart. We claim that θ_j is of transitory type. Indeed, we have $p(\theta_j)/q(\theta_j) = \deg \varphi_{\theta_j} / \deg \psi_{\theta_j} = \deg g / \deg f$ (by similarity of the previous parts of the Newton diagrams $\Gamma(f_i)$ and $\Gamma(g_i)$; thus $p(\theta_j)q(\theta_j) > 0$ and the third equation in (3.6) gives the equation

$$p(\theta_j)\varphi_{\theta_j}\psi'_{\theta_j} - q(\theta_j)\varphi'_{\theta_j}\psi_{\theta_j} \equiv 1. \quad (3.15)$$

Consider now the next chart θ_{j+1} , which is defined by means of a zero a_j of the polynomial $\varphi_{\theta_j}(u)$, or of $\psi_{\theta_j}(u)$; we admit here $a_j = 0$. Assume that it is a zero of φ_{θ_j} . Then, by (3.15) a_j is not a zero of ψ_{θ_j} and it is a simple zero of φ_{θ_j} . (We see that at this step the minimal resolutions of the rational functions $\hat{f}$ and $\hat{g}$ begin to differ.) Therefore the edge I_{j+1} contains only two points from $\text{Supp}(f_{j+1})$: $(p(\theta_{j+1}), 0)$ and $(a, 1)$ and J_{j+1} is reduced to the point $\{(q(\theta_j), 0)\}$, i.e. $\deg \varphi_{\theta_{j+1}} = 1$ and $\deg \psi_{\theta_{j+1}} = 0$. The equation $p(\theta_{j+1}) + q(\theta_{j+1}) + \gamma(\theta_{j+1}) - 1 = 0$ still holds true, but $p(\theta_{j+1})/q(\theta_{j+1}) \neq p(\theta_j)/q(\theta_j)$. Hence θ_{j+1} is of infinite type.

If $p(\theta_{j+1}) = 0$, then the resolution of the corresponding component $\mathcal{C}_\lambda^{(r)}$ of the generic curve C_λ stops. The divisor $D_{\theta_{j+1}}$ is sent to the point $(0 : 1 : 0) \in \mathcal{Y}_\infty$. If $p(\theta_{j+1}) > 0$, then we take a unique zero a_{j+1} of $\varphi_{\theta_{j+1}}$ and make a change θ_{j+2} defined by some edge I_{j+2} , which has the same properties as I_{j+1} . Again, either $p(\theta_{j+2}) = 0$ or $p(\theta_{j+2}) > 0$, etc. We see that $\theta_{j+1}, \theta_{j+2}, \dots$ are of infinite type.

Note that in each further step the quantity $q(\theta_m)$ remains constant while $p(\theta_m)$ is decreasing. So, there exists a moment l when $q(\theta_l) = p(\theta_l)$. The corresponding divisor D_{θ_l} is such that $\hat{P} \circ \pi(D_{\theta_l}) = \mathcal{Y}_\infty$. (It is possible that the edge I_l is reduced to a point; then $\varphi_{\theta_l} = \text{const} \cdot u$.) Such a divisor D_{θ_l} surely exists, because $\hat{P}(\mathcal{X}) = \mathcal{Y}$ and $\hat{P}(\mathcal{X}_\infty \setminus A_{\text{right}}) = \{(1 : 0 : 0)\}$. Therefore also transitory chart preceding such a chart of infinite type must exist. $\square$

The latter proof reveals the following additional properties of the N–P charts defined in Definition 3.13. They will be used in further sections.

Proposition 3.15. (a) For any N–P chart of Abhyankar type one has

$$\frac{p}{q} = \frac{\deg f}{\deg g}, \quad \psi(u) \equiv \varphi(u)^{q/p} \neq \text{const}, \quad \gamma > 1.$$

(b) For any N–P chart of non-properness type one has

$$\psi(u) \sim \varphi(u)^{q/p} \neq \text{const} \quad \text{as } u \rightarrow \infty.$$

(c) For any N–P chart of transitory type one has

$$\delta = \frac{\deg g}{\deg f} = \delta_0$$

and the equation

$$p\varphi\psi' - q\varphi'\psi \equiv 1,$$

is satisfied. The latter equation implies that zeros of the polynomials φ, ψ are simple and distinct; (if $\varphi(u_0) = 0$ then $\psi(u_0) \neq 0$ and vice versa).

(d) For any N–P chart of infinite type one has

$$\deg \varphi + \deg \psi = 1.$$

(e) The N–P resolutions of indeterminacies of f and of g differ only in the parts of $\mathfrak{N}(f)$ and $\mathfrak{N}(g)$ consisting of charts of infinite type. These parts form bamboos rooted at the charts of non-properness type.

The property (e) above is the reason that we use the name ‘non-properness’ for divisors which are sent to the finite plane. Orevkov calls them dicritical. But for us the notion of dicriticality is reserved for the resolution of indefiniteness of one polynomial (see Definition 2.1). Some divisors dicritical for f are not dicritical for g . But if a divisor of infinite type is dicritical for f (respectively for g) then $p = 0 < q$ (respectively $q = 0 < p$) and the divisor is non-dicritical for g (respectively for f).

Further plan of the paper. Here is a good place to outline the remaining results of the work as we promised in the introduction.

In the next section we exploit the Rationality Condition (see Proposition 3.6). Namely, the condition that the function $\tilde{\psi}_N(u)$ is rational implies that the integral $\int \varphi(v)^{(\gamma-1)/p}$ (of Schwarz–Christoffel type) should be a product of powers of linear factors (of Darboux type). In Propositions 4.1 and 4.2 we state numerical conditions for the above property. These propositions are proved in the Appendix. Next, we apply these results to the N–P charts associated with the edges of the Newton diagram $\Gamma(f)$. This gives simple proofs of Heitmann’s theorem about restrictions on the shape of $\Gamma(f)$ (Theorem 3.11) and of Appelgate–Onishi’s theorem which states that $\gcd(\alpha, \beta) \neq 1$ (Theorem 4.7). Further applications of Propositions 4.1 and 4.2 to N–P charts of depth $d = 1, 2, 3$ allow to improve these results. In particular, we show that $\gcd(\alpha, \beta) \neq 2$ (Theorem 4.12) and that $\alpha + \beta > 16$ (Theorem 4.16).

Section 5 is devoted to N–P charts in the target space, that is to the quasi-Puiseux expansions $\tilde{g} = b_0\tilde{f}^{\delta_0} + b_1\tilde{f}^{\delta_1} + \dots + b_{N-1}\tilde{f}^{\delta_{N-1}} + c_\theta(u)\tilde{f}^\delta + \dots$ associated with N–P charts θ in the source space. It turns out that some initial parts of two such expansions, associated with ‘adjacent’ charts θ and θ' , coincide (Proposition 5.2). This allows us to reach some conclusions about the components S_k of the non-properness set (see Theorem 3.14 and Corollary 5.4).

The first difference between the quasi-Puiseux expansions associated with adjacent right N–P charts θ, θ' is a source of a new constant term in one of these expansions. Namely, if $\tilde{g} = b_0\tilde{f}^{\delta_0} + b_1\tilde{f}^{\delta_1} + \dots + b_{N-1}\tilde{f}^{\delta_{N-1}} + c_{\theta'}(u)\tilde{f}^{\delta'} + \dots$ is associated with θ' and $\delta < \delta'$ then we have $\tilde{g} = b_0\tilde{f}^{\delta_0} + b_1\tilde{f}^{\delta_1} + \dots + b_N\tilde{f}^{\delta_N} + c_\theta(u)\tilde{f}^\delta + \dots$, where $\delta_N = \delta'$ and $b_N = c_{\theta'}(\infty)$.

(see Proposition 5.8). Moreover, the corresponding polynomial $\varphi_\theta(u)$ is a power $\chi(u)^\eta$, such that $\varphi_{\theta'}(u)$ is not a power of the same degree η (Proposition 5.11). This, in order, implies that for a path $\theta_0 - \theta_1 - \dots - \theta_N$, from the root θ_0 to a chart of transitory type θ_N , one has $\gcd(\alpha, \beta) = \eta_1 \dots \eta_{N-1}$, where $\eta_j > 1$ appear in the characteristic pairs associated with a suitable N–P chart θ in the target space (Theorem 5.13). The latter result is partly used in Section 4.

Concerning the quasi-Puiseux expansions associated with the left N–P charts we find that their constant parts are all the same (and are determined by the quasi-Puiseux expansions associated with the right N–P charts, as in Proposition 5.8) and the first non-constant coefficient has the form $c_\theta(u) = u^\kappa \times \text{polynomial}$ (Lemma 5.16 and Proposition 5.17).

In Section 6 we study the topological properties of the Jacobian maps. We estimate from below the degree $\deg_{\text{top}} P$ of such a (non-invertible) map. To this aim one can calculate the number of preimages of a point near a component S_k of the non-properness curve; it is Proposition 6.5 (due to Orevkov). In this way the cases with $\deg_{\text{top}} P = 2, 3$ are eliminated.

Another possibility is to count the number of preimages of a point near the line at infinity, or (more generally) near a divisor (in a compactification of the target space) associated with an N–P chart (in the source space) of another type (Lemma 6.8 and Proposition 6.9). We show that $\deg_{\text{top}} P$ is expressed via the N–P charts of transitory type (Proposition 6.10). In the case $\deg_{\text{top}} P = 4, 5$ there remain four cases which should be treated in detail (Corollary 6.11). In each case we construct a series of N–P charts in the target space $\Theta_0 : Y = UX^{\delta_0}$, $\Theta_1 : Y = b_0 X^{\delta_0} + UY^{\delta_1}$, $\dots$, $\Theta_N : Y = b_0 X^{\delta_0} + \dots + UX^{\delta_N}$. This series is associated with a path $\theta_0 - \theta_1 - \dots - \theta_N$ of N–P charts ending at a transitory chart, but in reverse order. Detailed study of the preimages of a generic point (X, U) in each of Θ_i allows us to eliminate the possibility $\deg_{\text{top}} P < 6$. This is Theorem 6.12 which improves Domrına–Orevkov's bound.

4. The rationality condition

Analogously as in Section 3 with any N–P chart θ of Abhyankar type or of transitory type we can associate the quasi-Puiseux expansion $\tilde{g} = b_0 \tilde{f}^{\delta_0} + b_1 \tilde{f}^{\delta_1} + \dots + b_{N-1} \tilde{f}^{\delta_{N-1}} + c(u) \tilde{f}^\delta + \dots$, by elimination of x (or y) from the expansions $\tilde{f}$ and $\tilde{g}$. The coefficients $b_0, \dots, b_{N-1}$ are constant and the first non-constant term $c(u) \tilde{f}^\delta = \tilde{\psi}_N(u) x^{\delta_N} + \dots$ is given by means of the integral formula

$$\tilde{\psi}_N(u) = \frac{\pm 1}{p} \varphi(u)^\delta \int^u \varphi(v)^{(\gamma-1)/p} dv.$$

Recall also that the function $\tilde{\psi}_N$ should be rational (see Proposition 3.6).

(In the case of a N–P chart of infinite type the quasi-Puiseux expansion takes the very simple form $\tilde{g} = c(u) \tilde{f}^\delta + \dots$, where $c(u) = \frac{\alpha u + \beta}{\gamma u + \delta}$ (see Theorem 3.14).)

Recall also that the Newton polygons $\Delta(f)$ and $\Delta(g)$ are similar (see Theorem 3.2). Thus there exists a **web Newton polygon** Δ_0 , with a **web Newton diagram** Γ_0 , such that

$$\Delta(f) = m\Delta_0, \quad \Gamma(f) = m\Gamma_0, \quad \Delta(g) = n\Delta_0, \quad \Gamma(g) = n\Gamma_0. \quad (4.1)$$

The diagram Γ_0 has the root vertex (α, β) (see Fig. 3 and Theorem 3.2). Therefore

$$\gcd(\deg f, \deg g) = \alpha + \beta. \quad (4.2)$$

For any edge $I \subset \Gamma_0$ and corresponding N–P chart θ we have

$$\tilde{f} = [x^{\beta_0} \chi(u)]^m + \dots \quad (4.3)$$

for a polynomial $\chi = \sqrt[p]{\varphi}$.

The next two propositions concern restrictions on the polynomials φ which arise from the latter condition. They are crucial for this section.

Proposition 4.1. *Let θ be a N–P chart of Abhyankar type such that*

$$\gamma \geq 1.$$

Then the function $\varphi^{(\gamma-1)/p}$ is either a polynomial or of the type $(u - u_0)^\vartheta \times \text{polynomial}$, $\vartheta \in \mathbb{Q}_+ \setminus \mathbb{Z}$.

Proposition 4.2. *Let θ be a N–P chart of Abhyankar type or of transitory type and such that*

$$\gamma = \frac{l}{k} < 1,$$

$k = k_1 \dots k_d$ (see Definition 2.5). Then we have three possibilities:

(a) $k_d = 1$ and $\varphi = \text{const} \cdot (u - u_0)^\epsilon$, $u_0 \neq 0$.

(b) $k_d = 1$ and $\varphi = \text{const} \cdot \prod_{i=1}^{s+1} (u - u_i)^{\epsilon_i}$, $s \geq 1$. Then

$$s - v \in \mathbb{Z}_- = \{0, -1, -2, \dots\},$$

where

$$v := (1 - \gamma) \frac{\deg \varphi}{p} - 1.$$

Moreover, when one fixes the exponents ϵ_i , there are only finitely many orbits of systems $(u_1, \dots, u_s)$, with respect to the action of the group $\text{Aff}(\mathbb{C})$ of changes $u \rightarrow \lambda u + \mu$, such that the Rationality Condition holds.

(c) $k_d > 1$ and $\varphi = \text{const} \cdot u^{\epsilon_0} \prod_{i=1}^s (u^{k_d} - w_i)^{\epsilon_i}$, $s \geq 1$. Then

$$s - v \in \mathbb{Z}_-,$$

where

$$v := \frac{1}{k_d} \left[(1 - \gamma) \frac{\deg \varphi}{p} - 1 \right].$$

Moreover, there are only finitely many orbits of systems $(w_1, \dots, w_s)$, with respect to the action of the group $\mathbb{C}^*$ of changes $u \rightarrow \lambda u$, such that the Rationality Condition holds.

Proof. These results follow from Lemmas A.3–A.5 and A.7 and Proposition A.10 in the Appendix. $\square$

These propositions reveal an important asymmetry between the left and right parts of the Taylor expansions of f . Indeed, since in the left case we always have $\gamma \geq 1$, the restrictions on the coefficients of the polynomial φ are not very rigid. It is enough that $\varphi^{(\gamma-1)/p}$ is a polynomial or of Darboux form $(u - u_0)^\epsilon \times \text{polynomial}$. Here is some scope in seeking a counterexample.

On the right hand side, when $0 \leq \gamma < 1$, the case (b) or (c) of Proposition 4.2 occurs; it follows from Theorem 3.14. Here the choice of φ is very limited (modulo some obvious changes). So this side would be tricky in working on a counterexample. Probably the coefficients of φ cannot be chosen to be rational; they would be given by means of some hypergeometric functions.

The conditions given in Proposition 4.2 are especially attractive when the depth $d = 1$. We shall work with the web Newton diagram Γ_0 (instead of $\Gamma(f)$ or $\Gamma(g)$, see (4.1)).

Let θ be a left N–P chart associated with a left edge $l \subset \Gamma_0$ with right endpoint (a, b) and the slope $-1 < \gamma = \frac{l}{k} < 1$, as in Fig. 4(a). Here we do not assume the restrictions stemming from Heitmann's theorem, and γ can also be negative. Then we have

$$\deg \chi = a, \quad p_0 = b - \frac{l}{k}a, \quad k_d = k$$

and

$$\begin{aligned} v &= \frac{1}{k_d} \left[(1 - \gamma) \frac{\deg \varphi}{p} - 1 \right] = \frac{1}{k} \left[\left(1 - \frac{l}{k} \right) \frac{\deg \chi}{p_0} - 1 \right] \\ &= \frac{a - b}{kp_0} = \frac{a - b}{kb - la}. \end{aligned}$$

Therefore Proposition 4.2 implies the following

Lemma 4.3. In the situation of Fig. 4(a) we have

$$\frac{a - b}{kp_0} = \frac{a - b}{kb - la} \in \mathbb{Z}_+ \setminus 0.$$

On the right side we have Fig. 4(b) and the following

Lemma 4.4. If θ is a right N–P chart associated with a right edge $l \subset \Gamma_0$ with upper endpoint (a, b) and the inverse slope $\gamma = \frac{l}{k} < 1$ (see Fig. 4(b)) then

$$v = \frac{b - a}{ka - lb} \in \mathbb{Z}_+ \setminus 0. \quad (4.4)$$

As an application we present the following

Proof of Theorem 3.11. Recall that the theorem states that the web Newton polygon Δ_0 lies in the quadrangle with vertices $(0, 0)$, $(\alpha, 0)$, (α, β) , $(0, \beta - \alpha)$. Thus it consists of two statements:

- (i) the open triangle bounded by the lines $i = 0$, $i + j = \alpha + \beta$, $j - i = \beta - \alpha$ does not contain points from Δ_0 ,
- (ii) the open triangle bounded by $j = 0$, $i = \alpha$, $i + j = \alpha + \beta$ is disjoint from Δ_0 (see Fig. 3).

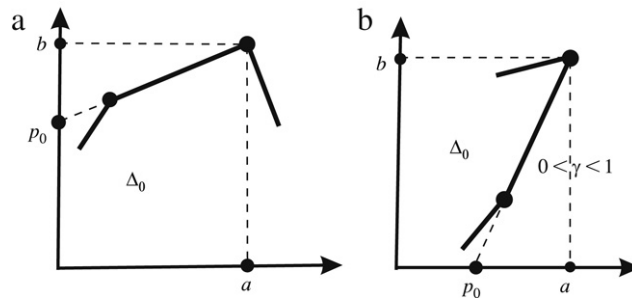


Fig. 4.

When property (i) fails, we take the first left edge $l \subset \Gamma_0$ in the mentioned triangle and incident to the root. We associate with it a N-P chart θ . We apply Lemma 4.3 with $a = \alpha$, $b = \beta$. We get the condition

$$\frac{\alpha - \beta}{kp_0} = \frac{\alpha - \beta}{\beta k - \alpha l} \in \mathbb{Z}_+ \setminus 0.$$

Because $p_0 > 0$ and $\alpha < \beta$ (see Theorem 3.2), the above expression cannot be a positive integer (even for $l = 0$).

(ii) Suppose that the first right edge $l \subset \Gamma(f)$ has negative slope $\frac{1}{\gamma} = \frac{k}{-l}$, $k, l > 0$. Applying Lemma 4.4 with $a = \alpha$, $b = \beta$ we arrive at the condition

$$\frac{\beta - \alpha}{\alpha k + \beta l} \in \mathbb{Z}.$$

This cannot be true. $\square$

In the following we assume the restrictions on Δ_0 as in Theorem 3.11.

Remark 4.5. There are examples which show that our proof of Theorem 3.11 cannot give further restrictions on the Newton polygons. In the case of a left edge with slope $\gamma = 1$ one has the rational (not polynomial) Jacobian pairs $(\varphi(xy)y^p, xy\varphi^{-1}(xy)y^{-p})$, i.e. $\tilde{\psi}_N = u/\varphi(u)$. In the case of a right vertical edge (with the inverse slope $\gamma = 0$) one has the rational Jacobian pairs $(x^{2m}y^m(1-y)^{3m}, x^{1-2m}y^{1-m}(1-y)^{1-3m})$.

Let us explore the Eq. (4.4). It can be rewritten in the form

$$\frac{b}{a} = \frac{\nu k + 1}{\nu l + 1}. \quad (4.5)$$

Let

$$a = a_0 r, \quad b = b_0 r, \quad r = \gcd(a, b). \quad (4.6)$$

Then (4.5) implies

$$\nu k + 1 = b_0 t, \quad \nu l + 1 = a_0 t \quad (4.7)$$

for some t relatively prime to ν . From this we find that

$$\nu(k - l) = t(b_0 - a_0) \quad (4.8)$$

and hence

$$k = l + wt, \quad b_0 = a_0 + w\nu \quad (4.9)$$

for some w . Now the number $p_0 = a - \frac{l}{k}b$ (see Fig. 4(b)) is calculated as follows

$$p_0 = \frac{rw}{k}. \quad (4.10)$$

Expressing w from the first formula in (4.9) we get the following useful

Lemma 4.6. If θ is a right N-P chart associated with a right edge $l \subset \Gamma_0$ with upper endpoint (a, b) and the inverse slope $\gamma = \frac{l}{k} < 1$ (see Fig. 4(b)), then

$$p_0 = \frac{r}{t} \left(1 - \frac{l}{k} \right).$$

As a corollary we shall prove the following

Theorem 4.7 (Appelgate, Onishi [2]).

$$\gcd(\alpha, \beta) \neq 1.$$

This implies that the Jacobian Conjecture is true for maps whose algebraic degree is a product of at most two primes.

Proof. Suppose $\alpha = a$ and $\beta = b$ are relatively prime, thus $r = 1$ in (4.6). Therefore Lemma 4.6 implies that $p_0 \leq 1$. The case $p_0 < 1$ is impossible, because $\Delta_0 \cap \{j = 0\}$ is a nontrivial interval (by Theorem 3.2).

The possibility $p_0 = 1$ means $\frac{1}{k} = 0$ and $t = 1$. By (4.7) we get $\alpha = a_0 = 1$. Since the point $(\alpha, 0) \notin \Delta_0$ (see (3.4) in Theorem 3.2), we also get a contradiction.

The second statement follows from the fact that transformations leading to the ‘normal form’ of the Jacobian map from Theorem 3.2 are elementary maps, like $(x, y) \rightarrow (x, y + R(x))$ and $(X, Y) \rightarrow (X, Y + S(X))$. $\square$

H. Appelgate and H. Onishi published this theorem in 1985. In fact, Appelgate and Onishi’s proof was not correct, but Nagata [20,21] and Nowicki with Nakai [24] completed it (see also [13]).

In the next section we shall give another proof of the Appelgate–Onishi theorem (following some of Heitmann’s argument).

Corollary 4.8. Whenever we have a situation as in Fig. 4(b) then $\gcd(a, b) > 1$ and $a < b$.

Proposition 4.2 can be used each time when we have a N–P chart θ with slope $\gamma < 1$ and which is of Abhyankar type or of transitory type. In the next section we study such charts from a different point of view. We look how far from the root θ_0 in the graph $\mathfrak{N}(P)$ the chart θ is located. It turns out that the longest paths consisting of charts of Abhyankar or transitory type with $\gamma < 1$ are those which end at a transitory chart. Any such path has the form

$$\theta_0 - \theta_1 - \theta_2 - \dots - \theta_N, \quad (4.11)$$

where the vertices θ_i can be nodes and θ_N transitory; in general i is not the depth of θ_i . In Theorem 5.13 (in Section 5) we prove that in such a case

$$\gcd(\alpha, \beta) = \eta_1 \dots \eta_{N-1}, \quad \eta_i \geq 2. \quad (4.12)$$

Since there exists at least one transitory chart in $\mathfrak{N}(P)$ (by Theorem 3.14) and the chart θ_1 , associated with the first right edge $I_1 \subset \Gamma_0$, is of Abhyankar type (Theorem 3.2), we have $N \geq 2$ and the product in (4.12) is nontrivial. It may consist of one factor η_1 .

But very often one can show that $N \geq 3$ in (4.11); here $\gcd(\alpha, \beta)$ is a product of at least two primes.

Lemma 4.9. When Γ_0 contains two right edges with inverse slope < 1 , then $\gcd(\alpha, \beta)$ is a product of at least two primes. This holds, in particular, when Γ_0 contains a (nontrivial) vertical edge.

Proof. Let I_1 (incident to the root (α, β)) and I_2 (adjacent to I_1) be the edges in Γ_0 with the inverse slopes $\gamma_1, \gamma_2 < 1$ and let θ_1, θ_2 be the corresponding N–P charts. If θ_2 lies in a branch in $\mathfrak{N}(P)$ leading to a transitory chart, then $N = 3$ in (4.11) and we are done.

So, there should be a path $\theta_0 - \theta_1 - \theta_3$ in $\mathfrak{N}(P)$ such that θ_3 is of transitory type. From the results of Section 5 (Lemmas 5.5, 5.9, Proposition 5.14) it would follow that $\delta(\theta_3) = \delta_0 = \frac{n}{m} > \delta(\theta_2) > \delta(\theta_1)$. By Proposition 5.8 (in the next section) the edge $\theta_1 - \theta_3$ implies that there are no terms between $\tilde{f}^{\delta(\theta_1)}$ and $\tilde{f}^{\delta_0}$ in the quasi-Puiseux expansion associated with θ_1 . But the edge $\theta_1 - \theta_2$ implies existence of the term $\tilde{f}^{\delta(\theta_2)}$.

If there is a vertical edge I_1 in Γ_0 , then the expansion of $\tilde{f}$ has the form $[x^\alpha \chi(y)]^m + \dots$. By a change $y \rightarrow y + \text{const}$ we can ensure that the next right edge I_2 is such that the corresponding chart θ_2 lies on a path leading to a transitory chart. $\square$

Lemma 4.10. When the inverse slope $\gamma = \frac{1}{k} < 1$ and $k \leq 6$ for some right edge in Γ_0 , then $\gcd(\alpha, \beta)$ is a product of at least two primes.

Proof. By Lemma 4.9 we can assume that it is the first right edge I_1 with the chart θ_1 and the next chart θ_2 on a road to transit has depth 2. So we have to show that θ_2 cannot be transitory.

θ_2 is associated with an edge $I_2 \subset \frac{1}{m}\Gamma(f_2)$ with vertices in the lattice $\frac{1}{k}\mathbb{Z} \times \mathbb{Z}$. It follows that $p_0(\theta_2) = p(\theta_2)/m \in \frac{1}{k}\mathbb{Z}$ and is $\geq \frac{1}{k}$. We have then the inequality

$$p(\theta_2) + q(\theta_2) + \gamma(\theta_2) > (m+n)p_0(\theta_2) + \gamma \geq 5 \cdot \frac{1}{k} + \frac{1}{k} = 1,$$

whereas for a transitory chart we should have $p + q + \gamma - 1 = 0$ (see Definition 3.13). $\square$

Lemma 4.11. $\alpha \neq 2$.

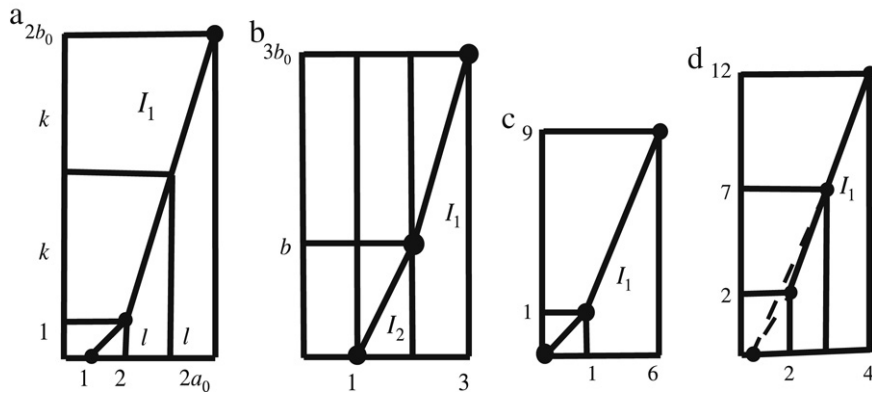


Fig. 5.

Proof. Suppose $\alpha = 2$. By Lemma 4.9 we can assume that Γ_0 does not contain vertical edges. Therefore there exists an edge joining the point $(1, 0)$ with $(a, b) = (2, \beta)$ (as in Fig. 4(b)).

But here we get $l = 1, k = \beta$ and (4.5) gives $\frac{\beta}{2} = \frac{\nu\beta+1}{\nu+1}$, i.e. $(\nu - 1)\beta + 2 = 0$. As $\nu \geq s \geq 1$ (Proposition 4.2), we get a contradiction. $\square$

We are in a state to improve the Appelgate–Onishi theorem.

Theorem 4.12. $\gcd(\alpha, \beta) \neq 2$.

It means that the Jacobian Conjecture is valid for Jacobian maps whose degree is a product of three primes one of whom is 2.

Proof. Let, as usual, I_1 be the first right edge with the inverse slope $0 \leq \gamma_1 = \frac{l}{k} < 1$. The case $\gamma_1 = 0$ implies $\alpha|\beta$ (see (4.5)) and $\alpha = 2$ (see Lemma 4.11).

Let $\gamma_1 > 0$. In terms of Fig. 4(b) we have $a = \alpha = 2a_0$ and $b = \beta = 2b_0$. Thus $r = 2$ in (4.6) and Lemma 4.6 implies that $p_0 = \frac{r}{t}(1 - \frac{l}{k}) < 2$. Therefore $\Delta_0 \cap \{j = 0\}$ is the interval between $(0, 0)$ and $(1, 0)$ and also $p_0 \geq 1$.

The case $p_0 = 1$ means $t = 1, l = 1, k = 2$. Since $k \leq 6$, we can use Lemma 4.10.

When $1 < p_0 < 2$ we have at least two right edges in Γ_0 . If their inverse slopes are < 1 , then we use Lemma 4.9.

The opposite occurs only when I_1 ends at $(2, 1)$ (then I_2 joins $(1, 0)$ with $(2, 1)$ and has $\gamma_2 = 1$). Here $2a_0 = 2 + nl, 2b_0 = 1 + nk$ for some n (see Fig. 5(a)). But the Eqs. (4.6) with $t = 1$ imply $a_0 = 1 + \nu l, b_0 = 1 + \nu k$. It is impossible. $\square$

The following lemmas prepare us to prove Theorem 4.16.

Lemma 4.13. $\alpha \neq 3$.

Proof. Suppose $\alpha = 3$. Firstly we agree that $\beta = 3b_0, b_0 \geq 2$ and that there is no vertical edge.

Suppose that the first right edge $I_1 \subset \Gamma_0$ joins the points $(2, b)$ and $(3, 3b_0)$, where $b = 0, 1, 2, 3$ (see Fig. 5(b)). Then its inverse slope $\gamma_1 = 1/(3b_0 - b)$ and the Eq. (4.5) implies

$$(2b_0 - b)\nu = b_0 - 1.$$

For $b = 0, 1, 2$ it is impossible. For $b = 3$ it holds only when $(\alpha, \beta) = (3, 6)$.

If $1/(3b_0 - 3) < \gamma_1 < 2/(3b_0)$ (i.e. $1 < p_0 < 2$ in Fig. 4(b)), then we have a right edge $I_2 \subset \Gamma_0$ with the inverse slope $\gamma_2 < 1$ and joining $(1, 0)$ with $(2, b), b > 3$. By arguments used in the proof of Lemma 4.11 it is impossible.

The case $\gamma_1 = 2/(3b_0)$ with b_0 odd leads (via (4.5)) to the false identity $b_0\nu = b_0 - 1$. But when $b_0 = 2b'$ we find $l = 1, k = 3b'$ and (4.5) gives $(2 - \nu)b' = 1$; again we arrive to the case $(\alpha, \beta) = (3, 6)$.

For $(\alpha, \beta) = (3, 6)$ we have $l = 1$ and $k = 3$ which is ≤ 6 (see Lemma 4.10). $\square$

Lemma 4.14. If $\beta = 2\alpha$ then $\alpha > 7$.

Proof. The Eq. (4.5) gives

$$(k - 2l)\nu = 1$$

for the inverse slope $\gamma_1 = \frac{l}{k}$ of the first right chart θ_1 . So, $\nu = 1, k = 2l + 1, r = \alpha$ and $t = l + 1$ in (4.6)–(4.7). Since $k \geq 7$, we have $l \geq 3$. By Lemma 4.6 we get

$$p_0 = \frac{\alpha}{2l + 1}.$$

If $\alpha < 7$, then $p_0 < 1$. If $\alpha = 7$, then $l = 3$, $k = 7$, i.e. $\gamma_1 = \frac{3}{7}$, and for the next chart θ_2 of depth 2 we get

$$p(\theta_2) + q(\theta_2) + \gamma_2 > \frac{5}{7} + \frac{3}{7} > 1.$$

So θ_2 cannot be of transitory type and $\gcd(\alpha, \beta) = \alpha$ should be a product of at least two primes, which is not the case. $\square$

Lemma 4.15. If $\frac{\beta}{\alpha} = \frac{3}{2}$ then $\alpha > 6$.

Proof. As above we start with the equation $(2k - 3l)v = 1$, which is solved as follows: $v = 1$, $k = 2 + 3v$, $l = 1 + 2v$. Here $v \geq 2$ and $l < \alpha$.

Note that $\gcd(\alpha, \beta)$ is a prime for $\alpha \leq 6$; hence Lemma 4.10 holds. If $\alpha \leq 6$, then we should take $k = 8$ and $l = 5$. Here $(\alpha, \beta) = (6, 9)$, the edge I_1 joins $(1, 1)$ with $(6, 9)$ and Γ_0 should have one-point intersection with the i axis (see Fig. 5(c)). $\square$

Heitmann proved that the situation with $\alpha + \beta < 16$ is impossible [14, Theorem 2.23]. To this he used two statements: $\alpha \geq 4$, and $b \geq \alpha + 8$ (where (α, b) , $b \leq \beta$, is an endpoint of the vertical edge). The first statement is proved in [14, Proposition 2.22].

Below we reprove and improve Heitmann's result.

Theorem 4.16. $\alpha + \beta > 16$.

Proof. In view of the above it is enough to consider the case $(a, b) = (\alpha, \beta) = (4, 12)$.

We have $a_0 = 1$, $b_0 = 3$, $r = 4$ and the Eq. (4.5) gives $v(k - 3l) = 2$.

If $v = 1$, then $k = 3l + 2$, $t = l + 1$ and $p_0 = \frac{8}{3l+2}$. If $l > 2$, then $p_0 < 1$. If $l = 1$, $k = 5$, then we have an edge I_2 with upper vertex $(2, 2)$ or $(3, 7)$ and with $\gamma_2 < 1$ (see Fig. 5(d)); by Corollary 4.8 this is impossible. If $l = 2$, then $k = 8$ is not relatively prime with l .

If $v = 2$, then $k = 3l + 1$, $t = 2l + 1$ and $p_0 = \frac{4}{3l+1}$. If $l > 1$, then $p_0 < 1$.

If $l = 1$, then $k = 4$ and $p_0 = 1$. Here we cannot apply Lemma 4.10 because $\gcd(\alpha, \beta) = 2 \cdot 2$ is a product of two primes. Note also that any chart θ_2 on the right of θ_1 in $\mathfrak{A}(P)$ cannot be transitory; this is because $p_0(\theta_2) \geq \frac{1}{4}$ and hence $p(\theta_2) + q(\theta_2) + \gamma(\theta_2) > 1$. Thus one expects that the path to transitory chart is as follows $\theta_0 - \theta_1 - \theta_2 - \theta_3$.

Now we again use results of the next section. Namely, Proposition 5.11 and Theorem 5.13 say that $\varphi_{\theta_2} = \chi_2^{\eta_0}$, $\varphi_{\theta_1} = \chi_1^{\eta_0\eta_1}$, $\varphi_{\theta_0} = \chi_0^{\eta_0\eta_1\eta_2}$. It should be $\eta_0 = m$, $\eta_1 = \eta_2 = 2$ and $\chi = \varphi_{\theta_1}^{1/m}$ should be of the form χ_1^2 .

But the above calculations give $\chi(u) = \text{const} \cdot (u^4 - w_1)^{\epsilon_1} (u^4 - w_2)^{\epsilon_2}$ with $w_1 w_2 \neq 0$ and $\epsilon_1 + \epsilon_2 = 3$. $\square$

Remark 4.17. Following the lines of the above proof one can consider more cases of (α, β) . This would allow one to obtain estimates from below on the degree of a Jacobian counterexample. Since $\deg P = \deg f = m(\alpha + \beta)$ and $m \geq 3$ we get $\deg P \geq 3(\alpha + \beta)$. From Theorem 4.16 we get $\deg P > 48$ for the counterexample.

The author has checked all the cases with $\alpha + \beta \leq 34$, i.e. $\deg P \leq 103$ (as 103 is prime), and did not encounter any Jacobian map. Moreover, the calculations are elementary and performed without any computer program. Recall that Moh [19] has checked the Jacobian Conjecture for $\deg P \leq 100$, using a computer.

5. Quasi-Puiseux expansions

In this section we compare quasi-Puiseux expansions associated with different N-P charts. This leads to interesting consequences. For example, the constant terms of the quasi-expansions associated with left charts are generated from non-compatibilities of the first non-constant terms of quasi-Puiseux expansions associated with right charts.

Consider two N-P charts θ and θ' which are **adjacent** in the following sense. Either θ, θ' are defined by means of adjacent edges $I, I' \subset \Gamma(f)$, I on the left of I' ; here I and/or I' can be right or left as well. Another possibility is that θ, θ' are defined by means of systems of the following type: $[(I_1, a_1), \dots, (I_{d-1}, a_{d-1}), I]$ and $[(I_1, a_1), \dots, (I_{d-1}, a_{d-1}), I']$, where I, I' are adjacent edges in $\Gamma(f_d)$ (see Algorithm 2.10).

The charts θ, θ' are compatible in the following sense. Suppose that $I, I' \subset \Gamma(f)$ are left edges and $\theta : x = uy^{-\gamma}$, $\theta' : x = u'y^{-\gamma'}$ (the other cases are analogous). We have a line (diagonal) $\mathcal{L}$ through $(0, 0)$ and the vertex $I \cap I'$ (see Fig. 6). If an integer point $(a, b) \in \mathcal{L} \cap \mathbb{Z}^2$, then it should be $x^a y^b = u^a y^{p(\theta)} = (u')^a y^{p(\theta')}$.

In the case $\theta = \theta(I)$, $I \subset \Gamma(f)$ the (right) N-P chart θ defines the **quasi-homogeneous gradations** in the ring $\mathbb{C}[x, y]$:

$$\deg_{\theta} u = -\gamma, \quad \deg_{\theta} y = 1. \quad (5.1)$$

Analogously θ' defines a gradation $\deg_{\theta'}$.

We use a new gradation $\deg_{\theta, \theta'}$ in the ring of polynomials in x and y . Let $I \cap I' = (c, d)$ (see Fig. 6). Let $J = \delta_0 I, J' = \delta_0 I'$ be the corresponding edges in $\Gamma(g)$. If a point (i, j) lies on the left of the line $\mathcal{L}$, then $\deg_{\theta, \theta'}(x^i y^j)$ is defined by means of the gradation $\deg_{\theta}$, i.e. we put it equal to the j coordinate of the intersection of the line parallel to I and through (i, j) with the line $\mathcal{L}$. If (i, j) lies on the right of the line $\mathcal{L}$, then $\deg_{\theta, \theta'}(x^i y^j)$ is defined by means of the gradation $\deg_{\theta'}$, i.e. we put it equal to the j coordinate of the intersection of the line parallel to I' and through (i, j) with the line $\mathcal{L}$. Such a gradation (called the 'Newton gradation') was used in the book [3] in the classification of singularities.

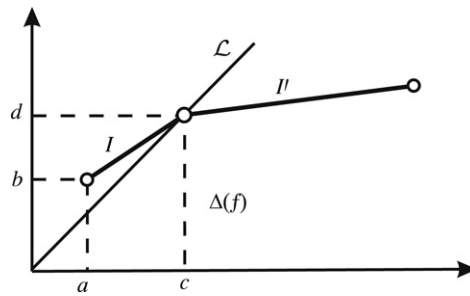


Fig. 6.

Analogously one defines gradations $\deg_\theta$, $\deg_{\theta'}$ and $\deg_{\theta, \theta'}$ in the ring of Puiseux polynomials of two variables in the case θ, θ' are associated with edges $I, I' \subset \Gamma(f_d)$. (The expansions (3.5) and (3.6) are expansions into homogeneous parts with respect to the $\deg_\theta$ -degree.)

Denote also $\delta = \delta(\theta)$, $\delta' = \delta(\theta')$.

Proposition 5.1. *The quasi-Puiseux expansions modulo $O(\tilde{f}^{\max(\delta, \delta')})$ associated with two adjacent quasi-Puiseux charts θ and θ' coincide.*

Proof. Consider only the case when θ, θ' are associated with $I, I' \subset \Gamma(f)$; the general case is proved in the same way.

We have the expansion $f = f^{(1)} + f^{(2)} + \dots$, where $f^{(i)}$ contain terms of the same degree $\deg_{\theta, \theta'}$; in particular, $f^{(1)} = \sum_{(i,j) \in I \cup I'} f_{ij} x^i y^j$. Similarly, $g = g^{(1)} + g^{(2)} + \dots$.

Let $f_i = Ax^a y^b \prod (u - v_i)$ and $f_{i'} = Bx^c y^d \prod (u' - w_i)$, $a < c$, $b < d$, be the leading terms associated with the gradations $\deg_\theta$ and $\deg_{\theta'}$. Then we have $f^{(1)} = Ax^a y^b \prod (u - v_i) \prod (u' - w_i) + \dots =: f_{i, i'} + \dots$ where the dots mean terms of lower degree $\deg_{\theta, \theta'}$ than of $f_{i, i'}$; so we write $f = f_{i, i'} + f^{(2)} + \dots$. Similarly we get the leading term $g_{j, j'}$ (for the expansion of g) in the form of a product. Because the parts f_i and g_j satisfy a homogeneous relation and the parts $f_{i'}$ and $g_{j'}$ satisfy the same relation, we obtain that $g_{j, j'} = (f_{i, i'})^{\delta_0}$.

As in the usual quasi-Puiseux expansion, we apply the change $g \rightarrow g_1 = g - b_0 f^{\delta_0}$. But here we must properly define the formal expansion of f^{δ_0} . More generally, if $f_{i, i'} = (\chi(x, y))^{\hat{p}}$, then we define $f^{1/\hat{p}} = \chi \left[1 + f^{(2)} \chi^{-\hat{p}} + \dots \right]^{1/\hat{p}} = \chi + \frac{1}{\hat{p}} f^{(2)} \chi^{1-\hat{p}} + \dots$ as a series whose general term has the form $h(x, y) / \chi^r$. Each numerator h is expanded as a sum of terms with fixed $\deg_{\theta, \theta'}$ -degree. Namely in this sense also other powers $f^{j/\hat{p}}$ appearing in the below series must be considered.

We have $f_{i, i'} = f_i \cdot \prod (u' - w_i) = f_i \cdot (1 + \text{l.o.t. in } \deg_\theta)$ and $f_{i, i'} = f_{i'} \cdot \prod (1 - v_i/u) = f_{i'} \cdot (1 + \text{l.o.t. in } \deg_{\theta'})$. Therefore also negative powers of $f_{i, i'}$ and χ behave well with respect to these two gradations, e.g. $(f_{i, i'})^{-1} = (f_i)^{-1} \cdot (1 + \text{l.o.t. in } \deg_\theta)$ and $(f_{i, i'})^{-1} = (f_{i'})^{-1} \cdot (1 + \text{l.o.t. in } \deg_{\theta'})$.

The series $g - f^{\delta_0}$ has smaller degree $\deg_{\theta, \theta'}$. It also has the leading term in the form of a product of powers of irreducible factors, which must be proportional to a rational power of $f_{i, i'}$, and so on. These changes give the first terms of the expansion $g = b_0 f^{\delta_0} + b_1 f^{\delta_1} + \dots$ which are constant and good for the $\deg_\theta$ -gradation as well as for the $\deg_{\theta'}$ -gradation. $\square$

There are some special curves in the target space $\mathcal{Y}^0$:

$$K_{\text{left}} : y \rightarrow P(0, y), \quad K_{\text{right}} : x \rightarrow P(x, 0) \quad (5.2)$$

and the curves $S_k = \hat{P} \circ \pi(D_{i_k} \setminus \infty)$, the images of divisors of non-properness type (see Theorem 3.14). It is interesting to compare the Puiseux expansions (at infinity) of the latter curves with our quasi-Puiseux expansions.

Proposition 5.2. (a) *Let I be a left-most left edge of $\Gamma(f)$ (respectively the right-most right edge) and θ the corresponding N-P chart. Then the part modulo $O(\tilde{f}^{\delta})$ of the corresponding quasi-Puiseux expansion coincides with the analogous part of the Puiseux expansion of the rational curve $K_{\text{left}} \subset \mathbb{C}^2$ at infinity (respectively of the curve K_{right}).*

(b) *Let θ be an N-P chart of Abhyankar type adjacent to a chart θ_k of non-properness type. Then the part modulo $O(\tilde{f}^{\delta})$ of the corresponding quasi-Puiseux expansion coincides with the analogous part of the Puiseux expansion of the suitable component S_k of the non-properness set $S(P)$.*

Proof. We consider only the case when θ is associated with the left-most left edge of $\Gamma(f)$; other cases are proved in the same way.

We have $\tilde{f}(u, y) = \varphi(u)y^p + \dots$, where $\varphi(0) \neq 0$ and $\tilde{f}(0, y) = f(0, y)$. Therefore the rational powers $\tilde{f}^{\delta_1}$ admit expansions into powers of y with (rational) coefficients, which are continuous at $u = 0$. Thus the quasi-Puiseux expansion is continuous in u at $u = 0$. For $u = 0$ we obtain the Puiseux expansion for the curve K_{left} . $\square$

The latter two propositions lead to interesting properties concerning the ‘left’ divisors of resolution and the left components S_k . Indeed, since $\gamma \geq 1$ for any left N–P chart, we have $\delta = \frac{1-\gamma}{p} - 1 \leq -1$ (see (3.10)). Therefore the parts modulo $O(\tilde{f}^{-1})$ of the quasi-Puiseux expansions associated with the left N–P charts of Abhyankar type are the same (do not depend on the chart). Moreover, they are the same as the mod $O(X^{-1})$ parts of the Puiseux expansions of the curves K_{left} and of the left curves S_k .

The following lemma says what happens when the Puiseux expansion of a rational curve is very simple.

Lemma 5.3. (a) *Let*

$$(X, Y) = (t^p + c_1 t^{p-1} + \cdots + c_p, t^q + d_1 t^{q-1} + \cdots + d_q),$$

$p > q$, $q \nmid p$, *be a parametric rational curve. Then its Puiseux expansion at infinity has the form*

$$Y = X + aX^{1/p-1} + O(X^{-1}) \quad (5.3)$$

if and only if

$$Y'X - \frac{q}{p}YX' \equiv (1 - p - q)a. \quad (5.4)$$

In particular, $a = 0$ iff the image is a quasi-homogeneous curve, $Y = X^{q/p}$.

(b) *If, instead of (5.4), we have the equation*

$$Y'X - \frac{q}{p}YX' \equiv bX^e, \quad b \neq 0, \quad (5.5)$$

then

$$Y = X^{q/p} + b_1 X^{1/p-1+e} + O(X^{-1+e}), \quad b_1 \neq 0.$$

Proof. Consider the function $H(t) = Y(t)/X^{q/p}(t)$. By the assumption (5.3), $H(t) = 1 + at^{1-p-q} + \cdots$ as $t \rightarrow \infty$. Thus $H'(t) = (1 - p - q)at^{-p-q} + \cdots$. But then the formula $H' = [Y'X - \frac{q}{p}YX']/X^{q/p+1}$ shows that the polynomial $Y'X - \frac{q}{p}YX'$ must be identically equal to the constant $(1 - p - q)a$. It is also clear that, the identity (5.4) implies the property (5.3).

The second statement is proved in the same way. $\square$

Corollary 5.4. *No left component S_{k_0} of the non-properness set of a Jacobian map is a simply connected curve.*

Proof. We use the following theorem of Zaidenberg and Lin [33] (in the smooth case this is the Abhyankar–Moh–Suzuki theorem):

any simply connected algebraic curve in $\mathbb{C}^2$ can be reduced, via a composition of elementary transformations, to a quasi-homogeneous curve.

We apply this theorem to the simply connected component S_{k_0} .

(Note that the properties of the Jacobian map described in Theorem 3.2(d) become preserved when applying the elementary transformations in the target space. In fact, the assumption $\frac{\deg f}{\deg g} \notin \mathbb{Z}$ implies that S_{k_0} cannot be smooth.)

Hence we can assume that the Puiseux expansion of S_{k_0} is $Y = X^{q/p} + O(X^{-1})$. Then by Proposition 5.2 the Puiseux expansion of K_{left} is also of this form. By Lemma 5.3(a) we find that the curve $K_{\text{left}} = P(\{x = 0\})$ is quasi-homogeneous, and hence singular. This contradicts the local regularity of the Jacobian map P . $\square$

From Proposition 5.1 we know how to compare the quasi-Puiseux expansions mod $O(\tilde{f}^{\max(\delta, \delta')})$ associated with adjacent charts θ and θ' ; here $\delta = \delta(\theta)$ and $\delta' = \delta(\theta')$. Now we shall look more carefully at the terms with $\tilde{f}^{\max(\delta, \delta')}$.

But before that we take a look at the structure of the non-constant part $\tilde{g}_N$ of the quasi-Puiseux expansion.

Lemma 5.5. *Let $\theta : (x, u) \rightarrow (x, y)$ be a right N–P chart of Abhyankar type or of transitory type and with inverse slope $\gamma < 1$ and let $\tilde{g} = b_0 \tilde{f}^{\delta_0} + \cdots + b_{N-1} \tilde{f}^{\delta_{N-1}} + \tilde{g}_N(x, u)$, $\tilde{g}_N(x, u) = c(u) \tilde{f}^\delta + \cdots$ be its quasi-Puiseux expansion. Then we have*

$$\deg c(u) = 0,$$

i.e. $c(u) \rightarrow \text{const} \neq 0, \infty$ as $u \rightarrow \infty$.

This means that the Newton's diagram $\Delta(\tilde{g}_N)$ of the function $\tilde{g}_N(x, u) = \tilde{\psi}_N(u) x^{\delta_N} + \cdots$ contains the vertex $B = \delta \cdot A$, $A = (p, \deg \varphi)$ and, in particular, $\delta \cdot \deg \varphi$ is integer.

Proof. This is stated explicitly in Lemmas A.5 and A.7 (in the Appendix). Fig. 7 presents this situation in the case $\theta = \theta(I)$ is associated with a right edge $I \subset \Gamma(f)$. $\square$

Remark 5.6. In the case θ is a chart of transitory type one has $\tilde{g}_N = \tilde{g}$ and $\delta = \frac{q}{p} = \delta_0$ is positive. Thus the first non-constant term of the corresponding quasi-Puiseux expansion is just the first term of this expansion, $\tilde{g} = c(u) \tilde{f}^\delta + \cdots$. The corresponding point B lies in the same quadrant as the point A .

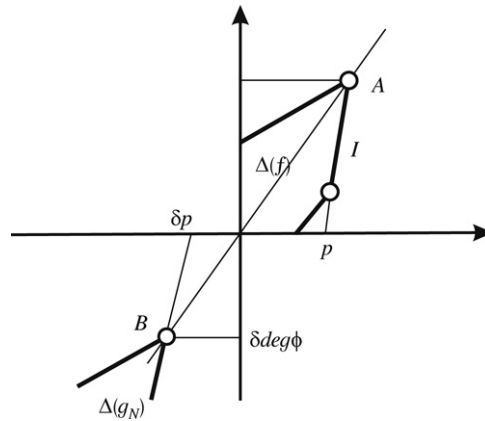


Fig. 7.

Recall that θ, θ' are associated with adjacent edges $I, I' \subset \Gamma(f_m)$, I on the left of I' . Recall also that $\Gamma(f_m)$ lies in a plane $\mathbb{R}^2$ obtained by means of the application of the transformation

$$T : (i, j) \rightarrow (i - \gamma_m j, j)$$

(in the right case), or $(i, j) \rightarrow (i, j - \gamma_m i)$ (in the left case). Let $\tilde{I}, \tilde{I}'$ be the intervals in $\mathbb{R}^2$ (the initial plane) obtained from I, I' by application of the inverse to the above transformation, i.e. $\tilde{I} = T^{-1}(I)$ etc.

Lemma 5.7. Let $\mathcal{L}$ be the line through the vertex $A := \tilde{I} \cap \tilde{I}'$ and the origin O . Then the slope of $\mathcal{L}$ is > 1 if and only if $\delta < \delta'$.

Proof. We consider only the right case. Using the quasi-homogeneous gradations: $\deg_{\theta} x = \deg_{\theta'} x = 1, \deg_{\theta} y = -\gamma, \deg_{\theta'} y = -\gamma', \deg_{\theta} \tilde{f} = p, \deg_{\theta'} \tilde{f} = p'$ we get that

$$\delta = \frac{q_N}{p} = \frac{\deg_{\theta}(xy/\tilde{f})}{\deg_{\theta}\tilde{f}}, \quad \delta' = \frac{\deg_{\theta'}(xy/\tilde{f})}{\deg_{\theta'}\tilde{f}},$$

(see (3.10)). Geometrically it means the following (see Fig. 8(a)). Take the point $-A$ and add to it the vector $(1, 1)$, you get a point $C = -A + (1, 1)$. We have two lines through the point C : L parallel to $\tilde{I}$ and L' parallel to $\tilde{I}'$. The line L , i.e. $L \cap T^{-1}(\Delta(\tilde{g}_N))$, 'supports' the part $\tilde{\psi}_N(u)x^{q_N}$ in the expansion of $\tilde{g}_N$. The analogous property is true for the line L' .

Now $\delta < \delta'$ iff the point δA lies on left of the point $\delta' A$ in the line $\mathcal{L}$ (see Fig. 8(a)). So $\delta < \delta'$ iff the segment $[-A, C]$ lies below the line $\mathcal{L}$, which is equivalent to the fact that the slope of $\mathcal{L}$ is > 1 . $\square$

Lemma 5.7 concerns non-compatibility of quasi-Puiseux expansions associated to different N–P charts. It turns out that this non-compatibility is a source of constant terms in the left quasi-Puiseux expansions.

Proposition 5.8. Assume the situation of **Lemma 5.7**, but such that θ, θ' are right N–P charts such that

$$\delta < \delta'.$$

Let

$$\tilde{g} = b_0 \tilde{f}^{\delta_0} + \dots + b_{N-1} \tilde{f}^{\delta_{N-1}} + c_{\theta'}(u') \tilde{f}^{\delta'} + \dots$$

be the quasi-Puiseux expansion associated with θ' .

Then the quasi-Puiseux expansion associated with θ has the form

$$\tilde{g} = (b_0 \tilde{f}^{\delta_0} + \dots + b_{N-1} \tilde{f}^{\delta_{N-1}}) + b_N \tilde{f}^{\delta_N} + c_{\theta}(u) \tilde{f}^{\delta} + \dots, \quad (5.6)$$

where $\delta_N = \delta'$ and $b_N = c_{\theta'}(\infty)$. Therefore the constant term $b_N \tilde{f}^{\delta_N}$ 'arises' from the non-constant term $c_{\theta'} \tilde{f}^{\delta'}$.

Proof. Note firstly that $\gamma, \gamma' < 1$ (because the slope of $\mathcal{L}$ is > 1). Therefore by **Lemma 5.5** Darboux function $c_{\theta'}(u')$ has degree 0, i.e. $c(\infty) \neq 0, \infty$. Thus the point $B = \delta' A$ in Fig. 7 corresponds to the point B on the line $\mathcal{L}$ in Fig. 8(b). Namely this point is responsible for the leading term $(u^{\epsilon_0} x^p)^{\delta'}$, $\epsilon_0 = \deg \varphi_{\theta'}$ (as $u \rightarrow 0$) in the gradation defined by means of a line L'' (parallel to the edge $\tilde{I}$). We have

$$(\tilde{g}_N)_{L''} = (u^{\epsilon_0} x^p)^{\delta'} \cdot [\text{const} + O(u)] \sim (b_N \tilde{f}^{\delta_N})_{L''} \quad \text{as } u \rightarrow 0, x \rightarrow \infty.$$

Therefore it remains to show that there are no constant terms in the expansion (5.6) between $b_N \tilde{f}^{\delta_N}$ and $c_{\theta}(u) \tilde{f}^{\delta}$.

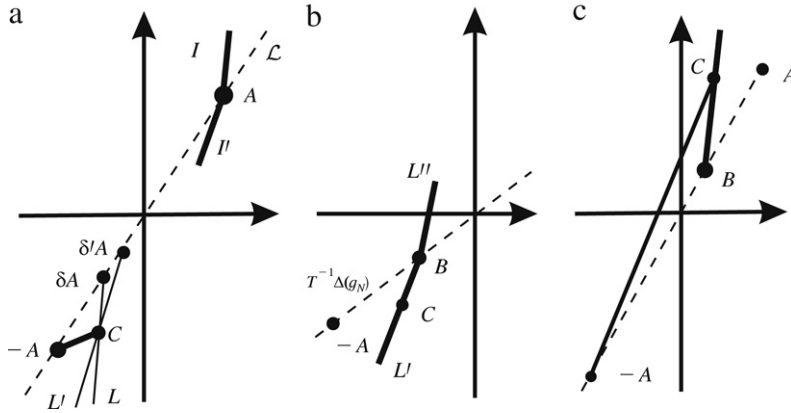


Fig. 8.

Introduce variables u' (defined in the chart θ') and u (defined in the chart θ). We have $u' = ux^{\gamma' - \gamma}$, where $\gamma' - \gamma > 0$, and we shall assume that $u' \rightarrow \infty$ and $u \sim O(1)$.

We can write

$$f \circ \theta' = x^{p'} [\varphi_{\theta'}(u') + u\chi(u, u')],$$

where $x^{p'} \varphi_{\theta'}(u') = (f \circ \theta)_{I'}$ is the part of the expansion of $f \circ \theta$ associated with the edge $\tilde{I}'$ and χ is a Puiseux series. Analogously we can write

$$\tilde{g}_N = x^{q_N} [\tilde{\psi}_N(u') + u\omega(u, u')],$$

where $x^{q_N} \tilde{\psi}_N(u')$ is the (leading) part $(\tilde{g}_N)_{L'}$ associated with the line L' (parallel to $\tilde{I}'$) and ω is a Puiseux series.

The functions $\varphi_{\theta'}(u')$ and $\tilde{\psi}_N(u')$ satisfy the equation $\tilde{\psi}_N \varphi_{\theta'} - \frac{q_N}{p'} \tilde{\psi}_N \dot{\varphi}_{\theta'} \equiv \frac{1}{p'}$, $\cdot = \frac{d}{du'}$ (see the Eq. (3.14)). Here $\tilde{\psi}_N$ is a rational function of the form $v(u')/\varphi_{\theta'}^m$ of degree $\delta \cdot \deg \varphi_{\theta'}$. The polynomial v satisfies the equation $\dot{v} \varphi_{\theta'} - \text{const} \cdot v \dot{\varphi}_{\theta'} = \text{const} \cdot \varphi_{\theta'}^m$, the same as (5.5) in Lemma 5.3. Using the latter lemma, after simple transformations, we get

$$\tilde{\psi}_N = b_N (\varphi_{\theta'})^{\delta_N} + O((\varphi_{\theta'})^{1/\deg \varphi_{\theta'} - 1}).$$

This implies that $x^{q_N} \tilde{\psi}_N = x^{q_N} (\varphi_{\theta'})^{\delta_N} + O(x^\eta)$, where

$$\begin{aligned} \eta &= q_N + (\gamma' - \gamma)(1 - \deg \varphi_{\theta'}) \\ &= (1 - \gamma' - p') + (\gamma' - \gamma) - (\gamma' - \gamma) \deg \varphi_{\theta'} \\ &= 1 - \gamma - (p' + (\gamma' - \gamma) \deg \varphi_{\theta'}) \\ &= 1 - \gamma - p \\ &= \delta p. \end{aligned}$$

Therefore the quasi-Puiseux expansion of $\tilde{g}_N$, i.e. $b_N \tilde{f}^{\delta_N} + b_{N+1} \tilde{f}^{\delta_{N+1}} + \dots + b_N \tilde{f}^{\delta_{N'}} + c_\theta(u) \tilde{f}^\delta + \dots$, has the property that $b_{N+1}|_{u=0} = b_{N+2}|_{u=0} = \dots = b_{N'}|_{u=0} = 0$. Since these b_j 's are constants, the corresponding terms $b_j (f \circ \theta)^{\delta_j}$ are absent in (5.6). $\square$

Lemma 5.9. Assume the situation of Proposition 5.8, but such that θ' is a right chart of transitory type. Then the slope of the corresponding line $\mathcal{L}$ is > 1 , and hence $\delta < \delta'$ (like in Proposition 5.8).

Proof. We know that $\delta' = q'/p' = \delta_0 > 0$, i.e. $B = \delta'A$ lies in the first quadrant. If the slope of $\mathcal{L}$ were < 1 , then $C = -A + (1, 1)$ would also lie in the first quadrant (see Fig. 8(c)). It follows that the j coordinate of the point $A = (a, \epsilon_0) \in L$ should satisfy $0 < \epsilon_0 < 1$. But ϵ_0 is an integer. $\square$

Remark 5.10. There is an interesting observation stemming from Proposition 5.8 and Lemma 5.9. Let θ' be of transitory type and θ of Abhyankar type and adjacent to θ' . Then the expansion (5.6) takes the form $\tilde{g} = \tilde{f}^{\delta_0} + c_\theta(u) \tilde{f}^\delta + \dots$. Its terms depend on $p, p', \delta_0 = \deg g / \deg f, \gamma, \gamma', \varphi_\theta$ and $\varphi_{\theta'}$. In particular, a change $g \rightarrow g - \text{const}$ cannot change this expansion. It follows that $\delta > 0$ (compare the proof of Corollary 5.4).

We can reverse the roles of f and g and get the expansion $\tilde{f} = \tilde{g}^{\lambda_0} + e_\theta(u) \tilde{g}^\lambda + \dots$, $\lambda = \frac{1-\gamma}{q} - 1$. Since the elementary changes $f \rightarrow f - Q(g)$, $\deg Q \leq [p/q]$ (the integer part of p/q) do not change it, it follows that $\lambda > [p/q]$.

We have one more observation following from Fig. 8(b) (due to Heitmann [14, Proposition 1.10]).

Proposition 5.11. *Let us assume a situation as in Proposition 5.8. Let us represent $\delta_N = \delta'$ as an irreducible ratio*

$$\delta_N = \frac{\rho}{\eta}.$$

Then the polynomial φ_θ is a power of order η of another polynomial $\varphi_\theta = \chi^\eta$, while $\varphi_{\theta'}$ is not.

Moreover, for any other chart θ'' defined by a vertex-edge $I \cap I'$ with the inverse slope $\gamma < \gamma'' < \gamma'$ the same statement as above is true. Thus $\varphi_{\theta''} = \text{const} \cdot u^{\deg \varphi_{\theta'}}$ is a power of an order for which $\varphi_{\theta'}$ is not.

Proof. Since the term $b_N (f \circ \theta)^{\delta_N}$ is constant, φ_θ must be an η -th power. If $\varphi_{\theta'}$ were such a power, then exponents in Darboux function $\varphi_{\theta'}^{-\delta'-1}$, in the Schwarz–Christoffel integral $\int \varphi_{\theta'}^{-\delta'-1}$ (see (3.13)), would be negative integers. But this contradicts Lemma A.7 in the Appendix. $\square$

Remark 5.12 (Heitmann's Proof of the Appelgate–Onishi Theorem). Consider the situation as in the second part of Proposition 5.11, where θ' is associated with the first right edge $I' \subset \Gamma(f)$ and $\theta'' = \theta_0$ (the root). Then $I'' = \{(m\alpha, m\beta)\}$, $f_{I''} = (x^\alpha y^\beta)^m$ and $f_{I'} = (f_0)^m$ for a polynomial f_0 . Proposition 5.11 implies that $x^\alpha y^\beta = (x^{\alpha_0} y^{\beta_0})^r$, where $r > 1$ is such that f_0 is not an r -th power. So, $\gcd(\deg f, \deg g) = (\alpha_0 + \beta_0)r$ is a product of at least two primes.

The situation as in Proposition 5.11 arises each time we encounter an edge on a branch of the Newton–Puiseux graph $\mathfrak{N}(P)$ from the root θ_0 to a transitory chart.

Theorem 5.13. *If there is a path*

$$\theta_0 - \theta_1 - \theta_2 - \dots - \theta_N$$

in $\mathfrak{N}(P)$ with a transitory chart θ_N , then

$$\gcd(\alpha, \beta) = \eta_1 \dots \eta_{N-1}, \quad \eta_j > 1.$$

The factors η_j appear in the characteristic pairs (ξ_j, η_j) associated with the N–P chart $\theta : Y = X^{\delta_0} + b_1 X^{\delta_1} + \dots + U X^{\delta_N}$ (in the target space) and defined by means of the quasi-Puiseux expansion associated with the N–P chart θ_0 . We have $\varphi_{\theta_N} \equiv \chi_N^{\eta_0}$, $\varphi_{\theta_{N-1}} \equiv \chi_{N-1}^{\eta_0 \eta_1}$, $\dots$, $\varphi_{\theta_0} \equiv \chi_0^{\eta_0 \dots \eta_N}$. In the next section we present examples, see also Remark 5.18.

Let us continue to compare the quasi-Puiseux expansions for adjacent N–P charts θ, θ' .

Proposition 5.14. *A situation as in Lemma 5.7 with $\gamma, \gamma' < 1$, but*

$$\delta \geq \delta'$$

(the inequality opposite to the one in Proposition 5.8) cannot hold.

This means that whenever the edge $\tilde{I}$ (associated with $I \subset \Gamma(f_0)$) has the lower endpoint below the line $j = i$, then the N–P charts on the right of $\theta(I)$ in $\mathfrak{N}(P)$ have inverse slope $\gamma \geq 1$; in particular, any right-most such N–P chart in $\mathfrak{N}(P)$ is of non-properness type.

Proof. The situation with $\delta > \delta'$ is presented at Fig. 9(a) (compare with Fig. 8(a)). Here the j coordinates of the points $-A, B, C$ should be integer. But $C = -A + (1, 1)$ and $j(B)$ should lie between $j(-A)$ and $j(C)$, which is impossible.

Suppose that $\delta = \delta'$, i.e. the slope of the line $\mathcal{L}$ equals 1. Here the points B and C should coincide.

This common point should correspond to the term $(x^p u^{\epsilon_0})^\delta$ in the series $\tilde{g}_N = x^{q_N} \tilde{\psi}_N(u) + \dots$, associated with the chart θ , where $\varphi_\theta(u) = u^{\epsilon_0}(a + \dots)$, $a \neq 0$ as $u \rightarrow 0$. The same point should also correspond to the term $(x^{p'}(u')^{\epsilon_0})^\delta$ in $\tilde{g}'_N = x^{q'_N} \tilde{\psi}'_N(u') + \dots$, associated with the chart θ' , where $\varphi_{\theta'}(u') \sim \text{const} \cdot (u')^{\epsilon_0}$ as $u' \rightarrow \infty$. It follows that $c_\theta(0) = c_{\theta'}(\infty) \neq 0, \infty$ in the corresponding quasi-Puiseux expansions.

But we have $c_\theta(u) = \frac{1}{p} \int \varphi^{(\gamma-1)/p} = u^{1+\epsilon_0(\gamma-1)/p} \cdot (b + \dots)$ as $u \rightarrow 0$, where $1 + \epsilon_0(\gamma-1)/p \neq 0$ (since there are no logarithms in c_θ). So it is not true that $c_\theta(0) \neq 0, \infty$ and we have arrived at a contradiction. $\square$

Proposition 5.15. *If $\gamma < 1$, $\gamma' \geq 1$, $\delta \geq \delta'$ (for adjacent right N–P charts θ, θ') and*

$$\tilde{g} = b_0 \tilde{f}^{\delta_0} + \dots + b_R \tilde{f}^{\delta_R} + c_\theta(u) \tilde{f}^\delta + \dots$$

is the quasi-Puiseux expansion associated with θ , then the quasi-Puiseux expansion associated with θ' has the form

$$\tilde{g} = b_0 \tilde{f}^{\delta_0} + \dots + b_R \tilde{f}^{\delta_R} + O(\tilde{f}^{-1}).$$

Proof. Under the assumption $\gamma' \geq 1$ we have $\delta' < -1$, i.e. the point $-A$ lies below the line L' (see Fig. 9(b)). The point C is a vertex of the Newton diagram $T^{-1}\Gamma(\tilde{g}_N)$, because the asymptotic of $\tilde{\psi}_N(u)$ as $u \rightarrow 0$ agrees with the asymptotic of $\tilde{\psi}'_N(u')$ as $u' \rightarrow \infty$ (see Proposition 4.1). It follows that there are no points from $\text{Supp}(\tilde{g}_N)$ below the line L' . Therefore $\tilde{g}_N = O(\tilde{f}^{-1})$. $\square$

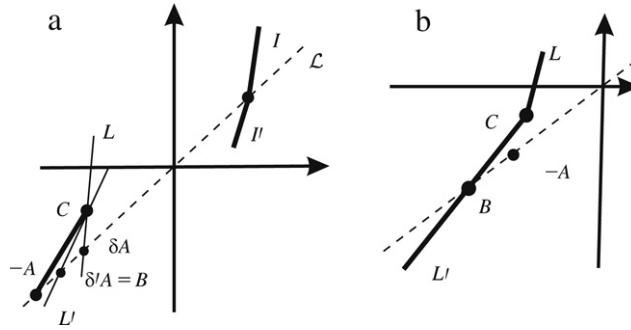


Fig. 9.

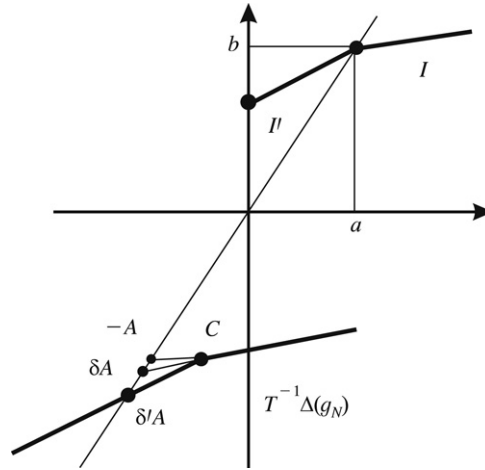


Fig. 10.

Let us change sides, i.e. now we consider the left N–P charts θ of Abhyankar type. Recall that there the slopes $\gamma > 1$. Recall also that Proposition 4.1 says that the Schwarz–Christoffel integral $\int \varphi^{(\gamma-1)/p}$ (appearing in the definition of the coefficient $c_\theta(u)$, see (3.13)) is such that the subintegral function is either a polynomial or a product $(u - u_0)^v$ times a polynomial. But the constant of integration was not fixed; (note that in the case of charts with $\gamma < 1$ we have $c_\theta(u) = c_\theta(\infty) + \frac{1}{p} \int_\infty^u \varphi^{(\gamma-1)/p}$).

Lemma 5.16. *In the case of a N–P chart (left or right) of Abhyankar type with*

$$\gamma \geq 1$$

we have

$$\varphi^{(\gamma-1)/p} = u^{v_0} \times \text{polynomial} \quad (5.7)$$

and

$$c_\theta(u) = \pm \frac{1}{p} \int_0^u \varphi^{(\gamma-1)/p}.$$

Proof. Let θ be associated with a left edge $I \subset \Gamma(f)$ with the slope γ and let (a, b) be the left endpoint of I . Either (a, b) is incident to an edge I' with slope $\gamma' > \gamma$ or lies on the vertical axis (we say then that $\gamma' = +\infty$). Consider a N–P chart θ'' defined by the edge-vertex (a, b) and slope $\gamma < \gamma'' < \gamma'$. The quasi-Puiseux expansion associated with θ'' is such that its initial (constant) part, modulo $\tilde{f}^{\delta(\theta'')}$, coincides with the corresponding part for θ . The initial term $c_{\theta''}(u) \tilde{f}^{\delta(\theta'')}$ of the non-constant part is the monomial $x^{1-a} y^{1-b} = u^{\kappa} \tilde{f}^{\delta(\theta'')}$, $\tilde{f} = x^a y^b$. Here there is no term $\text{const} \cdot \tilde{f}^{\delta(\theta'')}$, because it would be equal to $\text{const} \cdot u^{\kappa_1} y^{\kappa_2}$, where κ_1 would be non-integer (for suitable choice of γ''). This would contradict the Rationality Condition. It follows that the Newton diagram of $\tilde{g}_N$ is like Fig. 10. The part $\tilde{g}_N|_L$ of $\tilde{g}_N$ takes the form $x^{q_N} \cdot u^{\delta a+1} (1 + \dots)$. This implies (5.7).

If $\varphi^{(\gamma-1)/p} = (u - u_0)^v \times \text{polynomial}$, $u_0 \neq 0$ and $v \notin \mathbb{Z}$, then $c_\theta(u)$ (defined via (5.7)) could not be of Darboux type.

Analogous arguments work in the case of N–P charts of depth $d > 1$ and/or in the case of right charts. $\square$

As a corollary we have the following

Proposition 5.17. *In the situation of Fig. 10 the constant parts of the quasi-Puiseux expansions associated with the charts θ and θ' are the same.*

In particular, the constant parts of the quasi-Puiseux expansions associated with all the left charts of Abhyankar type are all the same.

Also the same are the constant parts of the quasi-Puiseux expansions associated with all the right N–P charts, of Abhyankar type with $\gamma \geq 1$, which belong to a subtree of $\mathfrak{A}(P)$ rooted in a fixed N–P chart of Abhyankar type with $\gamma < 1$.

Remark 5.18. The above results reveal the following interesting picture. If θ_N is a N–P chart of transitory type, then

$$\tilde{g} = c_{\theta_N}(u)\tilde{f}^{\delta_0} + \dots, \quad \delta_0 = \delta(\theta_N).$$

If θ_{N-1} is left and of Abhyankar type adjacent to θ_N , then

$$\tilde{g} = b_0\tilde{f}^{\delta_0} + c_{\theta_{N-1}}(u)\tilde{f}^{\delta_1} + \dots, \quad \delta_1 = \delta(\theta_{N-1}), b_0 = c_{\theta_N}(\infty) = 1.$$

If θ_{N-2} is left and of Abhyankar type adjacent to θ_{N-1} on the left, then

$$\tilde{g} = b_0\tilde{f}^{\delta_0} + b_1\tilde{f}^{\delta_1} + c_{\theta_{N-2}}(u)\tilde{f}^{\delta_2} + \dots, \quad \delta_2 = \delta(\theta_{N-2}), b_1 = c_{\theta_{N-1}}(\infty),$$

etc.

The above sequence of N–P charts in the source space has its equivalent in the target space:

$$\Theta_0 : (X, Y) = (X, U_0X^{\delta_0}), \quad \Theta_1 : Y = b_0X^{\delta_0} + U_1X^{\delta_1}, \dots;$$

thus N is the depth of Θ_N . Let (ξ_i, η_i) be the characteristic pairs associated with Θ_i . Then $(\xi_0, \eta_0) = (n, m)$. Moreover, one has

$$\varphi_{\theta_N} = \chi_N^{\eta_0}, \quad \varphi_{\theta_{N-1}} = \chi_{N-1}^{\eta_0\eta_1}, \dots$$

for some polynomials $\chi_N, \chi_{N-1}, \dots$.

In particular, if θ is the first left N–P chart (i.e. defined by means of the first left edge of $\Gamma(f)$ incident to the root $(\alpha m, \beta m)$), then the constant part

$$\tilde{g} = b_0\tilde{f}^{\delta_0} + \dots + b_L\tilde{f}^{\delta_L}$$

of its quasi-Puiseux expansion is obtained from the first non-constant terms of the quasi-Puiseux expansions for right N–P charts from a chain of adjacent right N–P charts which end at a chart of transitory type. Moreover, the result does not depend on the chain. The same is true for other left N–P charts.

In the case of the right N–P charts θ of Abhyankar type with $\gamma \geq 1$ the mod $O(\tilde{f}^{\delta})$ parts of the corresponding quasi-Puiseux expansions take the forms

$$\tilde{g} = b_0\tilde{f}^{\delta_0} + \dots + b_R\tilde{f}^{\delta_R},$$

where $R =: R(\theta) \leq L$ and the terms $b_j\tilde{f}^{\delta_j}, j \leq R$ are the same as the first R terms of the constant parts of the quasi-Puiseux expansions associated with the left N–P charts. But in the right case there can be fewer terms than in the left case.

6. Topology of Jacobian maps

Definition 6.1. A continuous map $F : X \rightarrow Y$ is **proper** if the preimage of any compact subset of Y is compact. The **non-properness set** of F , $S(F)$ consists of points $y_0 \in Y$ which have a neighborhood U with compact closure $\bar{U}$ and such that $F^{-1}(\bar{U})$ is not compact. In other words, any such $y_0 \in S(F)$ is approximated by points $y_n \in Y \setminus S(F)$ such that at least one of the preimage sequences $x_n \in F^{-1}(y_n)$ tends to infinity, i.e. the sequence $\{x_n\}$ does not have a convergent subsequence.

From Theorem 3.14 it follows that for the Jacobian map P (as described in Section 3) the non-properness set equals

$$S(P) = \bigcup_{k=1}^{M_1} S_k \cup \bigcup_{k=M_1+1}^M S_k, \quad (6.1)$$

where the curves $S_k = \widehat{P} \circ \pi(D_{i_k} \setminus \infty)$ are images of the non-properness divisors D_{i_k} of the resolution of indefiniteness of P . The curves $S_k, k \leq M_1$ are called the **left components** of $S(P)$, the remaining ones are called the **right components** of $S(P)$.

Existence of the (non-empty) non-properness curve $S(P)$ for the Jacobian map $P = (f, g)$ constitutes an obstacle to the invertibility of P . Otherwise P would be a (unramified) covering between connected and simply connected spaces.

For the proof of invertibility of P it would be enough to show that the map $\widehat{P} \circ \pi$ (see Definition 3.3) is non-ramified near any non-properness divisor D . Then P would be a diffeomorphism between the simply connected space $\mathbb{C}^2$ and the space $\mathbb{C}^2 \setminus S(P)$ which is not simply connected. Indeed, if $S(P) = \{h(X, Y) = 0\}$ is the reduced algebraic definition of $S(P)$ and $\beta \notin \mathbb{Q}$, then the function $h^\beta(X, Y)$ on $\mathbb{C}^2 \setminus S(P)$ has infinite cyclic monodromy group.

We note one more property of the components S_k .

Lemma 6.2. *The preimages $P^{-1}(S_k)$ are non-empty algebraic curves and the image of P is $\mathbb{C}^2 \setminus \text{finiteset}$.*

Proof. If S_k is given by a polynomial equation $h_k(X, Y) = 0$, then $G_k = P^{-1}(S_k)$ is given by the equation $h_k \circ P(x, y) = 0$. If this equation had an empty solution, then the function $h_k \circ P$ would be constant on $\mathbb{C}^2$, with nonzero value c , and $P(\mathbb{C}^2)$ would be a 1-dimensional set $h_k(X, Y) = c$. This contradicts the fact that f is a local diffeomorphism.

The fact that G_k is non-empty does not imply that $P(G_k)$ is the whole S_k . Some points of intersections $S_k \cap S_l$, or of self-intersection of S_k , may have preimages only at infinity. $\square$

Recall the resolution of indeterminacies of $\hat{P}, \pi : \mathcal{Z} \rightarrow \mathcal{X}$ (see Definition 3.3). Define the topological space $\tilde{\mathcal{Z}} = \mathcal{Z} / \sim$, where the equivalence relation $\sim$ denotes squeezing to one point $[\mathcal{Z}_\infty] \in \tilde{\mathcal{Z}}$ all the divisors in $\mathcal{Z}$ which are sent to $\mathcal{Y}_\infty$ by $\hat{P} \circ \pi$ and by squeezing to points all divisors in $\mathcal{Z}$ which are sent to points in $\mathcal{Y}^0 = \mathbb{C}^2$. (The first are divisors from $\mathfrak{A}_\infty(P)$ and the other are divisors from the bamboos $\mathfrak{A} \subset \mathfrak{A}(P)$, defined in Theorem 2.2.) If $D \subset \mathcal{Z}$ is a divisor, then we denote $\tilde{D} = D / \sim$ a corresponding ‘divisor’ in $\tilde{\mathcal{Z}}$. In fact, only divisors D of non-properness type give nontrivial (i.e. of positive dimension) quotient divisors $\tilde{D}$ (homeomorphic to D); they have distinguished points $0, \infty \in \tilde{D}$ and are still called divisors of non-properness type.

Define also the topological space $\tilde{\mathcal{Y}} = \mathcal{Y} / \sim$ by squeezing the line at infinity $\mathcal{Y}_\infty$ to one point $[\mathcal{Y}_\infty]$.

The map $\hat{P} \circ \pi$ induces a continuous map $\tilde{P} : \tilde{\mathcal{Z}} \rightarrow \tilde{\mathcal{Y}}$.

Definition 6.3. For $z \in \tilde{\mathcal{Z}}$ we denote $\mu_z \tilde{P} = \max \{ \tilde{P}^{-1}(y_\varepsilon) \} \cap U$, where U is a sufficiently small neighborhood of z and y_ε is a small deformation of $\tilde{P}(z)$. We call it the **multiplicity of $\tilde{P}$ at z** . If $D = D_{i_k}$ is a non-properness divisor, then we denote by $\mu_D = \mu_D \tilde{P}$ the multiplicity of a typical point of $\tilde{D}$ and call it the **typical ramification index of D** .

Generally, for any divisor $D \subset \mathcal{Z}$, $D \in \mathfrak{A}_\infty(P)$ we define the **typical ramification index of D** , $\mu_D = \mu_D \hat{P} \circ \pi$ as the maximal number of preimages near a typical point $z \in D$ of a point $y_\varepsilon, y_\varepsilon \rightarrow \mathcal{Y}_\infty$. The maximal number of preimages of y_ε near D is called the **multiplicity of the divisor D** and is denoted by $\deg_{\text{top}} D$.

Remark 6.4. If D is a divisor corresponding to a N-P chart θ in $\mathfrak{A}(P)$, then the corresponding N-P alteration chart $\tilde{\theta} : (u, v) \rightarrow (x, y)$ defines a ‘chart’ ϑ near the quotient divisor $\tilde{D} \setminus \infty$ via the following commutative diagram

$$\begin{array}{ccc} \mathbb{C} \times (\mathbb{C}, 0) & \xrightarrow{\tilde{\theta}} & \mathcal{X} \\ \searrow \vartheta & & \nearrow \pi \\ (\tilde{D} \setminus \infty) \times (\mathbb{C}, 0) & & \end{array}$$

Here $\vartheta(\{v = 0\}) = (\tilde{D} \setminus \infty) \times 0$ and $\vartheta(0, 0) = (0, 0)$.

By Definition 2.5 ϑ is ramified along $v = 0$ and $\mu_{(u,0)} \vartheta = k_1 \dots k_{d-1}$ for $u \neq 0$ and $\mu_{(0,0)} \vartheta = k$; here $k = k_1 \dots k_d$ and k_j are components of the characteristic pairs (r_j, k_j) .

Recall the following result of Orevkov [27].

Proposition 6.5. (a) *We have the following formula for the topological degree*

$$\deg_{\text{top}} P = 1 + \sum_D \left[\mu_D + \sum_{z \in \tilde{D} \setminus \infty} (\mu_z \tilde{P} - \mu_D) \right],$$

where the summation is over divisors of non-properness type and each entry in the second sum is non-negative.

(b) *Let D be a divisor of non-properness type and let $z_0 \in \tilde{D} \setminus \infty$. Then $\mu_{z_0} \tilde{P} > \mu_D$ iff the image point $y_0 = \tilde{P}(z_0)$ is a singular point of the immersed curve $S = \tilde{P}(\tilde{D} \setminus \infty)$. If $\mu_D = 1$, then S is smooth. If $k_d > 1$, then S is singular at $\tilde{P}(0)$.*

(Here by singularity of the immersed curve $t \rightarrow (X, Y)(t)$ we mean a point $y_0 = (X, Y)(t_0)$ such that $(X, Y)'(t_0) = 0$; an intersections of smooth local components is not treated as a singularity).

The following examples are illustrative:

- $(x, y) \rightarrow (x, \frac{1}{3}y^3 - xy)$ (the Whitney map) with the ramification curve $D : x = y^2$ and its image $S : y \rightarrow (y^2, -\frac{2}{3}y^3)$, here $\mu_D = 2$ and $\mu_{(0,0)} = 3$.
- $(x, y) \rightarrow (y^4 + 4xy, y^6 + 6y^3x + 6x^2)$ with the ramification divisor $D : x = 0$, but with the two-fold map $y \rightarrow (y^4, y^6)$ from D to S , here $\mu_D = 3$ and $\mu_{(0,0)} = 8$.

Proof of Proposition 6.5. (a) The above formula for $\deg_{\text{top}} P = \deg_{\text{top}} \tilde{P}$ follows from a generalization of the Riemann–Hurwitz formula for the map $\tilde{P} : \tilde{\mathcal{Z}} \rightarrow \tilde{\mathcal{Y}}$ (relation between the Euler characteristics $\chi(\tilde{\mathcal{Y}})$ and $\chi(\tilde{\mathcal{Z}})$). For details we refer to [27].

(b) The Orevkov’s proof of this point relies upon deep results (of Mumford and Milnor) about the fundamental group of the complement of an analytic curve. Here we present a direct proof relying on the notion of deck transformation.

Assume firstly that $z_0 \neq 0$ in $\tilde{D}$ and $\mu_{z_0}\tilde{P} = \mu_D$. Then $\mu_{z_0}\tilde{P} = \mu_{z_0}\hat{P} \circ \pi$.

Consider some neighborhood $U \subset \mathbb{Z}$ of the point z_0 in the form of a component of the set $(\hat{P} \circ \pi)^{-1}(V)$, where V is a neighborhood of y_0 . We shall define a map $F : U \setminus D \rightarrow U \setminus D$ which associates to a point z some point $z' = F(z)$ such that $\hat{P} \circ \pi(z') = \hat{P} \circ \pi(z)$. We choose the map F as locally continuous. Then it defines (in general) a holomorphic multivalued map.

We shall show that this map is, in fact, single-valued and has prolongation to a holomorphic map defined on a whole set U . In other words, F is a deck transformation of the covering $U \setminus (\hat{P} \circ \pi)^{-1}(S) \rightarrow V \setminus S$ and this covering is a Galois covering.

Indeed, near a point $t_0 \in U \cap (D \setminus z_0)$ the map $\hat{P} \circ \pi$ is equivalent to a map $(x, y) \rightarrow (x, y^\mu)$, $\mu = \mu_D$, and hence we choose the map F as (locally) equivalent to the map $(x, y) \rightarrow (x, e^{2\pi i/\mu}y)$. Thus, if $s_0 = \hat{P} \circ \pi(t_0) \in S$ and s_ϵ is its small deformation, then near D we have μ distinct points $t_\epsilon^{(1)}, \dots, t_\epsilon^{(\mu)}$ and F acts on them via cyclic permutation. It turns out that the points $t_\epsilon^{(1)}, \dots, t_\epsilon^{(\mu)}$ are all points of $(\hat{P} \circ \pi)^{-1}(s_\epsilon) \cap U$. It is because otherwise the point z_0 would have multiplicity greater than μ . The map F is not ramified outside D ; its ramification could correspond to gluing of some points in $(\hat{P} \circ \pi)^{-1}(s)$, $s \in V \setminus S$, whereas any such gluing may occur only in D . Moreover, the property of the cyclic permutation on the μ -element sets $\{t_\epsilon^{(1)}, \dots, t_\epsilon^{(\mu)}\}$ near $t_0 \in D \setminus z_0$ is preserved as the point t_0 moves along a loop around z_0 . So F can be prolonged along loops in $U \setminus z_0$ to a single-valued holomorphic map. By Hartog's theorem F becomes holomorphic on the whole of U .

Introduce local holomorphic coordinates (ρ, σ) near z_0 such that $D = \{\rho = 0\}$ and $\rho(z_0) = \sigma(z_0) = 0$. Then F is periodic with the period μ and takes the form $(\rho, \sigma) \rightarrow (e^{2\pi i/\mu}\rho + O(\rho^2), \sigma + O(\rho))$. Thus it is a diffeomorphism. Therefore the space U/F of orbits of F is a smooth complex manifold diffeomorphic to $(\mathbb{C}, 0) \times (\mathbb{C}, 0)$. The map $\hat{P} \circ \pi$ induces the map $[\hat{P} \circ \pi] : U/F \rightarrow \mathbb{C}^2$, which is holomorphic. The latter map is not ramified along $(U \cap D \setminus z_0)/F$. It is also unramified outside D/F . So it is a diffeomorphism and the curve $[\hat{P} \circ \pi](D \cap U/F) \subset S$ is smooth at the point y_0 .

Assume now that $z_0 = 0 \in \tilde{D}$. Here we shall work with the coordinates u, v . Thus we consider the map $\hat{P} \circ \pi \circ \vartheta = P \circ \tilde{\theta}$ (see Remark 6.4).

If $k_d = 1$, then the map ϑ is uniformly ramified along $v = 0$ (with the ramification index $k_1 \dots k_{d-1} = k$). Here we simply repeat the above arguments.

If $k_d \neq 1$, then the multiplicity $\mu_{(u,0)}\vartheta$ jumps at $(0, 0)$. The covering η , as well as $P \circ \tilde{\theta}$, is not Galois; (the monodromy group is larger than the deck transformation group). But we can define a deck transformation $F : \tilde{U} \rightarrow \tilde{U}$, $\tilde{U} = (P \circ \tilde{\theta})^{-1}(V)$, such that near any point $(u, 0)$, $u \neq 0$, it is a rotation in the v direction by the angle $2\pi/\mu k_1 \dots k_{d-1}$ (in linear approximation). This construction is completely analogous to the case $z_0 \neq 0$. We find a regular map $\tilde{U}/F \rightarrow \mathbb{C}^2$ with smooth image $S = P \circ \tilde{\theta}(\{v = 0\})$ near $Y_0 = P \circ \tilde{\theta}(0, 0)$.

Therefore we have proved the implication $\mu_{z_0}\tilde{P} = \mu_D \implies S$ smooth at Y_0 . The reverse implication follows from a local calculation of the Jacobian and is elementary.

The remaining statements of the proposition are obvious. $\square$

Corollary 6.6. *The situations with $\deg_{\text{top}} P = 2$ and $\deg_{\text{top}} P = 3$ cannot occur for a Jacobian map P .*

If $\deg_{\text{top}} P = 4$ (respectively $\deg_{\text{top}} P = 5$), then there are at most 2 (respectively 3) divisors of non-properness type; (recall that at least one of them is a left divisor).

Proof. In the case $\deg_{\text{top}} P = 2$ by the formula for $\deg_{\text{top}} P$ we would have $\mu_D = 1$ for a unique such D , i.e. the map would be an embedding.

Assume the case $\deg_{\text{top}} P = 3$. Then Proposition 6.5 allows us only two possibilities: either there are two divisors of non-properness type, moreover unramified (then P is injective); or there is one divisor D of non-properness type with $\mu_z\tilde{P} = 2$ for all $z \in \tilde{D} \setminus \infty$. Thus $S = \tilde{P}(\tilde{D} \setminus \infty)$ is a smooth immersed line. If it had a double point, then a near point would have four preimages near $\tilde{D}$. The case when S is simply connected falls into the Zaidenberg–Lin theorem, which states that S can be reduced (via elementary transformations) to a quasi-homogeneous curve. Thus S would be singular (like in the proof of Corollary 5.4 in the previous section).

(Orevkov [27] uses here a slightly different argument. By the Abhyankar–Moh theorem S can be rectified and hence the space $\mathcal{Y}^0 \setminus S$ is homotopically equivalent to the circle S^1 . As such, it admits only one covering of degree 3 (a Galois covering). That covering is a restriction of a ramified covering with unique ramification divisor of multiplicity 3.)

The second statement of the corollary follows from the formula for $\deg_{\text{top}} P$. $\square$

One can obtain some information about the non-properness divisors only from their N–P charts.

Proposition 6.7. *Let $D \subset \mathbb{Z}$ be a divisor of non-properness type with its N–P chart θ of the form $x = a_1y^{-\gamma_1} + \dots + uy^{-\gamma}$ (or $y = a_1x^{-\gamma_1} + \dots + ux^{-\gamma}$), where $\gamma_j = l_j/k$, $\gamma = l/k$ and $\gcd(l_1, \dots, l_{d-1}, l, k) = 1$. Then the typical ramification index of D equals*

$$\mu_D = l - k.$$

Proof. Assume that θ is a left chart (the right case is analogous). Then the corresponding N–P alteration chart $\tilde{\theta} : (x, y) = (a_1v^{l_1} + \dots + uv^l, v^{-k})$ has the Jacobian $\text{Jac } \tilde{\theta} = v^{l-k-1}$. Since $\text{Jac } P = 1$, the map $P \circ \tilde{\theta}$ is ramified along $v = 0$ with typical ramification index $\mu_{\{v=0\}}P \circ \tilde{\theta} = l - k$.

On the other hand, Lemma 2.11 says that the variable v is a primitive parametrization of local components of the curves $\hat{f} \circ \pi = \lambda$. Since the lines $X = \lambda$ are typically transversal to the curve S , the map $\{\hat{f} \circ \pi = \lambda\} \rightarrow \{X = \lambda\}$ is ramified at $v = 0$ with multiplicity $\mu_D = l - k$.

(One can also calculate μ_D by expanding $\tilde{f}$, $\tilde{g}$ and $(\text{Jac } P) \circ \theta$. One finds that $\tilde{f} = \varphi \circ \tilde{u}(u, y) + O(y^{1-\gamma})$, $\tilde{g} = \psi \circ \tilde{u}(u, y) + O(y^{1-\gamma})$, where $\tilde{u} = u + \chi_1(u)y^{-\kappa_1} + \dots$ with rational coefficients $\chi_j(u)$.) $\square$

Lemma 6.8. Let $D \subset \mathcal{Z}$ be a divisor of infinite type, with its N–P chart θ as above, and such that $\hat{P} \circ \pi(D) = \mathcal{Y}_\infty$; thus $\tilde{f} \sim x^p$, $\tilde{g} \sim x^p$. Then $\mu_D = kp$.

Proof. The lines $Y = vX$ are transversal to the line $\mathcal{Y}_\infty$. Their preimages $(\hat{g}/\hat{f}) \circ \pi = v$ (near D) are primitively parametrized by v (like the curves $\hat{f} \circ \pi = \lambda$). So the arguments are the same as in the previous proof. $\square$

Further we shall need the following generalization of the latter two results. Let $D \subset \mathcal{Z}$ be a (right) divisor, which is not of non-properness type, and let $\theta : y = a_1x^{-\gamma_1} + \dots + ux^{-\gamma}$ be the corresponding N–P chart. We have $\tilde{f} \sim x^p$.

Let $(r_1, k_1), \dots, (r_d, k_d)$ be the corresponding characteristic pairs, thus $\gamma_1 = \frac{r_1}{k_1}$, $\Delta\gamma_2 = \gamma_2 - \gamma_1 = \frac{r_2}{k_1k_2}$, $\dots$, $\Delta\gamma_d = \gamma - \gamma_{d-1} = \frac{r_d}{k_1 \dots k_d}$ and let $k = k_1 \dots k_d$ (see Definition 2.5 and Algorithm 2.10).

Let $\Theta : Y = b_0X^{\delta_0} + \dots + UX^{\delta}$ be a N–P chart near a divisor $E \subset \mathcal{T}$ of resolution of $\mathcal{Y}$, which is associated with the quasi-Puiseux expansion related to θ .

Let $(\xi_0, \eta_0), \dots, (\xi_N, \eta_N)$ be the characteristic pairs associated with Θ and let $\eta = \eta_0 \dots \eta_N$.

Proposition 6.9. (a) The typical ramification index for the induced map from a neighborhood of D to a neighborhood of E equals

$$\mu_D = \frac{k}{\eta} p.$$

(b) The induced map $D \rightarrow E$ has (topological) degree equal to the topological degree $\deg_{\text{top}} \tilde{c}$ of the following rational function $\tilde{c} : \mathbb{CP}^1 \rightarrow \mathbb{CP}^1$,

$$\tilde{c}(z) = [c(u)]^{\eta_N}, \quad z = u^{k_d}.$$

Therefore the multiplicity $\deg_{\text{top}} D$ of D (see Definition 6.3) equals

$$\deg_{\text{top}} D = \mu_D \cdot \deg_{\text{top}} \tilde{c}.$$

Proof. Item (a) is proved using the primitive parametrizations of the curves $\{U = U_0\}$ and $(\hat{P} \circ \pi)^{-1}\{U = U_0\}$ defined by $X = V^{-\eta}$ and $x = v^{-k}$. Item (b) follows from the fact that natural parameters on the curves D and E are u^{k_d} and $U_0^{\eta_N}$ respectively (see Section 2).

For example, when D is like in Lemma 6.8 $\mu_D = kp$, $\deg_{\text{top}} \tilde{c} = 1$ and $\deg_{\text{top}} D = kp$. $\square$

We can get another formula for the topological degree. It is given in terms of charts of transitory type.

Proposition 6.10. We have

$$\deg_{\text{top}} P = \sum_{\theta: \text{transitory}} k_1 \dots k_{d-1} \cdot q \cdot \deg \varphi,$$

where $q = q(\theta)$, $\varphi = \varphi_\theta$ and $k_1, \dots, k_d$ are defined via the characteristic pairs for θ .

Proof. We calculate the number of preimages of a point near $\mathcal{Y}_\infty$. So we concentrate on divisors D which are sent to $\mathcal{Y}_\infty$ and on their multiplicities (computed in Lemma 6.8). Any such divisor has a N–P chart θ_D of infinite type. This chart lies on a bamboo of the Newton–Puiseux graph $\mathfrak{N}(P)$ which starts at a N–P chart θ of transitory type and consists of N–P charts of infinite type (see the proof of Theorem 3.14 and Fig. 11).

Let us calculate the contribution to $\deg_{\text{top}} P$ arising from one such N–P chart θ . In the corresponding N–P alteration chart $\tilde{\theta} : (x, y) = (v^{-k}, a_1v^{l_1} + \dots + uv^{l_j})$ we have

$$P \circ \tilde{\theta} = (v^{-kp}\varphi(u) + \dots, v^{-kq}\psi(u) + \dots),$$

where $p + q - 1 + \gamma = 0$, $\frac{p}{q} = \frac{\deg \varphi}{\deg \psi}$ and $\deg \varphi > \deg \psi$ (see Definition 3.13).

The bamboo $\theta - \theta_1 - \dots - \theta_D - \dots$ which contains θ_D ‘starts’ at a (simple) zero a of $\varphi(u)$ (see Algorithm 2.10 and the proof of Theorem 3.14). The changes leading from θ_i to θ_{i+1} are of the form $u_i = a_i + u_{i+1}x^{\Delta\gamma_{i+1}}$ where $\Delta\gamma_{i+1} = p(\theta_{i+1}) - p(\theta_i)$. Because φ_{θ_i} have degree 1 (as the zero a is simple, see Proposition 3.15 (c)), it follows that $p(\theta_{i+1}), p(\theta_i), \Delta\gamma_{i+1} \in \frac{1}{k}\mathbb{Z}$ and one has

$$k(\theta_D) = k(\theta) = k = k_1 \dots k_d.$$

Thus the number of preimages near D of a typical point near $\mathcal{Y}_\infty$ equals $\deg_{\text{top}} D = kq$, where $k = k(\theta)$ and $q = q(\theta)$ are the quantities defined for the chart θ (not for θ_D).

If $k_d = 1$, then we are done.

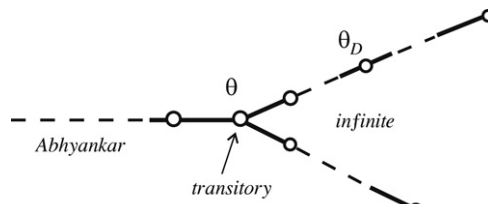


Fig. 11.

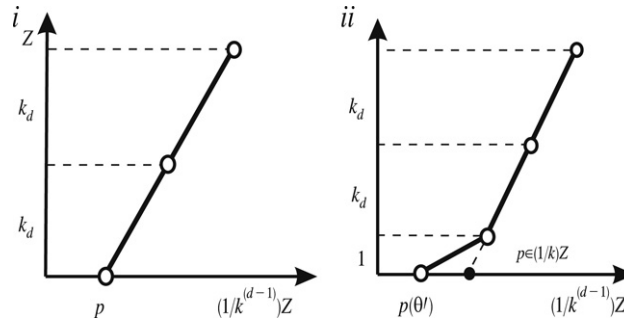


Fig. 12.

Suppose that $k_d > 1$. Then either

- (i) $\varphi = \text{const} \prod (u^{k_d} - w_i)$, or
- (ii) $\varphi = \text{const} \cdot u \prod (u^{k_d} - w_i)$

(see Fig. 12). Here we use the fact that the zeros of φ and ψ are simple and distinct (by Proposition 3.15(c)).

In case (i) there are $\deg \varphi / k_d$ divisors D (as above) arising from θ with multiplicity $\mu_D = kq$ each; so the proposition follows.

In case (ii) there are $(\deg \varphi - 1) / k_d$ divisors D (corresponding to zeros $a_j \neq 0$ of φ) with multiplicity $\mu_D = kq$ each. There is also one divisor D' with $\theta_{D'}$ on the bamboo in $\mathfrak{A}(P)$ which 'starts' at the zero $u = 0$ of φ with multiplicity $\mu_{D'} = k(\theta_{D'})q$; here $k(\theta_{D'}) = k^{(d-1)}$,

$$k^{(d-1)} := k_1 \dots k_{d-1}.$$

This is because the edge I_1 , associated with θ_1 and adjacent to $I \subset \Gamma(f_{d-1})$ (associated with θ) has the endpoints $(p(\theta_1), 0)$ and $(*, 1)$ in the lattice $(\frac{1}{k^{(d-1)}}\mathbb{Z}) \times \mathbb{Z}$ (see Fig. 12). Therefore we get the contribution $kq \frac{\deg \varphi - 1}{k_d} + k^{(d-1)}q = k^{(d-1)}q \deg \varphi$ as in the statement of the proposition. $\square$

Corollary 6.11. *If a Jacobian map P had $\deg_{\text{top}} P \leq 5$, then its resolution of indeterminacies would contain only one N-P chart θ (of depth d) of transitory type and satisfying one of the following properties:*

Case 1. $k_d = 3$, $k^{(d-1)}p = 1$, $k^{(d-1)}q = \frac{2}{3}$, $\deg_{\text{top}} P = 4$ and

$$\varphi = a^3 \left(u^6 + 3bu^3 + \frac{3}{2}b^2 \right), \quad \psi = a^2 u(u^3 + 2b).$$

Case 2. $k_d = 3$, $k^{(d-1)}p = \frac{4}{3}$, $k^{(d-1)}q = 1$, $\deg_{\text{top}} P = 4$ and

$$\varphi = a^4 u(u^3 + 4b), \quad \psi = a^3 (u^3 + 3b).$$

Case 3. $k_d = 4$, $k^{(d-1)}p = \frac{5}{4}$, $k^{(d-1)}q = 1$, $\deg_{\text{top}} P = 5$ and

$$\varphi = a^5 u(u^4 + 5b), \quad \psi = a^4 (u^4 + 4b).$$

Case 4. $k_d = 2$, $k^{(d-1)}p = \frac{5}{2}$, $k^{(d-1)}q = 1$, $\deg_{\text{top}} P = 5$ and

$$\varphi = a^5 u \left(u^4 + 5bu^2 + \frac{15}{2}b^2 \right), \quad \psi = a^2 (u^2 + 2b).$$

Proof. If $k_d = 1$ for a summand $k^{(d-1)}q \deg \varphi$ (in the formula for $\deg_{\text{top}}$ from Proposition 6.10), then the numbers $k^{(d-1)}p$ and $k^{(d-1)}q$ would be integers with the ratio $\frac{p}{q} = \frac{\deg \varphi}{\deg \psi} > 1$ (but not integer). Thus $\deg \varphi \geq 3$, $k^{(d-1)}q \geq 2$ and $\deg_{\text{top}} P \geq 6$. Assume then that $k_d > 1$ for each summand.

Firstly we consider case (i) at Fig. 12

$$\varphi = \text{const} \cdot \prod_1^s (u^{k_d} - w_i), \quad \psi = \text{const} \cdot u \prod_1^r (u^{k_d} - v_i);$$

here $s \geq 2$ and $r \geq 1$ (since $1 < \frac{\deg \varphi}{\deg \psi} \notin \mathbb{Z}$). Then the number $k^{(d-1)}p$ is an integer (we have the vertex point $(p, 0) \in I \subset \Gamma(f_{d-1})$ (associated with θ) in the lattice $\frac{1}{k^{(d-1)}}\mathbb{Z} \times \mathbb{Z}$) and $k^{(d-1)}q =: \frac{e}{k_d} \in \frac{1}{k_d}\mathbb{Z}$. The corresponding contribution to $\deg_{\text{top}}$ equals

$$k^{(d-1)}q \deg \varphi = k^{(d-1)}q \cdot k_d s = es \leq 5.$$

Then the equation $\frac{\deg \varphi}{\deg \psi} = \frac{p}{q}$ gives $kq \frac{sk_d}{rk_d+1} = kp \in \mathbb{Z}$, i.e.

$$t := \frac{es}{rk_d+1} \in \mathbb{Z}.$$

Suppose that $e = 1$. When $s \leq 5$ and $k_d \geq 2$ the only possibility is that $t = 1$. But this leads to $\frac{\deg f}{\deg g} = k_d \in \mathbb{Z}$, in contradiction to our assumptions (see Theorem 3.2).

Therefore there remains the case $e = s = 2$ (as $es \leq 5$); then $t = 1$ and $r = 1$, $k_d = 3$. It is Case 1 of the corollary (with unique divisor of transitory type).

Consider now the case (ii) at Fig. 12

$$\varphi = \text{const} \cdot u \prod_1^s (u^{k_d} - w_i), \quad \psi = \text{const} \cdot \prod_1^r (u^{k_d} - v_i).$$

Here $k^{(d-1)}q \in \mathbb{Z}$ and it must be equal 1 (because $\deg \varphi \geq 3$) and $k^{(d-1)}p \in \frac{1}{k_d}\mathbb{Z}$. It is then clear that there is only one chart of transitory type with $\deg \varphi = 4, 5$. So we have the following possibilities for the pairs $(m/n, k_d)$, $m/n = p/q$:

$$(4/3, 3), (5/4, 4), (5/2, 2).$$

The polynomials φ and ψ satisfy the equation $p\varphi\psi' - q\varphi'\psi = 1$ (see Proposition 3.15(c)) and this determines their form. $\square$

Now we can formulate the principal result of this section. It improves the results of Orevkov [27] and of Domrina and Orevkov [7,8].

Theorem 6.12. Any Jacobian map P with $\deg_{\text{top}} P \leq 5$ is invertible.

Proof. A. The transitory chart. It is enough to consider the cases with $\deg_{\text{top}} P = 4, 5$. We shall obtain a contradiction by calculating the number of preimages of points $(X, Y) \in \mathcal{Y}^0$ which lie near some divisors of a compactification $\mathcal{T}$ of $\mathcal{Y}^0$. These divisors have corresponding N–P charts (X, U) related to suitable quasi-Puiseux expansions. In the proof of Corollary 6.11 the chart $(X, Y) = (X, UX)$ was used. Here the charts are more subtle.

We begin with the chart

$$\Theta_0 : (X, U_0) \rightarrow (X, U_0 X^{\delta_0}), \quad \delta_0 = \frac{\deg g}{\deg f} = \frac{n}{m}$$

(of depth 1). This chart is related to the quasi-Puiseux expansion $\tilde{g} = c_{\theta_N}(u)\tilde{f}^{\delta_0}$, $u = u_N$, associated with a N–P chart $\theta_N : y = a_1 x^{-\gamma_1} + \dots + u x^{-\gamma_d}$ (of depth $d = d(\theta_N)$) of transitory type and with corresponding divisor $D_N = D_{\theta_N}$. (The justification for using the subscript N will be given in the next point). By Corollary 6.11 we can assume that there is only one such chart. Θ_0 has one characteristic pair

$$(\xi_0, \eta_0) = (kq, kp) = (n, m),$$

where $p = p_N = p(\theta_N)$, $q = q_N = q(\theta_N)$ and $k = k(\theta_N) = k_1 \dots k_d$ are like in Corollary 6.11.

By Proposition 6.9 we get $\deg_{\text{top}} D_N = \mu_{D_N} \cdot \deg_{\text{top}} \tilde{c}_{\theta_N}$ preimages of a point Y laying on a curve $Y = (X, U_0 X^{\delta_0})$, $U_0 = \text{const}$, $X \rightarrow \infty$ (near a divisor E_{Θ_0} of $\mathcal{T}$). Since $\mu_D = \frac{k}{\eta_0} p = 1$, they are given by solutions to the equation $\tilde{c}_{\theta_N}(z) = W$, where $W = U_0^{\eta_0}$, $z = u^{k_d}$, $\tilde{c}_{\theta_N}(z) = [c_{\theta_N}(u)]^{\eta_0}$.

For example, in Case 1 of Corollary 6.11 we have $c_{\theta_N} = \psi \varphi^{-2/3} = u(u^3 + 2b)(u^6 + 3bu^3 + \frac{3}{2}b^2)^{-2/3}$, $\delta_0 = \frac{2}{3}$ and

$$\tilde{c}_{\theta_N} = \frac{z(z + 2b)^3}{(z^2 + 3bz + \frac{3}{2}b^2)^2}.$$

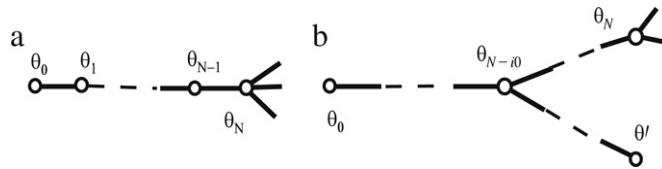


Fig. 13.

The equation $\tilde{c}_{\theta_N}(z) = W$ has $4 = \deg_{\text{top}} P$ finite solutions for generic W (as expected). Note also that the equation $\tilde{c}_{\theta_N}(z) = \tilde{c}_{\theta_N}(\infty)$ (where $\tilde{c}_{\theta_N}(\infty) = 1$) has only one finite solution $z = -\frac{9}{4}b$.

In the other cases of Corollary 6.11 the corresponding numbers $\deg_{\text{top}} \tilde{c}_{\theta_N}$ and $\deg_{\infty} c_{\theta_N}$ of solutions to the equations $\tilde{c}_{\theta_N}(z) = W$ and $\tilde{c}_{\theta_N}(z) = 1$ are: $\deg_{\text{top}} \tilde{c}_{\theta_N} = 4$ and $\deg_{\infty} \tilde{c}_{\theta_N} = 2$ in Case 2; 5 and 3 in Case 3; 5 and 2 in Case 4.

We see that using θ_0 we cannot get a contradiction, we have $\deg_{\text{top}} D_N = \deg_{\text{top}} P$.

B. A chain of N - P charts. In the next step one should look for preimages of a typical point near a divisor E_{θ_1} (in the compactification $\mathcal{T}$) with a N - P chart $\theta_1 : Y = X^{\delta_0} + U_1 X^{\delta_1}$ (of depth 2), where $\delta_0 = \delta(\theta_N)$; thus in terms of θ_0 and θ_N we have $U_0 = 1$ and $u_N = \infty$. The preimages are located near a divisor $D_{N-1} = D_{\theta_{N-1}}$ with a N - P chart θ_{N-1} (of depth $d-1$ or d). For example, in Case 1 there should exist 3 preimages near D_{N-1} , the fourth preimage lies near D_N . In calculation of these preimages one expects to get a contradiction.

Unfortunately, life is not that easy. In order to get a contradiction we have to consider the whole path $\theta_0 - \theta_1 - \dots - \theta_N$ in $\mathfrak{N}(P)$ from the root chart θ_0 to transitory chart θ_N . The charts $\theta_1, \dots, \theta_{N-1}$ in the source $\mathcal{X}^0$ have their relatives $\theta_{N-1}, \dots, \theta_1$ in the target $\mathcal{Y}^0$. The charts θ_j have depth $j+1$.

Denote $\gamma_j = \gamma(\theta_j)$, $p_j = p(\theta_j)$, $q(\theta_j) = q_j$, $\varphi_j = \varphi_{\theta_j}$, $D_j = D_{\theta_j}$, $c_j(u) = c_{\theta_j}(u)$, $\tilde{c}_j(z) = \tilde{c}_{\theta_j}(z)$, $\delta_j = \delta(\theta_{N-j})$ and (r_i, k_i) , (ξ_i, η_i) the characteristic pairs for the sequences $\{\theta_i\}$ and $\{\theta_i\}$ respectively.

At this moment we cannot claim that the pairs (r_i, k_i) are exactly characteristic pairs of some of the θ_j 's. They are defined via $\gamma_i - \gamma_{i-1} = r_i/k^{(i)}$, $k^{(i)} = k_1 \dots k_i$, but the depth $d(\theta_i)$ may be $\neq i$.

Some vertices on the path can be nodes. For example, in Case 1 $\deg_{\text{top}} P = 4$ and by Corollary 6.6 there are at most two N - P charts of non-properness type; moreover, at least one of them is a left chart. Therefore we have two possibilities for the right part of the Newton–Puiseux graph $\mathfrak{N}(P)$, which are presented in Fig. 13.

In the case (a) of Fig. 13 we have

$$\varphi_{N-i} = \text{const} \cdot [(z-w)^{\epsilon}]^{\eta_0 \dots \eta_{i-1}}, \quad (6.2)$$

where $\epsilon = 2$ and $z = u^{k_{N-i}}$, $w = w_i \neq 0$. For example, $\varphi_{N-1}(u) = \text{const} \cdot (z-w)^{\alpha}$, where $\alpha = w^{1/k_{N-1}}$ is the zero of φ corresponding to the unique edge on the right of θ_{N-1} and the power $\alpha = \deg \varphi_N = 6 = 2 \cdot \eta_0$.

Due to the results of Section 5 there are no charts such that $\varphi_{N-i} = \text{const} \cdot u^{\zeta}$. Below we shall show that $\eta_i = k_{N-i}$ and $d(\theta_i) = i$.

In the case (b) we should have (6.2) for $i < i_0$ and $i > i_0$. For $i = i_0$ we have either

$$\varphi_{N-i_0} = \text{const} \cdot [u^{\epsilon}(z-w)^2]^{\eta_0 \dots \eta_{i_0-1}} \quad (6.3)$$

or

$$\varphi_{N-i_0} = \text{const} \cdot [u^2(z-w)^{\epsilon}]^{\eta_0 \dots \eta_{i_0-1}} \quad (6.4)$$

or

$$\varphi_{N-i_0} = \text{const} \cdot [(z-w)^2(z-v)^{\epsilon}]^{\eta_0 \dots \eta_{i_0-1}}. \quad (6.5)$$

Here $\epsilon = \epsilon_i$ are integers and $w = w_i \neq 0$, $v \neq 0$. This follows from the fact that each φ_{N-i} is a power of order $\eta_0 \dots \eta_{i-1}$ such that φ_{N-i-1} is not such a power (see Proposition 5.11). The form of φ_{N-i_0} follows from the fact that θ_{N-i_0} is a nodal vertex in $\mathfrak{N}(P)$ and φ_{N-i_0} has two essentially different zeros.

In the case (6.4) passing from the chart θ_{N-i_0} to θ_{N-i_0+1} corresponds to passing from one edge to an adjacent edge in the same Newton diagram $\Gamma(f_m)$, so the 'depth' is not changed; in the cases (6.3) and (6.5) the depth changes. The relation between η_j 's and k_i 's is not direct.

Formulas similar to (6.2)–(6.5) hold in the other cases of Corollary 6.11.

C. Scheme of calculations. We will determine inductively the following quantities:

- $k_N = k_d$, p_N , q_N , $\deg \varphi_N$, $\delta_0 = \frac{n}{m}$, $\eta_0 = m$;
they are given in Corollary 6.11.
- $1 - \gamma_N = p_N + q_N$;
since θ_N is of transitory type (see Definition 3.13).

- $\Delta\gamma_{N-i} = \gamma_{N-i} - \gamma_{N-i-1} = r_{N-i}/k^{(N-i)}$,
 $\gcd(r_{N-i}, k_{N-i}) = 1$, where the property $1 - \gamma_{N-i-1} = 1 - \gamma_{N-i} + \Delta\gamma_{N-i} \in (1/k^{(N-i)})\mathbb{Z}$ must be preserved; this restricts the choice of r_{N-i} .
 - $p_{N-i-1} = p_{N-i} + \Delta\gamma_{N-i} \cdot \deg \varphi_{N-i-1}$.
 - $1 + \delta_i = (1 - \gamma_{N-i})/p_{N-i} = \rho_i/\eta_0 \dots \eta_i$,
 $\gcd(\rho_i, \eta_i) = 1$.
- We introduce also auxiliary quantities
- $\kappa = \kappa_i = (1 + \delta_i)\eta_0 \dots \eta_{i-1} = \rho_i/\eta_i$.

Next, postulating the possible form of φ_{N-i} we determine the factors k_{N-i} and η_i from the formula for multiplicity of the divisor D_{N-i} and from the condition $\deg c_{N-i} = 0$ (algebraic degree).

Below the different cases of [Corollary 6.11](#) are considered separately. But Case 1 is treated with greatest attention.

D. Calculations for θ_{N-1} in Case 1. From [Corollary 6.11](#) we have $k_N = k_d = 3$, $p_N = 1/k^{(N-1)}$, $q_N = 2/3k^{(N-1)}$, $\deg \varphi_N = 6$, $\delta_0 = \frac{2}{3}$, $\eta_0 = 3$. Therefore $1 - \gamma_N = p_N + q_N = 5/3k^{(N-1)}$.

Assume $\Delta\gamma_N = r_N/3k^{(N-1)}$; hence $1 - \gamma_{N-1} = (5 + r_N)/3k^{(N-1)} \in (1/3k^{(N-1)})\mathbb{Z}$ and we put $r_N = 1 + 3s_{N-1}$. Thus

$$1 - \gamma_{N-1} = \frac{2 + s_{N-1}}{k^{(N-1)}}, \quad p_{N-1} = \frac{3(1 + 2s_{N-1})}{k^{(N-1)}} \quad (6.6)$$

and

$$1 + \delta_1 = \frac{2 + s_{N-1}}{3(1 + 2s_{N-1})} = \frac{\rho_1}{\eta_0\eta_1}, \quad \kappa = \frac{2 + s_{N-1}}{1 + 2s_{N-1}}. \quad (6.7)$$

Here $\eta_1 = 1 + 2s_{N-1}$ if $3 \nmid 1 + 2s_{N-1}$ and $\eta_1 = 1 + 2t_{N-1}$ if $s_{N-1} = 1 + 3t_{N-1}$.

The typical ramification index $\mu_{D_{N-1}} = k^{(N-1)}p_{N-1}/\eta^{(1)}$ of the divisor D_{N-1} (see [Proposition 6.9](#)) is the following:

$$\begin{aligned} \mu_{D_{N-1}} &= 1 \quad \text{if } 3 \nmid 1 + 2s_{N-1}, \\ \mu_{D_{N-1}} &= 1 \quad \text{otherwise.} \end{aligned} \quad (6.8)$$

E. Calculation of c_{N-1} in the subcase (6.2) (Case 1). We have

$$c_{N-1}(u) = \frac{1}{p} \int \varphi^{-\delta-1} = \text{const} \int (z - w)^{-2\kappa} du,$$

where $z = u^{k_{N-1}}$ and $2\kappa = 1 + \frac{3}{1+2s_{N-1}}$. Recall that by the Rationality Condition c_{N-1} should be of Darboux type.

If $s_{N-1} = 1$, then we have an integral of a rational function and by [Lemma A.7](#) (in the [Appendix](#)) c_{N-1} cannot be Darbouxian. Otherwise [Lemma A.5](#) gives the form

$$c_{N-1}(u) = \frac{uR(z)}{(z - w)^{2\kappa-1}} = \frac{uR(z)}{(z - w)^{3/(1+2s_{N-1})}},$$

where $R(z)$ is a polynomial.

If $3 \nmid 1 + 2s_{N-1}$, then $\tilde{c}_{N-1}(z) = [uR(z)]^{1+2s_{N-1}}/(z - w)^3$. Since $\deg_{\text{top}} D_{N-1} = \mu_{D_{N-1}} \cdot \deg_{\text{top}} \tilde{c}_{N-1} = 3$, we have $R(z) \equiv \text{const}$. So the condition $\deg \tilde{c} = 0$ leads to the wrong identity $1 + 2s_{N-1} = 3k_{N-1}$.

Let then $3 \mid 1 + 2s_{N-1}$, i.e. $s_{N-1} = 1 + 3t_{N-1}$. Then $\tilde{c}_{N-1}(z) = \text{const} \frac{z}{z-w}$ with $\deg_{\text{top}} \tilde{c} = 1$ and $k_{N-1} = \eta_1 = 1 + 2t_{N-1}$. We see that $\deg_{\text{top}} D_{N-1} = 3$, as expected, and there is no contradiction. Moreover, $1 - \gamma_{N-1} = 3(1 + t_{N-1})/k^{(N-1)}$, $p_{N-1} = 3\eta_0/k^{(N-2)}$, $\deg \varphi_{N-1} = 2\eta_0\eta_1$.

F. Solution in the subcase (a) from [Fig. 13](#) (Case 1). Suppose that θ_{N-2} is also non-node. Then analogous calculations for θ_{N-2} are the following (with $k_{N-1} = 1 + 2t_{N-1}$):

$$\begin{aligned} \Delta\gamma_{N-1} &= \frac{r_N - 1}{k^{(N-1)}}, \quad r_{N-1} = t_{N-1} - 1 + s_{N-2}k_{N-1}, \\ 1 - \gamma_{N-2} &= \frac{2 + s_{N-2}}{k^{(N-2)}}, \quad p_{N-2} = \frac{\eta_0\eta_1(1 + 2s_{N-2})}{k^{(N-2)}}, \\ 1 + \delta_2 &= \frac{2 + s_{N-2}}{\eta_0\eta_1(1 + 2s_{N-2})}, \quad \kappa_2 = \frac{2 + s_{N-2}}{1 + 2s_{N-2}} = \frac{\rho_2}{\eta_2} \end{aligned}$$

(compare the formulas (6.6)–(6.7)). As in the previous case we arrive at the property $s_{N-2} = 1 + 3t_{N-2}$, which implies $k_{N-2} = \eta_2 = 1 + 2t_{N-2}$, $\mu_{D_{N-2}} = 3$, $\tilde{c}_{N-2} = \text{const} \frac{z}{z-w}$ and $1 - \gamma_{N-2} = 3(1 + t_{N-2})/k^{(N-3)}(1 + 2t_{N-2})$, $p_{N-2} = 3\eta_0\eta_1/k^{(N-3)}$, $\deg \varphi_{N-2} = 2k_Nk_{N-1}k_{N-2} = 2\eta_0\eta_1\eta_2$.

If we have the case (a) in [Fig. 13](#), then we can continue in this way. Eventually we get $1 - \gamma_1 = 3(1 + t_1)/k^{(0)}(1 + 2t_1) > 1$ (since $k^{(0)} = 1$). So $\gamma_1 < 0$ for the first right chart, in contradiction with [Theorem 3.2](#).

G. *The subcase (6.3) (Case 1).* Suppose that the node is θ_{N-1} ; the cases when the node is θ_{N-i_0} , $i_0 > 1$, differ only in notations. Therefore the polynomial φ_{N-1} takes one of the forms (6.3)–(6.5). Here we assume $\varphi_{N-1} = \text{const} \cdot [u^\epsilon(z-w)^2]^{\eta_0}$, i.e. the case (6.3).

By Lemma A.5 we find

$$c_{N-1}(u) = \text{const} \int u^{-\epsilon\kappa} (z-w)^{-2\kappa} = \frac{u^{1-\epsilon\kappa} R(z)}{(z-w)^{2\kappa-1}},$$

where κ is given in (6.7). The possibility $s_{N-1} = 1$ leads to $\kappa = 1$ and to an integral of a rational function (impossible by Lemma A.7).

The case $3 \nmid 1 + 2s_{N-1}$ gives $1 + 2s_{N-1} \geq 5$. So $R(z) \equiv \text{const}$ and $\epsilon\kappa < 1$, i.e. $\epsilon = 1$ (because $\deg_{\text{top}} \tilde{c} = 3$ as $\mu_{D_{N-1}} = 1$). Then $\tilde{c}_{N-1} = \text{const} \cdot u^{s_{N-1}-1} (z-w)^3$, because $1 - \kappa = (s_{N-1} - 1)/(1 + 2s_{N-1})$. It should have degree 0, which implies $s_{N-1} = 1 + 3k_{N-1}$ (contradiction).

In the case $3 \mid 1 + 2s_{N-1}$, i.e. $s_{N-1} = 1 + 3t_{N-1}$, $t_{N-1} \geq 1$, we find the possibility $R \equiv \text{const}$, $\epsilon = 1$, $k_{N-1} = t_{N-1}$, $\eta_1 = 1 + 2t_{N-1}$ and $\tilde{c}_{N-1} = \text{const} \cdot z/(z-w_1)$. Here $\deg_{\text{top}} D_{N-1} = 3$ (no contradiction). We have also

$$1 - \gamma_{N-1} = \frac{3(1 + k_{N-1})}{k^{(N-1)}}, \quad p_{N-1} = \frac{3\eta_0\eta_1}{k^{(N-1)}}.$$

Let us estimate the inverse slope $\gamma' = l'/k'$ of a chart θ' which is the non-properness chart in the bamboo in $\mathfrak{A}(P)$ rooted at θ_{N-1} (and which 'starts' at the zero $u = 0$ in D_{N-1} of order $\text{ord}_0 \varphi_{N-1} = \epsilon\eta_0 = 3$). We have

$$\Delta\gamma' = \gamma' - \gamma_{N-1} \geq \frac{p_{N-1}}{\text{ord}_0 \varphi_{N-1}} = \frac{3(1 + 2k_{N-1})}{k^{(N-1)}};$$

(this inequality follows from the fact that we join the vertex $(p_{N-1}, \text{ord}_0 \varphi_{N-1}) \in \Gamma(f_m)$ with the origin using a broken line and $\Delta\gamma'$ is the inverse slope of the last segment). But then

$$\gamma' - 1 = \frac{l' - k'}{k'} \geq \frac{3}{k^{(N-2)}}.$$

Since $k_1, \dots, k_{N-2}$ appear in the characteristic pairs for θ' , it follows that $l' - k' \geq 3$. By Proposition 6.7 the typical ramification index $\mu_{D_{\theta'}} \geq 3$ and Proposition 6.5 would give $\deg_{\text{top}} P \geq 5$; recall that it must be $\deg_{\text{top}} P = 4$.

(One can also use arguments as in case (a), i.e. find a general formula for $1 - \gamma_{N-j}$ and obtain $\gamma_1 < 0$.)

H. *The subcase (6.4) (Case 1).* Here we get $c_{N-1} = R(z)(z-w_1)^{1-\epsilon\kappa}/u^{2\kappa-1}$ and, by eliminating the possibility $s_{N-1} = 1$ (Lemma A.7) and calculating $\deg_{\text{top}} D_{N-1}$ (see (6.8)), we land at $R \equiv \text{const}$ and $\epsilon = 1$. So $c_{N-1} = \text{const}[(z-w)^{s_{N-1}-1}/u^3]^{1/(1+2s_{N-1})}$. The condition $\deg c_{N-1} = 0$ gives $k_{N-1}(s_{N-1} - 1) = 3$. We have two possibilities: either $s_{N-1} = 2$, $k_{N-1} = 3$ or $s_{N-1} = 4$, $k_{N-1} = 1$.

In the first case $\mu_{D_{N-1}} = 1$ and $\deg_{\text{top}} \tilde{c} = \deg_{\text{top}} (z-w)/z = 1$, which contradicts $\deg_{\text{top}} D_{N-1} = 3$.

In the second case $\mu_{D_{N-1}} = 3$ and $\deg_{\text{top}} \tilde{c} = 3$ (no contradiction). Moreover, $1 - \gamma_{N-1} = 6/k^{(N-2)}$, $p_{N-1} = 27/k^{(N-2)}$.

Now we act as in the second part of the point G. With γ' as above we find that $\gamma' - 1 = \Delta\gamma' - (1 - \gamma_{N-1}) \geq \frac{p_{N-1}}{\text{ord}_0 \varphi_{N-1}} - \frac{6}{k^{(N-1)}} = \frac{3}{k^{(N-2)}}$. This, in order, implies that $\deg_{\text{top}} P \geq 5$.

I. *The subcase (6.5) (Case 1).* Here we have the following (and unique) possibility: $s_{N-1} = 3$, $k_{N-1} = \eta_1 = 7$, $\tilde{c}_{N-1} = \text{const} \cdot z(z - \frac{3}{2}w)^2/(z-w)^3$. Moreover, $\mu_{D_{N-1}} = 1$, $\deg_{\text{top}} \tilde{c} = 3$ and $\deg_{\infty} \tilde{c} = 1$. Also $1 - \gamma_{N-1} = 5/7k^{(N-2)}$ and $p_{N-1} = 3/k^{(N-2)}$.

We cannot have $N - 1 = 1$, since then $3\alpha = p_1 + \gamma_1 \deg \varphi_1 = 21 = 3\beta$.

Let us make calculations for θ_{N-2} ; now we expect that $\deg_{\text{top}} D_{N-1} = 2$. We find $r_{N-1} = 2 + 7s_{N-2}$, $1 - \gamma_{N-2} = (1 + s_{N-2})/k^{(N-2)}$, $p_{N-2} = \eta_0\eta_1(1 + 3s_{N-2})/k^{(N-2)}$, $1 + \delta_{N-2} = (1 + s_{N-2})/\eta_0\eta_1(1 + 3s_{N-2})$, $\kappa = 1 + 2/(1 + 3s_{N-2})$. One shows that it must be $s_{N-2} = 1 + 2t_{N-2}$, giving $k_{N-2} = \eta_2 = 2 + 3t_{N-2}$, $\mu_{D_{N-2}} = 2$ and $\tilde{c}_{N-2} = \text{const} \cdot z/(z-w)$. So, it is still OK.

But further calculations give nothing new: we get $1 - \gamma_{N-i} = (2 + 2t_{N-i})/k^{(N-i-1)}(2 + 3t_{N-i})$, $p_{N-i} = 2\eta_0 \dots \eta_{i-1}/k^{(N-i-1)}$, $\deg \varphi_{N-i} = 3\eta_0 \dots \eta_{i-1}$.

For $i = N - 1$ we find $\gamma_1 = t_1/(2 + 3t_1)$, $3\beta = \deg \varphi_1 = 3\eta_0 \dots \eta_{N-1}$ and $3\alpha = p_1 + \gamma_1 \deg \varphi_1 = (2 + 3t_1)\eta_0 \dots \eta_{N-2}$, where $2 + 3t_1 = \eta_{N-1}$. This means that $\frac{\beta}{\alpha} = 3$. Therefore the slope of a left N-P chart θ' of non-properness type must be > 3 . By Proposition 6.7 $\mu_{D_{\theta'}} \geq 3$ and by Proposition 6.5 $\deg_{\text{top}} P \geq 5$ (a contradiction).

J. *Remarks about Cases 2, 3, 4.* The analysis of these cases partly runs together. Namely, it turns out that the path in $\mathfrak{A}(P)$ from θ_0 to θ_N is a chain, whose vertices are non-nodes. It means that $\varphi_{N-1} = \text{const} \cdot (z-w)^{\eta_0}$, $\varphi_{N-2} = \text{const} \cdot (z-w)^{\eta_0\eta_1}$, $\dots$ and $k_{N-1} = \eta_1$, $k_{N-2} = \eta_2$, $\dots$. Thus there is no case (b) from Fig. 13 (or its analogue) and there are no right N-P charts of non-properness type. We shall justify this statement later.

K. *Case 2.* In Corollary 6.11 it was established that

$$k_N = 3, \quad p_N = \frac{4}{3k^{(N-1)}}, \quad q_N = \frac{1}{k^{(N-1)}}, \quad \deg \varphi_N = 4, \quad \delta_0 = \frac{3}{4}, \quad \eta_0 = 4.$$

The condition for θ_N to be transitory gives $1 - \gamma_{N-1} = 7/3k^{(N-1)}$. Putting $\Delta\gamma_N = r_N/3k^{(N-1)}$ with (necessarily) $r_N = 3s_{N-1} - 1$ we get

$$1 - \gamma_{N-1} = \frac{2 + s_{N-1}}{k^{(N-1)}}, \quad p_{N-1} = \frac{\eta_0 s_{N-1}}{k^{(N-1)}},$$

$$1 + \delta_1 = \frac{2 + s_{N-1}}{\eta_0 s_{N-1}}, \quad \kappa = \kappa_1 = 1 + \frac{2}{s_{N-1}}.$$

Recall also that $\deg_{\text{top}} D_{N-1} = 2$ (see the point A).

The polynomial φ_{N-1} (modulo a constant) has one of the following four forms:

$$(\alpha)(z - w)^{\eta_0}, (\beta)[u^\epsilon(z - w)]^{\eta_0}, (\gamma)[u(z - w)^\epsilon]^{\eta_0}, (\delta)[(z - w)(z - v)^\epsilon]^{\eta_0}.$$

The coefficient $c_{N-1}(u)$ takes the corresponding forms: $R(z)u/(z - w_1)^{k-1}$, $R(z)/[u^{\epsilon k-1}(z - w)^{k-1}]$, $R(z)/[u^{k-1}(z - w)^{\epsilon k-1}]$, $R(z)u/[(z - w)^{k-1}(z - v)^{\epsilon k-1}]$.

We know that κ cannot be integer (Lemma A.7). It follows that $\eta_1 = s_{N-1} \geq 3$ if s_{N-1} is odd and $\eta_1 = t_{N-1} \geq 2$ if $s_{N-1} = 2t_{N-1}$ is even. It is now clear that in the cases (β) , (γ) , (δ) the denominator in $\tilde{c}_{N-1}$ must have degree > 2 (as $\epsilon \geq 1$). Since we expect $\deg_{\text{top}} D_{N-1} = 2$, there are no cases (β) – (δ) .

In the case (α) we find that $R(z) \equiv \text{const}$. The condition $\deg c_{N-1} = 0$ leads to $s_{N-1} = 2k_{N-1}$ if s_{N-1} is odd (a contradiction) and to $k_{N-1} = \eta_1 = t_{N-1}$ if $s_{N-1} = 2t_{N-1}$. We have also $1 - \gamma_{N-1} = 2(1 + t_{N-1})/k^{(N-2)}t_{N-1}$ and $p_{N-1} = 2\eta_0/k^{(N-2)}$.

Repetition of these calculations gives $1 - \gamma_{N-i} = 2(1 + t_{N-i})/k^{(N-2)}t_{N-i}$. For $N - i = 1$ we get $\gamma_1 < 0$ (contradiction).

L. Case 3. It is established in Corollary 6.11 that $k_N = 4$, $p_N = 5/4k^{(N-1)}$, $q_N = 1/k^{(N-1)}$, $\deg \varphi_N = 5$, $\delta_0 = \frac{4}{5}$, $\eta_0 = 5$. Here $1 - \gamma_N = 9/4k^{(N-1)}$ and with $r_N = 4s_{N-1} - 1$ we get $1 - \gamma_{N-1} = (2 + s_{N-1})/k^{(N-1)}$, $p_{N-1} = \eta_0 s_{N-1}/k^{(N-1)}$, $1 + \delta_1 = (2 + s_{N-1})/5s_{N-1}$. As in Case 2 we conclude that $s_{N-1} = 2t_{N-1}$, $k_{N-1} = \eta_1 = t_{N-1}$, $\mu_{D_{N-1}} = 3$, $\varphi_{N-1} = \text{const} \cdot (z - w)^{\eta_0} \tilde{c}_{N-1} = \text{const} \cdot z/(z - w)$ and that there are no cases corresponding to (β) – (δ) above. We have also

$$1 - \gamma_{N-1} = \frac{2(1 + t_{N-1})}{k^{(N-2)}t_{N-1}}, \quad p_{N-1} = \frac{2\eta_0}{k^{(N-2)}}.$$

These calculations can be repeated and eventually lead to $1 - \gamma_1 = 2(1 + t_1)/t_1 > 1$.

M. Case 4. Recall that $k_N = 2$, $p_N = 5/2k^{(N-1)}$, $q_N = 1/k^{(N-1)}$, $\deg \varphi_N = 5$, $\delta_0 = \frac{2}{5}$, $\eta_0 = 5$ and we expect 3 preimages near D_{N-1} . Here $1 - \gamma_N = 7/2k^{(N-1)}$ and with $r_N = 2s_{N-1} - 1$ we get $1 - \gamma_{N-1} = (3 + s_{N-1})/k^{(N-1)}$, $p_{N-1} = \eta_0 s_{N-1}/k^{(N-1)}$, $1 + \delta_1 = (3 + s_{N-1})/5s_{N-1}$ and

$$\kappa = \kappa_1 = 1 + \frac{3}{s_{N-1}}.$$

We consider only the possibility (α) , i.e. $\varphi_{N-1} = \text{const} \cdot (z - w)^{\eta_0}$ and $c_{N-1} = uR(z)/(z - w)^{3/s_{N-1}}$. (It is easy to check that the possibilities (β) – (δ) lead to a contradiction.) We have two possibilities: either $3 \nmid s_{N-1}$ or $s_{N-1} = 3t_{N-1}$.

If $3 \nmid s_{N-1}$ then $\mu_{D_{N-1}} = 1$ and hence $\deg_{\text{top}} \tilde{c}_{N-1} = 3$. The only possibility is $s_{N-1} = k_{N-1} = \eta_1 = 2$ and $\tilde{c}_{N-1} = \text{const} \cdot z(z - 3w/2)^2/(z - w)^3$ with $\deg_{\infty} \tilde{c} = 1$. Here $1 - \gamma_{N-1} = 5/2k^{(N-2)}$, $p_{N-1} = \eta_0/k^{(N-2)}$ and $\deg \varphi_{N-1} = \eta_0\eta_1$. Since $5/2 > 1$, we cannot have $N - 1 = 1$.

The calculations for θ_{N-2} are the following: $\Delta\gamma_{N-1} = (2s_{N-2} - 1)/2k^{(N-2)}$, $1 - \gamma_{N-2} = (2 + s_{N-2})/k^{(N-2)}$, $p_{N-2} = \eta_0\eta_1s_{N-2}/k^{(N-2)}$, $1 + \delta_2 = (2 + s_{N-2})/\eta_0\eta_1s_{N-2}$ and $\kappa = \kappa_2 = 1 + \frac{2}{s_{N-2}}$. Moreover, $\deg_{\text{top}} D_{N-2} = 2$. Here also only the case (α) does not lead to a contradiction in estimating of $\deg_{\text{top}} D_{N-2}$; thus $\varphi_{N-2} = \text{const} \cdot (z - w)^{\eta_0\eta_1}$ and $c_{N-2} = \text{const} \cdot u/(z - w)^{2/s_{N-2}}$. If s_{N-2} is odd and ≥ 3 , then the condition $\deg c_{N-2} = 0$ provides a contradiction. Otherwise $s_{N-2} = 2t_{N-2}$, $k_{N-2} = \eta_2 = t_{N-2}$, $1 - \gamma_{N-2} = 2(1 + t_{N-2})/t_{N-2}k^{(N-3)}$, $p_{N-2} = 2\eta_0\eta_1/k^{(N-3)}$ and we can repeat these calculations for θ_{N-3} , θ_{N-4} , ... For θ_1 we get $1 - \gamma_1 = 2(1 + t_1)/t_1 > 1$ (a contradiction).

Let $s_{N-1} = 3t_{N-1}$. Then $\mu_{D_{N-1}} = 3$ and we find $k_{N-1} = \eta_1 = t_{N-1}$ and $\tilde{c}_{N-1} = \text{const} \cdot \frac{z}{z-w}$. Also $1 - \gamma_{N-1} = 3(1 + t_{N-1})/k^{(N-2)}t_{N-1}$, $p_{N-1} = 3\eta_0/k^{(N-2)}$. These calculations can be repeated and eventually lead to $1 - \gamma_1 = 3(2 + t_1)/(1 + t_1) > 1$. $\square$

We shall finish this section by estimating the degree of the non-properness set $S(F)$. Let D_{ik} , $k \leq M_1$ be a left divisor of non-properness type with associated left N-P chart θ and the component S_k of $S(P)$ (see (6.1)). Since $\deg S_k = \max(\deg \varphi, \deg \psi) = \deg \varphi$ (because $\deg f > \deg g$, see Theorem 3.2) and $\deg \varphi \leq \deg_{\mathcal{E}_k} f = \alpha m$ (this holds for any left N-P chart), it follows that

$$\deg S_k \leq \alpha m, \quad k \leq M_1. \quad (6.9)$$

The analogous estimate

$$\deg S_k \leq \beta m = \deg_y f, \quad M_1 < k \leq M, \quad (6.10)$$

holds for any right divisor of non-properness type.

It is also clear that

$$\deg \varphi \leq \beta m \quad (6.11)$$

for any N-P chart of transitory type. Generally, the following result generalizes the latter three inequalities.

Proposition 6.13. *One has the following equalities:*

$$\sum_{\theta: \text{leftnonprop.}} k \cdot \deg \varphi = \alpha m, \quad (6.12)$$

$$\left(\sum_{\theta: \text{rightnonprop.}} + \sum_{\theta: \text{transit.}} \right) k \cdot \deg \varphi = \beta m;$$

here $k = k(\theta)$, $\varphi = \varphi_\theta$.

This implies that

$$\deg S(P) \leq \deg P. \quad (6.13)$$

Proof. Consider the left case. Firstly we shall show formula (6.12).

To see it, we must recall the definition of the Newton–Puiseux resolution of indeterminacy of f (Algorithm 2.10). Each chart in (6.12) is associated with a system $[(I_1, a_1), \dots, (I_{d-1}, a_{d-1}), I_d]$ of edges $I_j \subset \Gamma(f_j)$ and zeros $a_j \neq 0$ of the polynomials $\varphi_{\theta_j}(u)$. Each edge I_{j+1} is related to ‘resolution’ at a_j and lies on the left to the edge $I'_j \subset \Gamma(f_{j+1})$ associated to I_j .

Thus we have $\deg \varphi_{\theta_d} = \text{ord}_{a_{d-1}} \varphi_{\theta_{d-1}}$ (the order of zero of $\varphi_{\theta_{d-1}}$ at a_{d-1}). Next, we have $\text{ord}_{a_{d-2}} \varphi_{\theta_{d-2}} = \sum \text{ord}_{a_{d-1}} \varphi_{\theta_{d-1}}$, the sum over all zeros a_{d-1} of $\varphi_{\theta_{d-1}}$ (taking into account zeros which differ by a root of 1 of order k_{d-1} and including $u = 0$) with fixed $(I_1, a_1), \dots, (I_{d-2}, a_{d-2})$. This sum equals $\sum_{a_{d-1}}^{k_{d-1}} k_{d-1} \cdot \text{ord}_{a_{d-1}} \varphi_{\theta_{d-1}} + \text{ord}_0 \varphi_{\theta_{d-1}}$.

Analogous identities hold for charts of lower depth. Thus summing them up one arrives at the formula (6.12), where $\alpha m = \deg \varphi_{\bar{\theta}}$ for the left chart $\bar{\theta}$ associated with the first left edge of $\Gamma(f)$.

In the right case the proof is practically the same. The only difference is that we finish the Newton–Puiseux resolution process either when arriving at a chart of non-properness type or of transitory type. $\square$

Remark 6.14. Since the N–P charts can be used on any polynomial maps (e.g. not Jacobian) the inequalities like (6.9), (6.10), (6.13) can be applied to such maps. In this way one can obtain estimates for the degree of the non-properness set, analogous to Jelonek’s estimate $\deg S(P) \leq (\deg f \cdot \deg g - \deg_{\text{top}} P) / \min(\deg f, \deg g)$ (see [5,15]).

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Appendix. Schwarz–Christoffel integrals

Definition A.1. By a **Schwarz–Christoffel integral** we mean an integral of the form

$$\Psi(u) = \int_{u_0}^u \prod_{i=1}^{s+1} (v - u_i)^{\vartheta_i} dv =: \int_{u_0}^u \Phi(v) dv. \quad (A.1)$$

Here the exponents ϑ_i are complex numbers, $u_i \neq u_j$ for $i \neq j$ and the functions Φ and Ψ are treated as multi-valued holomorphic functions (with singularities at u_i).

By a **Darboux function** we mean a function of the form

$$\text{const} \cdot \prod_{i=1}^r (u - u_i)^{\beta_i}. \quad (A.2)$$

(Darboux functions $\prod f_i^{\beta_i}$, f_i – polynomials, appear often as first integrals of polynomial vector fields.)

Remark A.2. A Schwarz–Christoffel integral of the form $\int_{u_k}^u \prod_{i=0}^{s+1} (v - u_i)^{\vartheta_i}$ (i.e. with the constant u_0 equal to one of the singularities) is a particular case of a hypergeometric integral (see [6])

$$F_{j,k}(u_0, \dots, u_s, u_{s+1}) = \int_{u_j}^{u_k} \prod_{i=1}^{s+1} (v - u_i)^{\vartheta_i} dv, \quad (A.3)$$

where ϑ_i are (in general) complex numbers. (In (A.1) one has $u_0 = u$ and $\vartheta_0 = 0$).

When some exponents are < -1 (and are not integer), then the right-hand side of (A.3) is defined by means of the formula $(e^{2\pi i \vartheta_j} - 1)^{-1} (e^{2\pi i \vartheta_k} - 1)^{-1} \times \int_{C(u_j, u_k)} \prod_{i=0}^{s+1} (v - u_i)^{\vartheta_i}$, where $C(u_j, u_k)$ is a some special cycle (Pochhammer cycle) in $\mathbb{C} \setminus \{u_0, \dots, u_{s+1}\}$ (see [6]). If $\sum \vartheta_i$ is not an integer, then the point $u = \infty$ is also singular for the subintegral form; in that case one adds the corresponding integrals $\int_{u_j}^{\infty}$ to the collection (A.3).

We ask for conditions that the function (A.1) be a Darboux function modulo an additive constant.

Lemma A.3. *If Φ is a polynomial, then Ψ is also a polynomial, and if Φ is of the form $(u - u_0)^{\vartheta} \times \text{polynomial}$, $\vartheta \notin \mathbb{Z}$, then $\Psi = (u - u_0)^{\vartheta+1} \times \text{polynomial}$.*

From this point on we shall consider the cases when

$$s \geq 1, \quad \vartheta_i \in \mathbb{Q} \setminus \mathbb{Z}_+. \quad (\text{A.4})$$

We use the notations $\mathbb{Z}_+ = \{0, 1, 2, \dots\}$, $\mathbb{Z}_- = \{0, -1, -2, \dots\}$.

Lemma A.4. *The necessary and sufficient conditions that Ψ is a Darboux function are the following:*

- if some $\vartheta_i \in \mathbb{Z}_- \setminus 0$, then the residue $\text{Res}_{u_i} \Phi = 0$ (e.g. $\vartheta_i \neq -1$);
- all the ‘values’ $c_i = \Psi(u_i)$ are equal.

Then we have

$$\Psi(u) = R(u) \cdot \prod (u - u_i) \cdot \Phi(u) = R(u) \prod (u - u_i)^{\vartheta_i+1},$$

where $R(u)$ is a polynomial.

Proof. When $\vartheta_i > -1$ the ‘value’ $\Psi(u_i)$ is the usual value of a function at the point u_i . If $\vartheta_i < -1$, then we define the ‘value’ as follows. Assume that $u_i = 0$, $\vartheta_i = \vartheta$. Near $v = 0$ we have $\Phi(v) = v^{\vartheta} \tilde{\Phi}(v)$ with a holomorphic function $\tilde{\Phi}$. We perform the integration by parts $\int_{u_0}^u \Phi(v) dv = \frac{1}{\vartheta+1} v^{\vartheta+1} \tilde{\Phi}(v) \Big|_{u_0}^u - \frac{1}{\vartheta+1} \int_{u_0}^u v^{\vartheta+1} \tilde{\Phi}'(v) dv$ several times, until we obtain a convergent integral. (In general the constants c_i depend on the paths of integration. However the condition $c_1 = c_2 = \dots = c_s$ is well defined. Note that the differences $c_i - c_j = \int_{u_j}^{u_i} \Phi(v)$, treated as functions of $u_1, \dots, u_s$, are also hypergeometric functions.)

We get

$$\Psi(u) = c_i + (u - u_i)^{\vartheta_i+1} \times (\text{holomorphic function}) \quad (\text{A.5})$$

near each u_i .

(Thus any integral of the form $\int^u (v - v_0)^{\vartheta} \times \text{polynomial}$ is of Darboux type $(u - v_0)^{\vartheta+1} \times \text{polynomial}$; this shows also Lemma A.3 and justifies the restriction (A.4).) $\square$

In the paper we have often dealt with Schwarz–Christoffel integrals such that the subintegral function takes the form

$$\Phi(u) = u^{v_0} \prod_{i=1}^s (u^k - w_i)^{v_i}, \quad w_i \neq w_j, \quad (\text{A.6})$$

where k is an integer, $v_i \notin \mathbb{Z}_+$ for $i > 0$, and $s \geq 1$.

Lemma A.5. *If the integral (A.1) with Φ of the form (A.6) is of Darboux type, then it equals*

$$\Psi(u) = R(u^k) \cdot u \prod (u^k - w_i) \cdot \Phi(u),$$

where R is a polynomial.

Proof. The substitution $v = w^{1/k}$ gives the Schwarz–Christoffel integral $\frac{1}{k} \int w^{(v_0+1)/k-1} \prod (w - w_i)^{v_i} dw$. Now we apply Lemma A.4. $\square$

The following results were probably known already to Liouville.

Lemma A.6. *If the Schwarz–Christoffel integral (A.1) with the assumption (A.4) represents a Darboux function, then*

$$s + (1 + \deg \Phi) \in \mathbb{Z}_-,$$

where $\deg \Phi := \sum_{i=1}^s \vartheta_i$. In this case the polynomial R from Lemma A.4 is uniquely defined and has degree $\deg R = -s - \deg \Phi$, i.e. $\deg \Psi = 0$.

Proof. Assume the representation from the thesis of Lemma A.4, i.e. that for general lower limit u_0 in the integral (A.1) $\Psi(u) = c + R(u) \prod (u - u_i)^{\vartheta_i+1}$ for a constant $c = c(u_0)$.

Consider the case $d := \deg \Phi = \sum \vartheta_i > -1$. Then the integral diverges at infinity and we have

$$\Psi(u) \sim u^{1+d}, \quad u \rightarrow \infty.$$

But the function $c + R(u) \cdot \prod (u - u_i)^{\vartheta_i+1}$ should have the asymptotic $\sim u^{s+1+\deg R+d}$. Of course, this cannot be true for $s \geq 1$.

The case $d = -1$ gives the asymptotic $\sim \log u$, which cannot be the asymptotic for a Darboux function.

Consider the case $d < -1$. We have the expansion

$$\Psi(u) = c_1 + c_2 u^{1+d} \dots, \quad u \rightarrow \infty,$$

where $c_1 = \Psi(\infty)$ and $c_2 \neq 0$. For the function $c + R(u) \cdot \prod (u - u_i)^{\vartheta_i+1}$ we have the following possibilities of the asymptotic behavior:

- (a) $c_3 u^{s+1+\deg R+d} + \dots \rightarrow \infty$, $c_3 \neq 0$ if $s + 1 + \deg R + d > 0$, or
- (b) $c_3 + c_4 u^{s+1+\deg R+d} + \dots$, $c_3 = c$, $c_4 \neq 0$ if $s + 1 + \deg R + d < 0$, or
- (c) $c_3 + c_4 u^{-1} + \dots$ if $s + 1 + \deg R + d = 0$. (In this case we have: $R = du^{\deg R} + \dots$, the function $c + R \prod$ is analytic near $u = \infty$ and $c_3 = c + d$.)

We see that only in the case (c) are the above two expansions compatible. $\square$

Lemma A.7. In the assumptions of Lemma A.6 one has that at least one $\vartheta_i \notin \mathbb{Z}$.

Proof. Suppose that all $\vartheta_i \in \mathbb{Z}_- \setminus 0$ and that the formula from Lemma A.4 holds true. Then in (A.1) the functions $\Phi(u)$ and $\Psi(u)$ are rational. Hence we can choose the constant of integration such that $\Psi(\infty) = 0$. But then $\Psi(u) = R(u) \prod (u - u_i)^{\vartheta_i+1}$ should behave like u^{1+d} at infinity, i.e. it should be $\deg R = -s < 0$. It is a contradiction. $\square$

Lemma A.8. If the Schwarz–Christoffel integral (A.1) with the subintegral function (A.6) represents a Darboux function, then

$$s + \frac{1}{k}(1 + \deg \Phi) \in \mathbb{Z}_-.$$

In this case the polynomial R from Lemma A.5 is uniquely defined and has degree $\deg R = -s - \frac{1}{k}(1 + \deg \Phi)$, i.e. $\deg \Psi = 0$.

Proof. This repeats the proof of Lemma A.6. $\square$

Example A.9. Liouville has shown that the Chebyshev integral $\int_0^u v^{\alpha-1}(1-v)^{\beta-1} dv$ is elementary iff one of the numbers α , β , $\alpha + \beta$ is an integer. We assume that α , $\beta \notin \mathbb{Z}$ and ask whether it is of Darboux type (modulo a constant). The answer is: if and only if $\alpha + \beta \in \mathbb{Z}_-$.

Indeed, we have the condition $B(\alpha, \beta) = \int_0^1 v^{\alpha-1}(1-v)^{\beta-1} dv = 0$. Here, when $\alpha < -1$ or $\beta < -1$, then the principal value of the integral is defined as the analytic prolongation of the Beta-function $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$. The Gamma-function has no zeros and has poles only in $\mathbb{Z}_-$.

For example, we have $\int (1-v^2)^{-1/2} = \arcsin u$, but $\int (1-v^2)^{-3/2} = u(1-u^2)^{-1/2}$, $\int (1-v^2)^{-5/2} = (u-2u^3/3)(1-u^2)^{-3/2}$.

I do not know many examples of Schwarz–Christoffel integrals which are of Darboux form. The main examples known to the author rely on a calculation of the Chebyshev integral. On the other hand, one needs many such examples at hand when working on a counterexample to the Jacobian Conjecture.

Therefore there is a need to determine the number of degrees of freedom in the choice of the subintegral function $\Phi = \prod_{i=1}^{s+1} (u - u_i)^{\vartheta_i}$; more precisely, we fix the exponents ϑ_i and try to deform the zeros u_i .

The group $\text{Aff}(\mathbb{C})$ acts on the space of such Φ 's via the formula $(u_1, \dots, u_{s+1}) \rightarrow (\lambda u_1 + \mu, \dots, \lambda u_{s+1} + \mu)$. In the case of the form (A.6) of Φ we have only the action of $\mathbb{C}^*$. It turns out that there are no more degrees of freedom.

Proposition A.10. (a) Assume that $\vartheta_i < 0$ are fixed. Then there are only a finite number of $\text{Aff}(\mathbb{C})$ -orbits of systems $(u_1, \dots, u_{s+1})$ such that the integral (A.1) represents a Darboux function.

(b) In the case of the form (A.6) with fixed $\epsilon_i < 0$ there are only a finite number of $\mathbb{C}^*$ -orbits of systems $(w_1, \dots, w_s)$ such that the integral (A.1) represents a Darboux function.

Proof. (a) Assume that $R(u) = a \cdot \prod_{j=1}^r (u - v_j)$, $r = \deg R = -\sum (1 + \vartheta_i)$ in Lemma A.4. Then the condition $\Psi' = \Phi$ means

$$R'(u) \prod_i (u - u_i) + R(u) \cdot \sum_i (\vartheta_i + 1) \prod_{m \neq i} (u - u_m) \equiv 1. \quad (\text{A.7})$$

It implies that: $v_i \neq v_j$ for $i \neq j$ and $v_j \neq u_i$; of course, we have also $u_i \neq u_j$ for $i \neq j$.

The equation $\Psi' = \Phi$ (where $\Phi(u) \sim u^{\sum \vartheta_i} = u^{-r-s-1}$ as $u \rightarrow \infty$) can be also translated to the condition

$$\Psi(u) - a = bu^{-(r+s)} + O(u^{-r-s-1}), \quad u \rightarrow \infty,$$

where $b = 1/(r + s)$. We rewrite it as follows

$$\log \prod \left(1 - \frac{v_j}{u}\right) \prod \left(1 - \frac{u_i}{u}\right) = \frac{c}{u^{r+s}} + \dots,$$

$c = \frac{b}{a}$. This leads to the following system of algebraic equations

$$\begin{aligned} \sum v_j + \sum (\vartheta_i + 1)u_i &= 0 \\ \sum v_j^2 + \sum (\vartheta_i + 1)u_i^2 &= 0 \\ &\dots \\ \sum v_j^{r+s-2} + \sum (\vartheta_i + 1)u_i^{r+s-1} &= 0 \\ \sum v_j^{r+s-1} + \sum (\vartheta_i + 1)u_i^{r+s} &= c. \end{aligned} \tag{A.8}$$

We treat $v_1, \dots, v_r, u_1, \dots, u_{s+1}$ as unknown. The group $\text{Aff}(\mathbb{C})$ acts diagonally on these collections. We add the additional equation $u_1 = 0$ to the system (A.8), i.e. we fix a section to the orbits of the translation subgroup $\mathbb{C} \subset \text{Aff}(\mathbb{C})$. Then the multiplication group $\mathbb{C}^* \subset \text{Aff}(\mathbb{C})$ acts equivariantly on the system (A.8), i.e. on v_j 's, u_i 's and on c (i.e. on a). After fixing $c = 1$ we get $r + s$ equations on $r + s$ unknowns.

For $u_1 = 0$ the determinant of the linearization of the left-hand side of (A.8) is proportional to the Vandermonde determinant $\prod_{2 \leq i < j} (u_i - u_j) \prod_{i < j} (v_i - v_j) \prod_{i,j} (u_i - v_j)$. It is nonzero at the points corresponding to our rational function ψ (see above). This means that the solutions which are of interest for us are isolated and there are a finite number ($\leq (r + s)!$) of them.

(b) This proof is analogous with $R(u) = \prod (u^k - v_j)$ in (A.7) and with the use of the function $\log \prod (1 - v_j/u^k) \prod (1 - u_i/u^k)$. $\square$

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